Supplementary Material for "Model Averaging Approach for Analyzing Complex Survey Data"

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This Supplementary Material is organized as follows. Section S1 explains the motivation for the proposed weight selection criterion. Additional theoretical results are presented in Section S2, and Section S3 provides proofs of all theoretical results. Section S4 discusses the relationship between model averaging and high-dimensional regression in survey sampling. Section S5 presents additional numerical analyses.

S1 Motivation for Weight Selection Criterion

In frequentist model averaging, Mallows-type criteria are typically constructed from an unbiased estimator of the risk function and consist of a goodness-of-fit term and a penalty term (Hansen 2007, Wan et al. 2010). Based on this principle, we propose a Mallows-type criterion for complex survey data. Note that

$$E\left\|\boldsymbol{y}_s-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2=E\left\|\boldsymbol{\mu}_s-\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2+E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2$$

$$
\begin{aligned}
&+ 2E\left[\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}' \boldsymbol{\Phi}_s \{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}\right] \\
&+ 2E\left[\boldsymbol{e}_s' \boldsymbol{\Phi}_s \{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}\right] + E_\epsilon(\boldsymbol{e}_U' \boldsymbol{e}_U) \\
= {} & R_N(\boldsymbol{\omega}) + N\sigma^2 + 2E\left[\boldsymbol{e}_s' \boldsymbol{\Phi}_s \{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}\right] \\
&+ \left\{\tilde{R}_N(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega})\right\} + E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 \\
&+ 2E\left[\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}' \boldsymbol{\Phi}_s \{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}\right]. \qquad \text{(S.1)}
\end{aligned}
$$

According to (S.24)–(S.26) in the proof of Theorem 1, the last three terms in the above equation are asymptotically negligible relative to $R_N(\boldsymbol{\omega})$. Consequently,

$$
E\left\|\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 - 2E\left[\boldsymbol{e}_s' \boldsymbol{\Phi}_s \{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}\right] \approx R_N(\boldsymbol{\omega}) + N\sigma^2.
$$

Hence, a feasible penalty term can be constructed by estimating $2E\left[\boldsymbol{e}_s' \boldsymbol{\Phi}_s \{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}\right]$. When $E_\epsilon E_d(\cdot) = E_d E_\epsilon(\cdot)$, which holds under a non-informative sampling design, a design unrelated to the study variable (Särndal et al. 1978, Raghunath 1990), we have

$$
\begin{aligned}
&2E\left[\boldsymbol{e}_s' \boldsymbol{\Phi}_s \{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}\right] \\
&= 2E_\epsilon E_d\left[\boldsymbol{e}_s' \boldsymbol{\Phi}_s \boldsymbol{\mu}_s - \sum_{m=1}^{M} \omega_m \boldsymbol{e}_s' \boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s} \left(\boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s}\right)^{-1} \boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{y}_s\right] \\
&= 2E_d E_\epsilon\left[\boldsymbol{e}_s' \boldsymbol{\Phi}_s \boldsymbol{\mu}_s - \sum_{m=1}^{M} \omega_m \boldsymbol{e}_s' \boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s} \left(\boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s}\right)^{-1} \boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{\mu}_s\right] \\
&\quad - 2E_d E_\epsilon\left[\sum_{m=1}^{M} \omega_m \boldsymbol{e}_s' \boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s} \left(\boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s}\right)^{-1} \boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{e}_s\right] \\
&= -2\sigma^2 E_d\left[\sum_{m=1}^{M} \omega_m \operatorname{tr}\left\{\boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s} \left(\boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s}\right)^{-1} \boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s\right\}\right] \\
&\triangleq -2\sigma^2 E_d\left[k_s(\boldsymbol{\omega})\right]. \qquad \text{(S.2)}
\end{aligned}
$$

This indicates that $-2\sigma^2 k_s(\boldsymbol{\omega})$ is a design-unbiased estimator of $2E\left[\boldsymbol{e}_s' \boldsymbol{\Phi}_s \{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}\right]$. Therefore, the corresponding penalty term is $2\sigma^2 k_s(\boldsymbol{\omega})$, where

$$
k_s(\boldsymbol{\omega}) = \sum_{m=1}^{M} \omega_m \operatorname{tr}\{\boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s} (\boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s})^{-1} \boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s\}.
$$

Noting that the trace term $\operatorname{tr}\{\boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s} (\boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s})^{-1} \boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s\}$ in $k_s(\boldsymbol{\omega})$ is complicated, we replace each first-order inclusion probability π_i in the first $\boldsymbol{\Phi}_s$ by their population

mean, $\sum_U \pi_i/N = n/N = f$. A similar idea of replacing individual values by their average is used to construct the generalized cross-validation (GCV) criterion (Craven & Wahba 1979). This approximation yields the simplified penalty term $2f^{-1}\sigma^2 k(\boldsymbol{\omega})$, where $k(\boldsymbol{\omega}) = \sum_{m=1}^{M} \omega_m k_m$. The resulting weight selection criterion is

$$C_n(\boldsymbol{\omega}) = \Big(\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\Big)' \boldsymbol{\Phi}_s \Big(\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\Big) + 2f^{-1}\sigma^2 k(\boldsymbol{\omega}).$$

S2 Additional Theoretical Results

In this section, we present two additional theoretical results that follow as corollaries of Theorems 1 and 2.

Corollary 1 *If Conditions 1–2, $E_\epsilon E_d(\cdot) = E_d E_\epsilon(\cdot)$, and $k_M(f\xi_N)^{-1} = o(1)$ hold, then*

$$E\left(C_n(\boldsymbol{\omega})\right) = R_N(\boldsymbol{\omega}) + N\sigma^2 + o\left(R_N(\boldsymbol{\omega})\right).$$

Moreover, if $\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$ and $k_M^2 = O(N)$ also hold, then

$$E\left(C_n^*(\boldsymbol{\omega})\right) = R_N(\boldsymbol{\omega}) + N\sigma^2 + o\left(R_N(\boldsymbol{\omega})\right).$$

Proof. See Supplementary Material S3.3.

Corollary 1 shows that, under the interchangeability condition $E_\epsilon E_d(\cdot) = E_d E_\epsilon(\cdot)$, the proposed criteria $C_n(\boldsymbol{\omega})$ and $C_n^*(\boldsymbol{\omega})$ remain asymptotically unbiased for $R_N(\boldsymbol{\omega}) + N\sigma^2$ even when Condition 3 is relaxed to the weaker assumption $k_M(f\xi_N)^{-1} = o(1)$.

Corollary 2 *If Conditions 1–4 hold, then*

$$C_n(\boldsymbol{\omega}) = R_N(\boldsymbol{\omega}) + N\sigma^2 + o_p\left(R_N(\boldsymbol{\omega})\right).$$

Moreover, if $\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$ and $k_M^2 = O(N)$ also hold, then

$$C_n^*(\boldsymbol{\omega}) = R_N(\boldsymbol{\omega}) + N\sigma^2 + o_p\left(R_N(\boldsymbol{\omega})\right).$$

Proof. See Supplementary Material S3.5.

As shown in Corollary 2, the proposed criteria $C_n(\boldsymbol{\omega})$ and $C_n^*(\boldsymbol{\omega})$ are consistent for $R_N(\boldsymbol{\omega}) + N\sigma^2$ in a relative sense. Specifically, after normalization by N, we have $\bar{C}_n(\boldsymbol{\omega}) = \bar{R}_N(\boldsymbol{\omega}) + \sigma^2 + o_p(\bar{R}_N(\boldsymbol{\omega}))$ and $\bar{C}_n^*(\boldsymbol{\omega}) = \bar{R}_N(\boldsymbol{\omega}) + \sigma^2 + o_p(\bar{R}_N(\boldsymbol{\omega}))$, where $\bar{C}_n(\boldsymbol{\omega}) = N^{-1}C_n(\boldsymbol{\omega})$, $\bar{C}_n^*(\boldsymbol{\omega}) = N^{-1}C_n^*(\boldsymbol{\omega})$, and $\bar{R}_N(\boldsymbol{\omega}) = N^{-1}R_N(\boldsymbol{\omega})$ is typically bounded.

S3 Proofs of Theoretical Results

In this section, we provide proofs of all theoretical results in the main manuscript and the Supplementary Material S2. Throughout the proofs, we use C to denote a generic positive constant whose value may vary from place to place. For any $m \times 1$ vector b and $m \times m$ symmetric matrix A, we write $\|b\|_A^2 \triangleq b'Ab$.

S3.1 Lemmas for the proofs of Theorems 1–4

Lemma 1 *If Condition 1 holds, then*

$$E_\epsilon \left\| \tilde{\boldsymbol{\beta}}_{(m)} - \boldsymbol{\beta}_{(m)}^* \right\|^4 \leq CN^{-2}k_m^2.$$

Proof. By the definition of $\boldsymbol{\beta}_{(m)}^*$, we have

$$\sum_{i \in U} \boldsymbol{x}_{i(m)} \left(\mu_i - \boldsymbol{x}_{i(m)}' \boldsymbol{\beta}_{(m)}^* \right) = \mathbf{0}.$$

Since $e_{i(m)}^* - e_i = \mu_i - \boldsymbol{x}_{i(m)}' \boldsymbol{\beta}_{(m)}^*$, it follows that

$$\sum_{i \in U} \boldsymbol{x}_{i(m)} e_{i(m)}^* = \sum_{i \in U} \boldsymbol{x}_{i(m)} e_i.$$

Consequently, for each $j = 1, \ldots, k_m$,

$$E_\epsilon \left(\sum_{i \in U} x_{ij(m)} e_{i(m)}^* \right) = 0. \tag{S.3}$$

Applying Loève's c_r inequality, for any $q \geq 1$, yields

$$
\begin{aligned}
E_\epsilon \left\| \frac{1}{N} \sum_{i \in U} \boldsymbol{x}_{i(m)} e^*_{i(m)} \right\|^{2q} &= \frac{1}{N^{2q}} E_\epsilon \left| \sum_{j=1}^{k_m} \left(\sum_{i \in U} x_{ij(m)} e^*_{i(m)} \right)^2 \right|^q \\
&\leq \frac{C k_m^{q-1}}{N^{2q}} \sum_{j=1}^{k_m} E_\epsilon \left| \sum_{i \in U} x_{ij(m)} e^*_{i(m)} \right|^{2q} \\
&\leq \frac{C k_m^{q-1}}{N^{2q}} \sum_{j=1}^{k_m} \left\{ \sum_{i \in U} \left(E_\epsilon \left| x_{ij(m)} e^*_{i(m)} \right|^{2q} \right)^{\frac{1}{q}} \right\}^q, \qquad \text{(S.4)}
\end{aligned}
$$

where the last inequality follows from (S.3) and Theorem 2 of Whittle (1960). Then, by Condition 1, we have

$$
\begin{aligned}
E_\epsilon \left\| \tilde{\boldsymbol{\beta}}_{(m)} - \boldsymbol{\beta}^*_{(m)} \right\|^4 =& E_\epsilon \left\| \left(\sum_{i \in U} \boldsymbol{x}_{i(m)} \boldsymbol{x}'_{i(m)} \right)^{-1} \sum_{i \in U} \boldsymbol{x}_{i(m)} e^*_{i(m)} \right\|^4 \\
\leq& E_\epsilon \left\{ \lambda_{\min}^{-4} \left(\frac{1}{N} \boldsymbol{X}'_{(m)U} \boldsymbol{X}_{(m)U} \right) \cdot \left\| \frac{1}{N} \sum_{i \in U} \boldsymbol{x}_{i(m)} e^*_{i(m)} \right\|^4 \right\} \\
\leq& C N^{-2} k_m^2.
\end{aligned}
$$

This completes the proof of Lemma 1.

Lemma 2 *If Condition 2 holds, then*

$$
E \left(\frac{1}{N} \sum_{i \in s} \frac{z_i}{\pi_i} - \frac{1}{N} \sum_{i \in U} z_i \right)^2 \leq \frac{C}{nN} \sum_{i \in U} E_\epsilon \left(z_i^2 \right).
$$

Proof. Let $I_i = 1$ if $i \in s$ and $I_i = 0$ otherwise. By Condition 2, we have

$$
\begin{aligned}
E \left(\frac{1}{N} \sum_{i \in s} \frac{z_i}{\pi_i} - \frac{1}{N} \sum_{i \in U} z_i \right)^2 &= E_\epsilon \left[E_d \left\{ \frac{1}{N} \sum_{i \in U} \frac{z_i (I_i - \pi_i)}{\pi_i} \right\}^2 \right] \\
&= E_\epsilon \left(\frac{1}{N^2} \sum_{i \in U} \frac{1 - \pi_i}{\pi_i} z_i^2 + \frac{1}{N^2} \sum_{i \neq j \in U} \frac{\pi_{ij} - \pi_i \pi_j}{\pi_i \pi_j} z_i z_j \right) \\
&\leq E_\epsilon \left(\frac{C}{N^2} \sum_{i \in U} z_i^2 \right) + E_\epsilon \left(\frac{C}{nN^2} \sum_{i \neq j \in U} |z_i z_j| \right) \\
&\leq \frac{C}{nN} \sum_{i \in U} E_\epsilon \left(z_i^2 \right).
\end{aligned}
$$

This completes the proof of Lemma 2.

Lemma 3 *If Conditions 1–2 hold, then*

$$E\left\|\hat{\boldsymbol{\beta}}_{(m)}-\boldsymbol{\beta}^*_{(m)}\right\|^2 \leq Cn^{-1}k_m.$$

Proof. Condition 1 implies

$$E_\epsilon\left(x_{ij(m)}e^*_{i(m)}\right)^2 \leq C.$$

Applying Lemma 2, we have

$$E\left(\frac{1}{N}\sum_{i\in s}\frac{x_{ij(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}x_{ij(m)}e^*_{i(m)}\right)^2 \leq Cn^{-1}.$$

Combining this bound with (S.4) gives

$$\begin{aligned}
&E\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^2 \\
&= E\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}+\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^2 \\
&\leq 2E\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^2+2E\left\|\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^2 \\
&= 2\sum_{j=1}^{k_m}\left\{E\left(\frac{1}{N}\sum_{i\in s}\frac{x_{ij(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}x_{ij(m)}e^*_{i(m)}\right)^2\right\}+2E\left\|\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^2 \\
&\leq Cn^{-1}k_m. \qquad \text{(S.5)}
\end{aligned}$$

From Condition 1, it follows that

$$\begin{aligned}
E\left\|\hat{\boldsymbol{\beta}}_{(m)}-\boldsymbol{\beta}^*_{(m)}\right\|^2 =& E\left\|\left(\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}\boldsymbol{x}'_{i(m)}}{\pi_i}\right)^{-1}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^2 \\
\leq& E\left\{\lambda^{-2}_{\min}\left(\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)\cdot\left\|\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^2\right\} \\
\leq& CE\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^2. \qquad \text{(S.6)}
\end{aligned}$$

Combining (S.5) with (S.6) yields the desired result.

Lemma 4 *If Conditions 1–2 and $k_M(f\xi_N)^{-1}=o(1)$ hold, then*

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}}\frac{\left|\tilde{R}_N(\boldsymbol{\omega})-R_N(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})}=o(1).$$

Proof. Note that

$$\begin{aligned}\tilde{R}_N(\boldsymbol{\omega}) &= E\left\|\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) + \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\|^2 \\ &= R_N(\boldsymbol{\omega}) + E\left\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\|^2 + 2E\left\{\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}'\left\{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}.\end{aligned}$$

Thus, it suffices to show that

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{E\left\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\|^2}{R_N(\boldsymbol{\omega})} = o(1) \tag{S.7}$$

and

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{E\left|\left\{\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}'\left\{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}\right|}{R_N(\boldsymbol{\omega})} = o(1). \tag{S.8}$$

Lemmas 1 and 3, together with Condition 1, imply that

$$\begin{aligned}&E\left\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\|^2 \\ &= E\left|\sum_{m=1}^{M}\sum_{l=1}^{M}\omega_m\omega_l\left(\boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)} - \boldsymbol{X}_{(m)U}\tilde{\boldsymbol{\beta}}_{(m)}\right)'\left(\boldsymbol{X}_{(l)U}\hat{\boldsymbol{\beta}}_{(l)} - \boldsymbol{X}_{(l)U}\tilde{\boldsymbol{\beta}}_{(l)}\right)\right| \\ &\le \sum_{m=1}^{M}\sum_{l=1}^{M}\omega_m\omega_l E\left|\left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(l)U}\left(\hat{\boldsymbol{\beta}}_{(l)} - \tilde{\boldsymbol{\beta}}_{(l)}\right)\right| \\ &\le \left\{\sum_{m=1}^{M}\omega_m\cdot\lambda_{\max}^{1/2}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)\cdot\left(E\left\|\tilde{\boldsymbol{\beta}}_{(m)} - \hat{\boldsymbol{\beta}}_{(m)}\right\|^2\right)^{1/2}\right\}^2 \\ &\le Cf^{-1}k_M. \end{aligned}\tag{S.9}$$

Thus, (S.7) follows from (S.9) and $k_M(f\xi_N)^{-1} = o(1)$.

We next verify (S.8). By the Cauchy–Schwarz inequality and (S.7),

$$\begin{aligned}&\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{E\left|\left\{\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}'\left\{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}\right|}{R_N(\boldsymbol{\omega})} \\ &\le \sup_{\boldsymbol{\omega}\in\mathcal{W}}\left\{\frac{R_N(\boldsymbol{\omega})\cdot E\left\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\|^2}{R_N^2(\boldsymbol{\omega})}\right\}^{1/2} \\ &= o(1).\end{aligned}$$

This completes the proof of Lemma 4.

Lemma 5 *If Conditions 1–3 hold, then*

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left|\tilde{L}_N(\boldsymbol{\omega}) - L_N(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})} = o_p(1).$$

Proof. Note that

$$\begin{aligned}\tilde{L}_N(\boldsymbol{\omega}) &= \|\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) + \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\|^2 \\ &= L_N(\boldsymbol{\omega}) + \|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\|^2 + 2\left\{\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}'\left\{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}.\end{aligned}$$

Thus, it suffices to show that

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\|^2}{R_N(\boldsymbol{\omega})} = o_p(1) \tag{S.10}$$

and

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left|\left\{\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}'\left\{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}\right|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.11}$$

Using Bonferroni's inequality and Markov's inequality, for any $\delta > 0$, we have

$$\begin{aligned}&P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left\{R_N^{-1}(\boldsymbol{\omega})\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\|^2\right\} > \delta\right) \\ &\leq P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\|\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\|^2 > \delta\xi_N\right) \\ &= P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left\{\sum_{m=1}^{M}\sum_{t=1}^{M}\omega_m\omega_t\left(\tilde{\boldsymbol{\beta}}_{(m)} - \hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(t)U}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right\} > \delta\xi_N\right) \\ &\leq \sum_{m=1}^{M}\sum_{t=1}^{M}P\left(\left|\left(\tilde{\boldsymbol{\beta}}_{(m)} - \hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(t)U}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right| > \delta\xi_N\right) \\ &\leq \delta^{-1}\xi_N^{-1}\sum_{m=1}^{M}\sum_{t=1}^{M}E\left|\left(\tilde{\boldsymbol{\beta}}_{(m)} - \hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(t)U}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right| \\ &\leq M\delta^{-1}\xi_N^{-1}\sum_{m=1}^{M}\left\{\lambda_{\max}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)\cdot E\left\|\tilde{\boldsymbol{\beta}}_{(m)} - \hat{\boldsymbol{\beta}}_{(m)}\right\|^2\right\} \\ &= O\left(\frac{M^2k_M}{f\xi_N}\right),\end{aligned}$$

where the last bound follows from Lemmas 1 and 3, together with Condition 1. Condition 3 then implies (S.10).

We next verify (S.11). By Condition 1 and Lemma 3,

$$
\begin{aligned}
E\left\|\boldsymbol{\mu}_U - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)}\right\|^2 &= E\left\|\boldsymbol{e}^*_{(m)U} - \boldsymbol{e}_U + \boldsymbol{X}_{(m)U}\boldsymbol{\beta}^*_{(m)} - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)}\right\|^2 \\
&\leq 3E_\epsilon\left\|\boldsymbol{e}^*_{(m)U}\right\|^2 + 3E_\epsilon\left\|\boldsymbol{e}_U\right\|^2 + 3E\left\|\boldsymbol{X}_{(m)U}\boldsymbol{\beta}^*_{(m)} - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)}\right\|^2 \\
&\leq CN, \qquad \text{(S.12)}
\end{aligned}
$$

where $\boldsymbol{e}^*_{(m)U} = [e^*_{i(m)}]_{i\in U}$ and $\boldsymbol{e}_U = [e_i]_{i\in U}$. Therefore, for any $\delta > 0$,

$$
\begin{aligned}
&P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}} R_N^{-1}(\boldsymbol{\omega})\left|\{\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\}'\{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\}\right| > \delta\right) \\
&\leq P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left|\{\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\}'\{\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\}\right| > \delta\xi_N\right) \\
&= P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left\{\sum_{m=1}^M\sum_{t=1}^M \omega_m\omega_t\left|\left(\boldsymbol{\mu}_U - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}_{(t)U}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right|\right\} > \delta\xi_N\right) \\
&\leq \sum_{m=1}^M\sum_{t=1}^M P\left(\left|\left(\boldsymbol{\mu}_U - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}_{(t)U}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right| > \delta\xi_N\right) \\
&\leq \delta^{-1}\xi_N^{-1}\sum_{m=1}^M\sum_{t=1}^M E\left|\left(\boldsymbol{\mu}_U - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}_{(t)U}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right| \\
&\leq \delta^{-1}\xi_N^{-1}\sum_{m=1}^M\sum_{t=1}^M\left\{E\left\|\boldsymbol{\mu}_U - \boldsymbol{X}_{(m)U}\hat{\boldsymbol{\beta}}_{(m)}\right\|^2\cdot\lambda_{\max}\left(\boldsymbol{X}_{(t)U}\boldsymbol{X}'_{(t)U}\right)\cdot E\left\|\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right\|^2\right\}^{1/2} \\
&= O\left(\frac{M^2\sqrt{nk_M}}{f\xi_N}\right),
\end{aligned}
$$

where the last bound follows from Lemmas 1 and 3, (S.12), and Condition 1. Condition 3 then implies (S.11). This completes the proof of Lemma 5.

Lemma 6 *If Conditions 1–2 and $k_M(f\xi_N)^{-1} = o(1)$ hold, then*

$$
E\left\|\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\|^2 = E\left\|\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\|^2_{\boldsymbol{\Phi}_s} + o(\xi_N),
$$

where $\boldsymbol{\omega}^* = \arg\min_{\boldsymbol{\omega}\in\mathcal{W}}\{R_N(\boldsymbol{\omega})\}$.

Proof. Note that

$$
\begin{aligned}
E\left\|\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\|^2 &= E\left\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\|^2_{\boldsymbol{\Phi}_s} + E\left\|\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\|^2 \\
&\quad + 2E\left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\}'\left\{\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\}
\end{aligned}
$$

$$
\begin{aligned}
&= E\left\|\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\|_{\boldsymbol{\Phi}_s}^2 + E\left\|\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\|_{\boldsymbol{\Phi}_s}^2 \\
&\quad + 2E\left\{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\}' \boldsymbol{\Phi}_s \left\{\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\} \\
&\quad + 2E\left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\}' \left\{\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\} \\
&\quad + E\left\|\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\|^2.
\end{aligned}
$$

Thus, it suffices to show that

$$
\xi_N^{-1} E\left\|\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\|^2 = o(1), \tag{S.13}
$$

$$
\xi_N^{-1} E\left\|\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\|_{\boldsymbol{\Phi}_s}^2 = o(1), \tag{S.14}
$$

$$
\xi_N^{-1} \left| E\left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\}' \left\{\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\} \right| = o(1) \tag{S.15}
$$

and

$$
\xi_N^{-1} \left| E\left\{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\}' \boldsymbol{\Phi}_s \left\{\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\} \right| = o(1). \tag{S.16}
$$

Lemmas 1 and 3, together with Condition 1, imply that

$$
\begin{aligned}
E\left\|\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 &= E\left| \sum_{m=1}^{M}\sum_{l=1}^{M} \omega_m \omega_l \left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)' \boldsymbol{X}'_{(m)s} \boldsymbol{\Phi}_s \boldsymbol{X}_{(l)s} \left(\hat{\boldsymbol{\beta}}_{(l)} - \tilde{\boldsymbol{\beta}}_{(l)}\right) \right| \\
&\leq \sum_{m=1}^{M}\sum_{l=1}^{M} \omega_m \omega_l E\left| \left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)' \boldsymbol{X}'_{(m)s} \boldsymbol{\Phi}_s \boldsymbol{X}_{(l)s} \left(\hat{\boldsymbol{\beta}}_{(l)} - \tilde{\boldsymbol{\beta}}_{(l)}\right) \right| \\
&\leq \left\{ \sum_{m=1}^{M} \omega_m \cdot \lambda_{\max}^{1/2}\left(\boldsymbol{X}'_{(m)s} \boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s}\right) \cdot \left(E\left\|\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right\|^2 \right)^{1/2} \right\}^2 \\
&\leq C f^{-1} k_M.
\end{aligned} \tag{S.17}
$$

Using (S.9), (S.17), and $k_M (f\xi_N)^{-1} = o(1)$, we obtain (S.13) and (S.14).

By the Cauchy–Schwarz inequality,

$$
\left| E\left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\}' \left\{\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\} \right|
$$

$$
\begin{aligned}
&\leq \left\{ E \left\| \boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) \right\|^2 \cdot E \left\| \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) \right\|^2 \right\}^{1/2} \\
&= \left\{ \tilde{R}_N(\boldsymbol{\omega}^*) \cdot E \left\| \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) \right\|^2 \right\}^{1/2} \\
&= \left\{ \xi_N \cdot \frac{\tilde{R}_N(\boldsymbol{\omega}^*)}{R_N(\boldsymbol{\omega}^*)} \cdot E \left\| \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*) \right\|^2 \right\}^{1/2}.
\end{aligned}
$$

Therefore, Lemma 4, (S.9), and $k_M(f\xi_N)^{-1} = o(1)$ imply (S.15).

We now prove (S.16). Note that

$$
\begin{aligned}
E \left\| \boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\|_{\boldsymbol{\Phi}_s}^2 &= E \left\| \boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) + \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\|_{\boldsymbol{\Phi}_s}^2 \\
&\leq 2E \left\| \boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\|_{\boldsymbol{\Phi}_s}^2 + 2E \left\| \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\|_{\boldsymbol{\Phi}_s}^2 \\
&= 2\tilde{R}_N(\boldsymbol{\omega}) + 2E \left\| \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\|_{\boldsymbol{\Phi}_s}^2.
\end{aligned}
$$

By the Cauchy–Schwarz inequality,

$$
\begin{aligned}
&\left| E \left\{ \boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\}' \boldsymbol{\Phi}_s \left\{ \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\} \right| \\
&\leq \left(E \left\| \boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|_{\boldsymbol{\Phi}_s}^2 \cdot E \left\| \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|_{\boldsymbol{\Phi}_s}^2 \right)^{1/2} \\
&\leq CE \left\| \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|_{\boldsymbol{\Phi}_s}^2 + C \left\{ \tilde{R}_N(\boldsymbol{\omega}^*) \cdot E \left\| \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|_{\boldsymbol{\Phi}_s}^2 \right\}^{1/2} \\
&= CE \left\| \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|_{\boldsymbol{\Phi}_s}^2 + C \left\{ \xi_N \cdot \frac{\tilde{R}_N(\boldsymbol{\omega}^*)}{R_N(\boldsymbol{\omega}^*)} \cdot E \left\| \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*) \right\|_{\boldsymbol{\Phi}_s}^2 \right\}^{1/2}.
\end{aligned}
$$

Thus, (S.16) follows from (S.17), Lemma 4, and $k_M(f\xi_N)^{-1} = o(1)$. This completes the proof of Lemma 6.

Lemma 7 *If Conditions 1–2 and $\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$ hold, then*

$$
\sum_{m=1}^{M} \left(\boldsymbol{e}_s' \boldsymbol{\Phi}_s \hat{\boldsymbol{\mu}}_{(m)s} \right)^2 = O_p\left(f^{-1} N M k_M \right).
$$

Proof. By Condition 1 and $\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$,

$$
E\left(\boldsymbol{y}_s' \boldsymbol{\Phi}_s \boldsymbol{y}_s \right) = E_\epsilon \|\boldsymbol{y}_U\|^2 = E_\epsilon \left\| \boldsymbol{\mu}_U + \boldsymbol{e}_U \right\|^2 = O(N). \tag{S.18}
$$

Note that

$$
E \left\| \boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{e}_s \right\|^2 = E \left\| \sum_{i \in s} \frac{\boldsymbol{x}_{i(m)} e_i}{\pi_i} - \sum_{i \in U} \boldsymbol{x}_{i(m)} e_i + \sum_{i \in U} \boldsymbol{x}_{i(m)} e_i \right\|^2
$$

$$
\begin{aligned}
&\leq C\sum_{j=1}^{k_m} E\left\{\sum_{i\in U}\frac{(I_i-\pi_i)x_{ij(m)}e_i}{\pi_i}\right\}^2 + C\sum_{j=1}^{k_m} E\left(\sum_{i\in U} x_{ij(m)}e_i\right)^2 \\
&\leq \frac{CN}{n}\sum_{j=1}^{k_m}\sum_{i\in U} E_\epsilon\left(x_{ij(m)}^2 e_i^2\right) + C\sum_{j=1}^{k_m}\sum_{i\in U} E_\epsilon\left(x_{ij(m)}^2 e_i^2\right) \\
&\leq Cf^{-1}Nk_m, \qquad \text{(S.19)}
\end{aligned}
$$

where the third line follows from Condition 1, Lemma 2, and Theorem 2 of Whittle (1960). It follows from (S.18), (S.19), and Condition 2 that

$$
\begin{aligned}
\sum_{m=1}^{M}\left(\boldsymbol{e}_s'\boldsymbol{\Phi}_s\hat{\boldsymbol{\mu}}_{(m)s}\right)^2 &= \sum_{m=1}^{M}\left\{\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{y}_s\right\}^2 \\
&\leq \sum_{m=1}^{M}\left\{\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right\}\cdot(\boldsymbol{y}_s'\boldsymbol{\Phi}_s\boldsymbol{y}_s) \\
&\leq \frac{C}{N}\cdot\left(\sum_{m=1}^{M}\left\|\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right\|^2\right)\cdot(\boldsymbol{y}_s'\boldsymbol{\Phi}_s\boldsymbol{y}_s) \\
&= O_p\left(f^{-1}NMk_M\right).
\end{aligned}
$$

This completes the proof of Lemma 7.

S3.2 Proof of Theorem 1

The proof of this theorem relies on Lemmas 1–4.

Proof. We begin by establishing the asymptotic unbiasedness of $C_n(\boldsymbol{\omega})$ for $R_N(\boldsymbol{\omega}) + N\sigma^2$. By direct calculation, we have

$$
\begin{aligned}
C_n(\boldsymbol{\omega}) &= \|\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2 + 2f^{-1}\sigma^2 k(\boldsymbol{\omega}) \\
&= \|\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2 + 2\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} + \boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s + 2f^{-1}\sigma^2 k(\boldsymbol{\omega}) \\
&= \|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2 + \|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2 \\
&\quad + 2\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} + 2f^{-1}\sigma^2 k(\boldsymbol{\omega}) \\
&\quad + 2\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} + 2\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} + \boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s. \qquad \text{(S.20)}
\end{aligned}
$$

Note that

$$E(\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s) = E_\epsilon E_d\left(\sum_{i\in s}\frac{e_i^2}{\pi_i}\right) = E_\epsilon\left(\sum_{i\in U} e_i^2\right) = N\sigma^2, \tag{S.21}$$

$$\begin{aligned} E\left\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 &= E_\epsilon E_d\left[\sum_{i\in s}\frac{\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega})\right\}^2}{\pi_i}\right] \\ &= E_\epsilon\left[\sum_{i\in U}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega})\right\}^2\right] \\ &= \tilde{R}_N(\boldsymbol{\omega}), \end{aligned} \tag{S.22}$$

and

$$\begin{aligned} E\left[2\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right] &= 2E_\epsilon E_d\left(\sum_{i\in s}\frac{e_i\mu_i}{\pi_i}\right) - 2E_\epsilon E_d\left\{\sum_{i\in s}\frac{e_i\tilde{\mu}_i(\boldsymbol{\omega})}{\pi_i}\right\} \\ &= 2E_\epsilon(\boldsymbol{e}_U'\boldsymbol{\mu}_U) - 2E_\epsilon\left\{\boldsymbol{e}_U'\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\} \\ &= -2\sum_{m=1}^{M}\left[\omega_m\cdot E_\epsilon\left\{\boldsymbol{e}_U'\boldsymbol{X}_{(m)U}\left(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}_{(m)U}'\boldsymbol{e}_U\right\}\right] \\ &= -2\sigma^2 k(\boldsymbol{\omega}). \end{aligned} \tag{S.23}$$

Combining Equations (S.20)-(S.23), we have

$$\begin{aligned} E\left(C_n(\boldsymbol{\omega})\right) = \tilde{R}_N(\boldsymbol{\omega}) &+ E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 \\ &+ 2E\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} \\ &+ 2E\left[\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right] + N\sigma^2 \\ &+ 2(f^{-1} - 1)\sigma^2 k(\boldsymbol{\omega}). \end{aligned}$$

Therefore, it suffices to show that

$$\frac{\left|\tilde{R}_N(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})} = o(1), \tag{S.24}$$

$$\frac{E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2}{R_N(\boldsymbol{\omega})} = o(1), \tag{S.25}$$

$$\frac{E\left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}' \boldsymbol{\Phi}_s \{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}\right|}{R_N(\boldsymbol{\omega})} = o(1), \tag{S.26}$$

$$\frac{E\left|\boldsymbol{e}_s' \boldsymbol{\Phi}_s \{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}\right|}{R_N(\boldsymbol{\omega})} = o(1), \tag{S.27}$$

and

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{2(1-f)\sigma^2 k(\boldsymbol{\omega})}{f R_N(\boldsymbol{\omega})} = o(1). \tag{S.28}$$

Condition 3 implies that $k_M(f\xi_N)^{-1} = o(1)$. Together with Lemma 4, this yields (S.24) and (S.28). For (S.25), Condition 1 and Lemmas 1 and 3 imply that

$$\begin{aligned}
E\left\|\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 &= E\left|\sum_{m=1}^{M}\sum_{l=1}^{M} \omega_m\omega_l \left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)' \boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{X}_{(l)s} \left(\hat{\boldsymbol{\beta}}_{(l)} - \tilde{\boldsymbol{\beta}}_{(l)}\right)\right| \\
&\le \sum_{m=1}^{M}\sum_{l=1}^{M} \omega_m\omega_l E\left|\left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)' \boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{X}_{(l)s} \left(\hat{\boldsymbol{\beta}}_{(l)} - \tilde{\boldsymbol{\beta}}_{(l)}\right)\right| \\
&\le \left\{\sum_{m=1}^{M} \omega_m \cdot \lambda_{\max}^{1/2}\left(\boldsymbol{X}_{(m)s}' \boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s}\right) \cdot \left(E\left\|\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right\|^2\right)^{1/2}\right\}^2 \\
&\le C f^{-1} k_M.
\end{aligned} \tag{S.29}$$

Since $\xi_N \le R_N(\boldsymbol{\omega})$, it follows from $k_M(f\xi_N)^{-1} = o(1)$ that

$$\frac{E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2}{R_N(\boldsymbol{\omega})} \le \frac{C k_M}{f\xi_N} = o(1).$$

Hence, (S.25) holds.

For (S.26), the Cauchy–Schwarz inequality implies

$$\begin{aligned}
&E\left|\left(\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right)' \boldsymbol{\Phi}_s \left(\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right)\right| \\
&\le \left(E\left\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 \cdot E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2\right)^{1/2} \\
&= \left(E_\epsilon\left\|\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\|^2 \cdot E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2\right)^{1/2} \\
&= \left(\tilde{R}_N(\boldsymbol{\omega}) \cdot E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2\right)^{1/2}.
\end{aligned}$$

Dividing both sides by $R_N(\boldsymbol{\omega})$ and using (S.24) and (S.25), we obtain (S.26).

For (S.27), applying the Cauchy–Schwarz inequality together with (S.29) yields

$$\begin{aligned} E\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right| &\leq \left(E\|\boldsymbol{e}_s\|_{\boldsymbol{\Phi}_s}^2\cdot E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2\right)^{1/2} \\ &= \left(N\sigma^2\cdot E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2\right)^{1/2} \\ &\leq Cf^{-1}(nk_M)^{1/2}. \end{aligned}$$

Dividing both sides by $R_N(\boldsymbol{\omega})$ and using Condition 3, we obtain (S.27). This completes the proof of the asymptotic unbiasedness of $C_n(\boldsymbol{\omega})$.

We next verify that $C_n^*(\boldsymbol{\omega})$ is asymptotically unbiased for $R_N(\boldsymbol{\omega})+N\sigma^2$. Comparing the definitions of $C_n^*(\boldsymbol{\omega})$ and $C_n(\boldsymbol{\omega})$, and using (S.20), it suffices to show that

$$\frac{k(\boldsymbol{\omega})\cdot E\left|\hat{\sigma}_{(M)}^2-\sigma^2\right|}{fR_N(\boldsymbol{\omega})}=o(1). \tag{S.30}$$

Note that

$$\left|\hat{\sigma}_{(M)}^2-\sigma^2\right|\leq\left|\hat{\sigma}_{(M)}^2-\tilde{\sigma}_{(M)}^2\right|+\left|\tilde{\sigma}_{(M)}^2-\sigma^2\right|,$$

where

$$\tilde{\sigma}_{(M)}^2=\frac{1}{N-k_M}\left\|\boldsymbol{y}_U-\boldsymbol{X}_{(M)U}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2.$$

Therefore, the proof of (S.30) reduces to verifying the following two relations:

$$\frac{k_M\cdot E\left|\hat{\sigma}_{(M)}^2-\tilde{\sigma}_{(M)}^2\right|}{f\xi_N}=o(1), \tag{S.31}$$

and

$$\frac{k_M\cdot E\left|\tilde{\sigma}_{(M)}^2-\sigma^2\right|}{f\xi_N}=o(1). \tag{S.32}$$

We first prove (S.31). Condition 1, Lemma 1 and $k_M^2=O(N)$ imply that

$$\begin{aligned} E_\epsilon\left(y_i-\tilde{\mu}_{i(m)}\right)^4 &= E_\epsilon\left(e_{i(m)}^*+\boldsymbol{x}_{i(m)}'\boldsymbol{\beta}_{(m)}^*-\boldsymbol{x}_{i(m)}'\tilde{\boldsymbol{\beta}}_{(m)}\right)^4 \\ &\leq CE_\epsilon\left(e_{i(m)}^{*4}\right)+Ck_m^2\cdot E_\epsilon\left\|\tilde{\boldsymbol{\beta}}_{(m)}-\boldsymbol{\beta}_{(m)}^*\right\|^4 \end{aligned}$$

$$= O(1) \tag{S.33}$$

uniformly in i and m. Then, by Condition 2, we have

$$\begin{aligned}
&E\left|\frac{1}{n-k_M}\left\|\boldsymbol{y}_s-\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2-\frac{1}{N-k_M}\left\|\boldsymbol{y}_U-\boldsymbol{X}_{(M)U}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2\right| \\
&= E\left|\sum_{i\in U}\left(\frac{I_i}{n-k_M}-\frac{1}{N-k_M}\right)\left(y_i-\tilde{\mu}_{i(M)}\right)^2\right| \\
&\leq \frac{C}{n-k_M}\cdot\sum_{i\in U}E_\epsilon\left(y_i-\tilde{\mu}_{i(M)}\right)^2 \\
&= O(1),
\end{aligned}$$

where the last equality follows from $f=\sum_U \pi_i/N \geq c_5$ and $k_M^2 = O(N)$. Moreover, Condition 1, Lemma 3, and (S.33) yield

$$E\left\|\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}-\boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\right\|^2 \leq E\left\{\lambda_{\max}\left(\boldsymbol{X}'_{(M)s}\boldsymbol{X}_{(M)s}\right)\left\|\tilde{\boldsymbol{\beta}}_{(M)}-\hat{\boldsymbol{\beta}}_{(M)}\right\|^2\right\} \leq Ck_M^2$$

and

$$\begin{aligned}
&E\left|\left(\boldsymbol{y}_s-\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}\right)'\left(\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}-\boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\right)\right| \\
&\leq \left(E\left\|\boldsymbol{y}_s-\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2\cdot E\left\|\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}-\boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\right\|^2\right)^{1/2} \\
&\leq C\left\{\sum_{i\in U}\pi_i E_\epsilon\left(y_i-\tilde{\mu}_{i(M)}\right)^2\cdot k_M^2\right\}^{1/2} \\
&= O(n^{1/2}k_M).
\end{aligned}$$

Using the preceding bounds and the triangle inequality, we obtain that

$$\begin{aligned}
&\frac{k_M\cdot E\left|\hat{\sigma}^2_{(M)}-\tilde{\sigma}^2_{(M)}\right|}{f\xi_N} \\
&\leq \frac{k_M}{f\xi_N}\cdot E\left|\frac{1}{n-k_M}\left\|\boldsymbol{y}_s-\boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\right\|^2-\frac{1}{N-k_M}\left\|\boldsymbol{y}_U-\boldsymbol{X}_{(M)U}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2\right| \\
&= \frac{k_M}{f\xi_N}\cdot E\left|\frac{1}{n-k_M}\left\|\boldsymbol{y}_s-\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2-\frac{1}{N-k_M}\left\|\boldsymbol{y}_U-\boldsymbol{X}_{(M)U}\tilde{\boldsymbol{\beta}}_{(M)}\right\|^2\right. \\
&\quad+\frac{2}{n-k_M}\left(\boldsymbol{y}_s-\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}\right)'\left(\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}-\boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\right) \\
&\quad\left.+\frac{1}{n-k_M}\left\|\boldsymbol{X}_{(M)s}\tilde{\boldsymbol{\beta}}_{(M)}-\boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\right\|^2\right|
\end{aligned}$$

$$= o(1),$$

where the last step follows from the conditions $k_M(f\xi_N)^{-1} = o(1)$ and $k_M^2 = O(N)$. This proves (S.31).

We now verify (S.32). Note that

$$\tilde{\sigma}^2_{(M)} = \frac{1}{N-k_M}\left\{\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{\mu}_U + \boldsymbol{e}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U \right.$$
$$\left. +2\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U\right\},$$

where $\boldsymbol{P}_{(M)U} = \boldsymbol{X}_{(M)U}\left(\boldsymbol{X}_{(M)U}'\boldsymbol{X}_{(M)U}\right)^{-1}\boldsymbol{X}_{(M)U}'$. Since

$$E\left\{\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U\right\} = 0$$

and

$$E\left\{\boldsymbol{e}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U\right\} = (N-k_M)\sigma^2,$$

it follows from Theorem 2 of Whittle (1960) that

$$E\left\{\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U\right\}^2 \leq C\left\|\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\right\|^2 = O(\|\boldsymbol{\mu}_U\|^2)$$

and

$$E\left\{\boldsymbol{e}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U - (N-k_M)\sigma^2\right\}^2 \leq C\,\mathrm{tr}\left\{\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)^2\right\} = O(N-k_M).$$

Then, using $\|\boldsymbol{\mu}_U\|^2 = O(N)$ and the fact that $\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}$ is symmetric and idempotent, we have

$$\begin{aligned}
&\frac{k_M \cdot E\left|\tilde{\sigma}^2_{(M)} - \sigma^2\right|}{f\xi_N} \\
&\leq \frac{k_M}{(N-k_M)f\xi_N}\left|\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{\mu}_U\right| + \frac{2k_M}{(N-k_M)f\xi_N}E\left|\boldsymbol{\mu}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U\right| \\
&\quad + \frac{k_M}{(N-k_M)f\xi_N}E\left|\boldsymbol{e}_U'\left(\boldsymbol{I}_N - \boldsymbol{P}_{(M)U}\right)\boldsymbol{e}_U - (N-k_M)\sigma^2\right| \\
&= O\left(\frac{Nk_M}{(N-k_M)f\xi_N}\right)
\end{aligned}$$

$$= o(1),$$

where the last step follows from the conditions $k_M(f\xi_N)^{-1} = o(1)$ and $k_M^2 = O(N)$. This proves (S.32) and completes the proof of Theorem 1.

S3.3 Proof of Corollary 1

The proof of this corollary relies on Lemmas 1–4.

Proof. By Equations (S.1) and (S.2), we have

$$\begin{aligned} E\left(C_n(\boldsymbol{\omega})\right) = R_N(\boldsymbol{\omega}) + N\sigma^2 + 2f^{-1}\sigma^2 k(\boldsymbol{\omega}) - 2\sigma^2 E_d\left[k_s(\boldsymbol{\omega})\right] \\ + \left\{\tilde{R}_N(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega})\right\} + E\left\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\|_{\boldsymbol{\Phi}_s}^2 \\ + 2E\left[\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}' \boldsymbol{\Phi}_s \left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right]. \end{aligned}$$

Following Equations (S.24)–(S.26) in the proof of Theorem 1, the absolute values of the last three terms on the right-hand side are all of order $o(R_N(\boldsymbol{\omega}))$ under the condition $k_M(f\xi_N)^{-1} = o(1)$. Moreover, Conditions 1 and 2 imply that

$$\begin{aligned} E_d\left[k_s(\boldsymbol{\omega})\right] &= E_d\left[\sum_{m=1}^{M} \omega_m \operatorname{tr}\left\{\boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s}\left(\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)^{-1}\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\right\}\right] \\ &= E_d\left[\sum_{m=1}^{M} \omega_m \operatorname{tr}\left\{\left(\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)^{-1}\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s^2\boldsymbol{X}_{(m)s}\right\}\right] \\ &= O(k_M). \end{aligned}$$

Since $\xi_N \leq R_N(\boldsymbol{\omega})$, it follows from $k_M(f\xi_N)^{-1} = o(1)$ that

$$\frac{\left|2f^{-1}\sigma^2 k(\boldsymbol{\omega}) - 2\sigma^2 E_d\left(k_s(\boldsymbol{\omega})\right)\right|}{R_N(\boldsymbol{\omega})} \leq \frac{Ck_M}{f\xi_N} = o(1).$$

Therefore, $E\left(C_n(\boldsymbol{\omega})\right) = R_N(\boldsymbol{\omega}) + N\sigma^2 + o\left(R_N(\boldsymbol{\omega})\right)$.

The result for $C_n^*(\boldsymbol{\omega})$ follows from Equation (S.30) in the proof of Theorem 1, together with the conditions $\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$ and $k_M^2 = O(N)$. This completes the proof of Corollary 1.

S3.4 Proof of Theorem 2

The proof of this theorem relies on Lemmas 1–5.

Proof. We first prove Equation (12) in Theorem 2. From Equation (S.20), we have

$$\begin{aligned} C_n(\boldsymbol{\omega}) = L_N(\boldsymbol{\omega}) &+ \left\{ \|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s} - L_N(\boldsymbol{\omega}) \right\} + \|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s} \\ &+ 2\boldsymbol{e}_s'\boldsymbol{\Phi}_s \left\{ \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\} + 2\left[\boldsymbol{e}_s'\boldsymbol{\Phi}_s \left\{ \boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\} + 2\sigma^2 k(\boldsymbol{\omega}) \right] \\ &+ 2\left\{ \boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\}' \boldsymbol{\Phi}_s \left\{ \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\} + \boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s + 2(f^{-1} - 1)\sigma^2 k(\boldsymbol{\omega}). \end{aligned} \tag{S.34}$$

By (S.28) and arguments similar to those in Li (1987) and Wan et al. (2010), it suffices to show that

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s}}{R_N(\boldsymbol{\omega})} = o_p(1), \tag{S.35}$$

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{|\boldsymbol{e}_s'\boldsymbol{\Phi}_s \left\{ \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\}|}{R_N(\boldsymbol{\omega})} = o_p(1), \tag{S.36}$$

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left|\left\{ \boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\}' \boldsymbol{\Phi}_s \left\{ \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\}\right|}{R_N(\boldsymbol{\omega})} = o_p(1), \tag{S.37}$$

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{|\boldsymbol{e}_s'\boldsymbol{\Phi}_s \left\{ \boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) \right\} + \sigma^2 k(\boldsymbol{\omega})|}{R_N(\boldsymbol{\omega})} = o_p(1), \tag{S.38}$$

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left| \|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s} - L_N(\boldsymbol{\omega}) \right|}{R_N(\boldsymbol{\omega})} = o_p(1), \tag{S.39}$$

and

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{|L_N(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega})|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.40}$$

Using Bonferroni's inequality and Markov's inequality, for any $\delta > 0$, we have

$$P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}} \left\{ R_N^{-1}(\boldsymbol{\omega}) \|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s} \right\} > \delta \right)$$

$$
\begin{aligned}
&\leq P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2>\delta\xi_N\right)\\
&=P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left\{\sum_{m=1}^{M}\sum_{t=1}^{M}\omega_m\omega_t\left(\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\tilde{\boldsymbol{\beta}}_{(t)}-\hat{\boldsymbol{\beta}}_{(t)}\right)\right\}>\delta\xi_N\right)\\
&\leq\sum_{m=1}^{M}\sum_{t=1}^{M}P\left(\left|\left(\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\tilde{\boldsymbol{\beta}}_{(t)}-\hat{\boldsymbol{\beta}}_{(t)}\right)\right|>\delta\xi_N\right)\\
&\leq\delta^{-1}\xi_N^{-1}\sum_{m=1}^{M}\sum_{t=1}^{M}E\left|\left(\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\tilde{\boldsymbol{\beta}}_{(t)}-\hat{\boldsymbol{\beta}}_{(t)}\right)\right|\\
&\leq\delta^{-1}\xi_N^{-1}E\left[\sum_{m=1}^{M}\left\{\lambda_{\max}^{1/2}\left(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)\cdot\left\|\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right\|\right\}\right]^2\\
&\leq M\delta^{-1}\xi_N^{-1}\sum_{m=1}^{M}\left\{\lambda_{\max}\left(\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)\cdot E\left\|\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right\|^2\right\}\\
&=O\left(\frac{M^2k_M}{f\xi_N}\right),
\end{aligned}
$$

where the last bound follows from Lemmas 1 and 3, together with Condition 1. Hence, (S.35) follows from Condition 3.

Similarly, to prove (S.36), for any $\delta>0$, we have

$$
\begin{aligned}
&P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left[R_N^{-1}(\boldsymbol{\omega})\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right|\right]>\delta\right)\\
&\leq P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})-\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right|>\delta\xi_N\right)\\
&\leq P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left\{\sum_{m=1}^{M}\omega_m\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right)\right|\right\}>\delta\xi_N\right)\\
&\leq\sum_{m=1}^{M}P\left(\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right)\right|>\delta\xi_N\right)\\
&\leq\delta^{-1}\xi_N^{-1}\sum_{m=1}^{M}E\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\left(\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right)\right|\\
&\leq\delta^{-1}\xi_N^{-1}\sum_{m=1}^{M}\left\{\lambda_{\max}\left(\boldsymbol{X}_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}'\right)\cdot E\left\|\tilde{\boldsymbol{\beta}}_{(m)}-\hat{\boldsymbol{\beta}}_{(m)}\right\|^2\cdot E\|\boldsymbol{e}_s\|_{\boldsymbol{\Phi}_s}^2\right\}^{1/2}\\
&=O\left(\frac{M\sqrt{nk_M}}{f\xi_N}\right),
\end{aligned}
$$

where the last bound follows from Lemmas 1 and 3, Condition 1, and the fact that $E\|\boldsymbol{e}_s\|_{\boldsymbol{\Phi}_s}^2=N\sigma^2$. Condition 3 then implies (S.36).

We now prove (S.37). By Lemma 1 and Conditions 1 and 4, we have

$$\begin{aligned} E_\epsilon\left(\mu_i - \tilde{\mu}_{i(m)}\right)^4 &= E_\epsilon\left(e^*_{i(m)} - e_i + \boldsymbol{x}'_{i(m)}\boldsymbol{\beta}^*_{(m)} - \boldsymbol{x}'_{i(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)^4 \\ &\leq 27E_\epsilon\left(e^{*4}_{i(m)}\right) + 27E_\epsilon\left(e_i^4\right) + 27k_m^2 \cdot E_\epsilon\left\|\tilde{\boldsymbol{\beta}}_{(m)} - \boldsymbol{\beta}^*_{(m)}\right\|^4 \\ &\leq C(1 + N^{-2}k_m^4) \end{aligned} \tag{S.41}$$

uniformly in i and m. It follows that

$$E\left\|\boldsymbol{\mu}_s - \boldsymbol{X}_{(m)s}\tilde{\boldsymbol{\beta}}_{(m)}\right\|^2_{\boldsymbol{\Phi}_s} = \sum_{i\in U}\left\{E_\epsilon\left(\mu_i - \tilde{\mu}_{i(m)}\right)^2\right\} \leq C(N + k_m^2). \tag{S.42}$$

Therefore, for any $\delta > 0$,

$$\begin{aligned} &P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}} R_N^{-1}(\boldsymbol{\omega})\left|\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right| > \delta\right) \\ &\leq P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left|\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}'\boldsymbol{\Phi}_s\left\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\}\right| > \delta\xi_N\right) \\ &= P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}}\left\{\sum_{m=1}^M\sum_{t=1}^M \omega_m\omega_t\left|\left(\boldsymbol{\mu}_s - \boldsymbol{X}_{(m)s}\tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right|\right\} > \delta\xi_N\right) \\ &\leq \sum_{m=1}^M\sum_{t=1}^M P\left(\left|\left(\boldsymbol{\mu}_s - \boldsymbol{X}_{(m)s}\tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right| > \delta\xi_N\right) \\ &\leq \delta^{-1}\xi_N^{-1}\sum_{m=1}^M\sum_{t=1}^M E\left|\left(\boldsymbol{\mu}_s - \boldsymbol{X}_{(m)s}\tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right)\right| \\ &\leq \delta^{-1}\xi_N^{-1}\sum_{m=1}^M\sum_{t=1}^M\left\{\lambda_{\max}\left(\boldsymbol{X}'_{(t)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\right)\cdot E\left\|\boldsymbol{\mu}_s - \boldsymbol{X}_{(m)s}\tilde{\boldsymbol{\beta}}_{(m)}\right\|^2_{\boldsymbol{\Phi}_s}\cdot E\left\|\tilde{\boldsymbol{\beta}}_{(t)} - \hat{\boldsymbol{\beta}}_{(t)}\right\|^2\right\}^{1/2} \\ &= O\left(\frac{M^2\sqrt{nk_M}}{f\xi_N}\right), \end{aligned}$$

where the last bound follows from (S.42), Lemmas 1 and 3, and Condition 1. Condition 3 then implies that the last term is $o(1)$, which proves (S.37).

To prove (S.38), it suffices to show that

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}}\frac{\left|\boldsymbol{e}'_U\left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\} + \sigma^2 k(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})} = o_p(1) \tag{S.43}$$

and

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}}\frac{\left|\boldsymbol{e}'_s\boldsymbol{\Phi}_s\left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} - \boldsymbol{e}'_U\left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}\right|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.44}$$

Note that

$$\boldsymbol{e}_U' \left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\} + \sigma^2 k(\boldsymbol{\omega}) = \boldsymbol{e}_U' \left\{\boldsymbol{I}_N - \boldsymbol{P}_U(\boldsymbol{\omega})\right\} \boldsymbol{\mu}_U + \left\{\sigma^2 k(\boldsymbol{\omega}) - \boldsymbol{e}_U' \boldsymbol{P}_U(\boldsymbol{\omega}) \boldsymbol{e}_U\right\},$$

where $\boldsymbol{P}_U(\boldsymbol{\omega}) = \sum_{m=1}^{M} \omega_m \boldsymbol{P}_{(m)U}$ and $\boldsymbol{P}_{(m)U} = \boldsymbol{X}_{(m)U}(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U})^{-1}\boldsymbol{X}_{(m)U}'$. Then, by Equations (A.1) and (A.2) of Wan et al. (2010), together with Condition 4, we have

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left|\boldsymbol{e}_U' \left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\} + \sigma^2 k(\boldsymbol{\omega})\right|}{\tilde{R}_N(\boldsymbol{\omega})} = o_p(1).$$

Furthermore, Lemma 4 and Condition 3 imply (S.43). Using Bonferroni's inequality, Markov's inequality, and Lemma 2, for any $\delta > 0$, we have

$$\begin{aligned}
&P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}} R_N^{-1}(\boldsymbol{\omega}) \left|\boldsymbol{e}_s' \boldsymbol{\Phi}_s \left\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\right\} - \boldsymbol{e}_U' \left\{\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega})\right\}\right| > \delta\right) \\
&\le P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}} \left|\sum_{i\in U} \frac{(I_i - \pi_i)\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega})\right\} e_i}{\pi_i}\right| > \delta\xi_N\right) \\
&\le P\left(\sup_{\boldsymbol{\omega}\in\mathcal{W}} \sum_{m=1}^{M} \omega_m \left|\sum_{i\in U} \frac{(I_i - \pi_i)\{\mu_i - \tilde{\mu}_{i(m)}\} e_i}{\pi_i}\right| > \delta\xi_N\right) \\
&\le \sum_{m=1}^{M} P\left(\left|\sum_{i\in U} \frac{(I_i - \pi_i)\{\mu_i - \tilde{\mu}_{i(m)}\} e_i}{\pi_i}\right| > \delta\xi_N\right) \\
&\le \delta^{-2}\xi_N^{-2} \sum_{m=1}^{M} E\left[\sum_{i\in U} \frac{(I_i - \pi_i)\{\mu_i - \tilde{\mu}_{i(m)}\} e_i}{\pi_i}\right]^2 \\
&\le f^{-1}\delta^{-2}\xi_N^{-2} \sum_{m=1}^{M} \sum_{i\in U} E_\epsilon\left[\{\mu_i - \tilde{\mu}_{i(m)}\}^2 e_i^2\right] \\
&= O\left(\frac{nM}{f^2\xi_N^2}\right),
\end{aligned}$$

where the last bound follows from the Cauchy–Schwarz inequality, (S.41), and Condition 4. Condition 3 then implies (S.44), and hence (S.38) follows.

Next, we establish (S.39) by verifying the following two relations:

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left|\tilde{L}_N(\boldsymbol{\omega}) - L_N(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})} = o_p(1) \tag{S.45}$$

and

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{\left|\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2 - \tilde{L}_N(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.46}$$

Lemma 5 directly implies (S.45). Using Bonferroni's inequality, Markov's inequality, and Lemma 2, for any $\delta > 0$, we have

$$
\begin{aligned}
& P\left(\sup_{\omega\in\mathcal{W}} R_N^{-1}(\boldsymbol{\omega})\left\{\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|_{\boldsymbol{\Phi}_s}^2 - \tilde{L}_N(\boldsymbol{\omega})\right\} > \delta\right) \\
& \le P\left(\sup_{\omega\in\mathcal{W}}\left|\sum_{i\in U}\frac{(I_i-\pi_i)\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega})\right\}^2}{\pi_i}\right| > \delta\xi_N\right) \\
& \le P\left(\sup_{\omega\in\mathcal{W}}\sum_{m=1}^{M}\sum_{t=1}^{M}\omega_m\omega_t\left|\sum_{i\in U}\frac{(I_i-\pi_i)\{\mu_i-\tilde{\mu}_{i(m)}\}\{\mu_i-\tilde{\mu}_{i(t)}\}}{\pi_i}\right| > \delta\xi_N\right) \\
& \le \sum_{m=1}^{M}\sum_{t=1}^{M} P\left(\left|\sum_{i\in U}\frac{(I_i-\pi_i)\{\mu_i-\tilde{\mu}_{i(m)}\}\{\mu_i-\tilde{\mu}_{i(t)}\}}{\pi_i}\right| > \delta\xi_N\right) \\
& \le \delta^{-2}\xi_N^{-2}\sum_{m=1}^{M}\sum_{t=1}^{M} E\left[\sum_{i\in U}\frac{(I_i-\pi_i)\{\mu_i-\tilde{\mu}_{i(m)}\}\{\mu_i-\tilde{\mu}_{i(t)}\}}{\pi_i}\right]^2 \\
& \le f^{-1}\delta^{-2}\xi_N^{-2}\sum_{m=1}^{M}\sum_{t=1}^{M}\sum_{i\in U} E_\epsilon\left[\{\mu_i-\tilde{\mu}_{i(m)}\}^2\{\mu_i-\tilde{\mu}_{i(t)}\}^2\right] \\
& = O\left(\frac{nM^2}{f^2\xi_N^2}\right),
\end{aligned}
$$

where the last bound follows from the Cauchy–Schwarz inequality and (S.41). By Condition 3, (S.46) holds, which further implies (S.39).

It remains to verify (S.40). Note that

$$
L_N(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega}) = L_N(\boldsymbol{\omega}) - \tilde{L}_N(\boldsymbol{\omega}) + \tilde{L}_N(\boldsymbol{\omega}) - \tilde{R}_N(\boldsymbol{\omega}) + \tilde{R}_N(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega}).
$$

By Lemmas 4 and 5, the verification of (S.40) reduces to showing that

$$
\sup_{\omega\in\mathcal{W}}\frac{\left|\tilde{L}_N(\boldsymbol{\omega}) - \tilde{R}_N(\boldsymbol{\omega})\right|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.47}
$$

By Condition 4 and Equation (A.3) of Wan et al. (2010), we have

$$
\sup_{\omega\in\mathcal{W}}\frac{\left|\tilde{L}_N(\boldsymbol{\omega}) - \tilde{R}_N(\boldsymbol{\omega})\right|}{\tilde{R}_N(\boldsymbol{\omega})} = o_p(1).
$$

Together with Lemma 4 and Condition 3, this establishes (S.47). Hence, (S.40) follows. This completes the proof of Equation (12) in Theorem 2.

We now turn to Equation (13) in Theorem 2. By (S.28) and (S.34), it suffices to establish

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{k(\boldsymbol{\omega}) \cdot \left|\hat{\sigma}^2_{(M)} - \sigma^2\right|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.48}$$

Note that

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{k(\boldsymbol{\omega}) \cdot \left|\hat{\sigma}^2_{(M)} - \sigma^2\right|}{R_N(\boldsymbol{\omega})} \leq \frac{k_M \cdot \left|\hat{\sigma}^2_{(M)} - \sigma^2\right|}{\xi_N}.$$

By (S.31), (S.32), and Markov's inequality, (S.48) follows. This completes the proof of Equation (13) in Theorem 2.

S3.5 Proof of Corollary 2

The proof of this corollary relies on Lemmas 1–5.

Proof. By Equation (S.34), we have

$$\begin{aligned}
&C_n(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega}) - N\sigma^2 \\
&= \{L_N(\boldsymbol{\omega}) - R_N(\boldsymbol{\omega})\} + \left\{\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s} - L_N(\boldsymbol{\omega})\right\} + \|\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\|^2_{\boldsymbol{\Phi}_s} \\
&\quad + 2\boldsymbol{e}_s'\boldsymbol{\Phi}_s\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\} + 2\left[\boldsymbol{e}_s'\boldsymbol{\Phi}_s\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\} + 2\sigma^2 k(\boldsymbol{\omega})\right] \\
&\quad + 2\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\}'\boldsymbol{\Phi}_s\{\tilde{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\} + \left(\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - N\sigma^2\right) + 2(f^{-1} - 1)\sigma^2 k(\boldsymbol{\omega}).
\end{aligned}$$

Following Equations (S.28), (S.35)–(S.40), and (S.48) in the proof of Theorem 2, it remains to show that

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}} \frac{|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - N\sigma^2|}{R_N(\boldsymbol{\omega})} = o_p(1). \tag{S.49}$$

By Lemma 2, we have

$$E\left(\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - N\sigma^2\right)^2 \leq 2E\left(\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - \boldsymbol{e}_U'\boldsymbol{e}_U\right)^2 + 2E\left(\boldsymbol{e}_U'\boldsymbol{e}_U - N\sigma^2\right)^2$$

$$\leq \frac{CN}{n}\sum_{i\in U} Ee_i^4 + C\sum_{i\in U}\left(Ee_i^4 - \sigma^4\right)$$
$$\leq CNf^{-1}.$$

It follows from Markov's inequality that

$$\left|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - N\sigma^2\right| = O_p\left(n^{1/2}f^{-1}\right).$$

Therefore,

$$\sup_{\boldsymbol{\omega}\in\mathcal{W}}\frac{|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - N\sigma^2|}{R_N(\boldsymbol{\omega})} \leq \frac{|\boldsymbol{e}_s'\boldsymbol{\Phi}_s\boldsymbol{e}_s - N\sigma^2|}{\xi_N} = o_p(1),$$

where the last equality follows from Condition 3. This completes the proof of Corollary 2.

S3.6 Proof of Theorem 3

The proof of this theorem relies on Lemmas 1–4 and 6–7.

Proof. Let $\epsilon = \xi_N^{1/2}N^{-1/2+\delta}$. To verify Equation (14) in Theorem 3, following Fan & Peng (2004) and Chen et al. (2018), it suffices to show that there exists a constant $C_0 > 0$ such that, for any M-dimensional vector $\boldsymbol{v} = (v_1, \ldots, v_M)'$,

$$\lim_{N\to\infty} P\left\{\inf_{\|\boldsymbol{v}\|=C_0,\, \boldsymbol{\omega}^*+\epsilon\boldsymbol{v}\in\mathcal{W}} C_n(\boldsymbol{\omega}^*+\epsilon\boldsymbol{v}) > C_n(\boldsymbol{\omega}^*)\right\} = 1.$$

This implies that there exists a minimizer $\breve{\boldsymbol{\omega}}$ in the set $\{\boldsymbol{\omega}^* + \epsilon\boldsymbol{v} \in \mathcal{W} : \|\boldsymbol{v}\| \leq C_0\}$ such that $\breve{\boldsymbol{\omega}} - \boldsymbol{\omega}^* = O_p(\epsilon)$.

Note that

$$\begin{aligned} C_n(\boldsymbol{\omega}^*+\epsilon\boldsymbol{v}) - C_n(\boldsymbol{\omega}^*) &= \|\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*+\epsilon\boldsymbol{v})\|_{\boldsymbol{\Phi}_s}^2 - \|\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\|_{\boldsymbol{\Phi}_s}^2 \\ &\quad + 2f^{-1}\sigma^2 k(\boldsymbol{\omega}^*+\epsilon\boldsymbol{v}) - 2f^{-1}\sigma^2 k(\boldsymbol{\omega}) \\ &= \|\hat{\boldsymbol{\mu}}_s(\epsilon\boldsymbol{v})\|_{\boldsymbol{\Phi}_s}^2 - 2\left\{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\}'\boldsymbol{\Phi}_s\hat{\boldsymbol{\mu}}_s(\epsilon\boldsymbol{v}) \end{aligned}$$

$$
\begin{aligned}
&\quad - 2\boldsymbol{e}_s'\boldsymbol{\Phi}_s\hat{\boldsymbol{\mu}}_s(\epsilon\boldsymbol{v}) + 2f^{-1}\sigma^2 k(\epsilon\boldsymbol{v}) \\
&\triangleq \Delta_1 + \Delta_2 + \Delta_3 + \Delta_4.
\end{aligned}
$$

Let $\hat{\boldsymbol{\mu}}_{(m)s} = [\hat{\mu}_{i(m)}]_{i\in s}$ and $\boldsymbol{V}_1 = (\hat{\boldsymbol{\mu}}_{(1)s}, \ldots, \hat{\boldsymbol{\mu}}_{(M)s})$. By Condition 5, we have

$$
\begin{aligned}
\Delta_1 &= \left(\sum_{m=1}^{M} \epsilon v_m \hat{\boldsymbol{\mu}}_{(m)s}\right)' \boldsymbol{\Phi}_s \left(\sum_{m=1}^{M} \epsilon v_m \hat{\boldsymbol{\mu}}_{(m)s}\right) \\
&= \epsilon^2 \boldsymbol{v}'\boldsymbol{V}_1'\boldsymbol{\Phi}_s\boldsymbol{V}_1\boldsymbol{v} \\
&\geq \epsilon^2 \lambda_{\min}\left(\boldsymbol{V}_1'\boldsymbol{\Phi}_s\boldsymbol{V}_1\right)\|\boldsymbol{v}\|^2 \\
&\geq CN\epsilon^2 \cdot \|\boldsymbol{v}\|^2 > 0
\end{aligned}
$$

in probability tending to one. Thus, to prove Equation (14) in Theorem 3, it remains to show that Δ_2, Δ_3, and Δ_4 are asymptotically negligible relative to Δ_1.

By Lemma 6 and Condition 6, we have

$$
\xi_N = E\left\|\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}^*)\right\|^2 = E\left\|\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\|_{\boldsymbol{\Phi}_s}^2 + o(\xi_N).
$$

Hence, $\left\|\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\|_{\boldsymbol{\Phi}_s} = O_p(\xi_N^{1/2})$. Using Condition 5, we obtain that

$$
\begin{aligned}
|\Delta_2| &= 2\epsilon\left|\left\{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\}'\boldsymbol{\Phi}_s\boldsymbol{V}_1\boldsymbol{v}\right| \\
&\leq 2\epsilon\left\|\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}^*)\right\|_{\boldsymbol{\Phi}_s} \lambda_{\max}^{1/2}(\boldsymbol{V}_1'\boldsymbol{\Phi}_s\boldsymbol{V}_1)\|\boldsymbol{v}\| \\
&= O_p\left(\xi_N^{1/2}N^{1/2}\epsilon\right)\cdot\|\boldsymbol{v}\|.
\end{aligned}
$$

Since $\epsilon = \xi_N^{1/2}N^{-1/2+\delta}$, we have

$$
\frac{|\Delta_2|}{\Delta_1} = O_p\left(\frac{\xi_N^{1/2}N^{1/2}\epsilon\cdot\|\boldsymbol{v}\|}{N\epsilon^2\cdot\|\boldsymbol{v}\|^2}\right) = O_p(N^{-\delta}) = o_p(1).
$$

Thus, Δ_2 is asymptotically negligible relative to Δ_1.

Next, we consider Δ_3. By Lemma 7, we have

$$
|\Delta_3| = 2\epsilon\left|\sum_{m=1}^{M}\left(v_m\boldsymbol{e}_s'\boldsymbol{\Phi}_s\hat{\boldsymbol{\mu}}_{(m)s}\right)\right|
$$

$$\leq 2\epsilon \left\{ \sum_{m=1}^{M} \left(\boldsymbol{e}_s' \boldsymbol{\Phi}_s \hat{\boldsymbol{\mu}}_{(m)s} \right)^2 \right\}^{1/2} \|\boldsymbol{v}\|$$
$$= O_p\left(f^{-1/2} (NMk_M)^{1/2} \epsilon \right) \cdot \|\boldsymbol{v}\|.$$

For Δ_4, it follows that

$$|\Delta_4| = 2f^{-1}\sigma^2 \epsilon \left| \sum_{m=1}^{M} k_m v_m \right| \tag{S.50}$$
$$= O_p\left(f^{-1} M^{1/2} k_M \epsilon \right) \cdot \|\boldsymbol{v}\|.$$

Thus, by Condition 6, Δ_3 and Δ_4 are asymptotically negligible relative to Δ_1.

We next prove Equation (15) in Theorem 3. Comparing $C_n(\boldsymbol{\omega})$ with $C_n^*(\boldsymbol{\omega})$, it suffices to replace σ^2 in (S.50) by $\hat{\sigma}^2_{(M)}$ and show that the resulting term is asymptotically negligible relative to Δ_1. By Conditions 1–2 and Lemma 3, we have

$$\begin{aligned}
\hat{\sigma}^2_{(M)} &= \frac{1}{n - k_M} \sum_{i \in s} \left(y_i - \boldsymbol{x}_{i(M)}' \hat{\boldsymbol{\beta}}_{(M)} \right)^2 \\
&\leq \frac{C}{n - k_M} \sum_{i \in s} \left\{ e^{*2}_{i(M)} + \left(\boldsymbol{x}_{i(M)} \boldsymbol{\beta}^*_{(M)} - \boldsymbol{x}_{i(M)}' \hat{\boldsymbol{\beta}}_{(M)} \right)^2 \right\} \\
&= O_p\left(\frac{n + k_M^2}{n - k_M} \right).
\end{aligned}$$

Since $k_M^2 = O(N)$ and $f \geq c_5$, it follows that

$$|\Delta_4'| \triangleq 2f^{-1}\epsilon \hat{\sigma}^2_{(M)} \left| \sum_{m=1}^{M} k_m v_m \right|$$
$$= O_p\left(f^{-1} M^{1/2} k_M \epsilon \right) \cdot \|\boldsymbol{v}\|.$$

Thus, by Condition 6, Δ_4' is asymptotically negligible relative to Δ_1. This completes the proof of Theorem 3.

S3.7 Proof of Theorem 4

Proof. We first prove Equation (16) in Theorem 4. Following Fan & Peng (2004) and Chen et al. (2018), it is sufficient to show that

$$\lim_{N\to\infty} P\left(\sup_{\check{\boldsymbol{\omega}}\in\mathcal{W}_1,\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2}\{C_n(\check{\boldsymbol{\omega}})-C_n(\tilde{\boldsymbol{\omega}})\}<0\right)=1.$$

For any $\check{\boldsymbol{\omega}}\in\mathcal{W}_1$ and $\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2$, we have

$$\begin{aligned}
C_n(\check{\boldsymbol{\omega}})-C_n(\tilde{\boldsymbol{\omega}}) &= \|\boldsymbol{y}_s-\hat{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s}-\|\boldsymbol{y}_s-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s}\\
&\quad+2f^{-1}\sigma^2k(\check{\boldsymbol{\omega}})-2f^{-1}\sigma^2k(\tilde{\boldsymbol{\omega}})\\
&=-\tilde{R}_N(\tilde{\boldsymbol{\omega}})+\|\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s}\\
&\quad+\left\{\tilde{R}_N(\tilde{\boldsymbol{\omega}})-\|\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s}\right\}\\
&\quad+2\left[\{\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s+f^{-1}\sigma^2k(\check{\boldsymbol{\omega}})\right]\\
&\quad-2\left[\{\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s+f^{-1}\sigma^2k(\tilde{\boldsymbol{\omega}})\right]\\
&\triangleq-\tilde{R}_N(\tilde{\boldsymbol{\omega}})+\tilde{\Delta}_1+\tilde{\Delta}_2+\tilde{\Delta}_3+\tilde{\Delta}_4.
\end{aligned}$$

Note that, for each correct model m, $\boldsymbol{\mu}_U=\boldsymbol{X}_{(m)U}\boldsymbol{\beta}_{(m)}$ and $\boldsymbol{\mu}_s=\boldsymbol{X}_{(m)s}\boldsymbol{\beta}_{(m)}$. Therefore, for any $\check{\boldsymbol{\omega}}\in\mathcal{W}_1$, whose weights are assigned only to correct models, we have

$$\begin{aligned}
\boldsymbol{\mu}_s-\tilde{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}}) &= \boldsymbol{\mu}_s-\sum_{m=1}^{M}\check{\omega}_m\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}'_{(m)U}\boldsymbol{y}_U\\
&=\boldsymbol{\mu}_s-\sum_{m=1}^{M}\check{\omega}_m\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}'_{(m)U}\left(\boldsymbol{X}_{(m)U}\boldsymbol{\beta}_{(m)}+\boldsymbol{e}_U\right)\\
&=-\sum_{m=1}^{M}\check{\omega}_m\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}'_{(m)U}\boldsymbol{e}_U\\
&\triangleq-\tilde{\boldsymbol{P}}_s(\check{\boldsymbol{\omega}})\boldsymbol{e}_U,
\end{aligned}$$

where $\tilde{\boldsymbol{P}}_s(\check{\boldsymbol{\omega}})=\sum_{m=1}^{M}\check{\omega}_m\boldsymbol{X}_{(m)s}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}'_{(m)U}$. It follows that

$$|\tilde{\Delta}_1|=\|\boldsymbol{\mu}_s-\hat{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s}$$

$$
\begin{aligned}
&= \|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) + \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} \\
&\leq 2\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} + 2\|\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} \\
&= 2\left\|\tilde{\boldsymbol{P}}_s(\breve{\boldsymbol{\omega}})\boldsymbol{e}_U\right\|^2_{\boldsymbol{\Phi}_s} + 2\|\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s}
\end{aligned}
$$

and

$$
\begin{aligned}
|\tilde{\Delta}_3| &= 2\left|\{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s + f^{-1}\sigma^2 k(\breve{\boldsymbol{\omega}})\right| \\
&= 2\left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) + \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s + f^{-1}\sigma^2 k(\breve{\boldsymbol{\omega}})\right| \\
&\leq 2\left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s + f^{-1}\sigma^2 k(\breve{\boldsymbol{\omega}})\right| + 2\left|\{\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right|.
\end{aligned}
$$

For any $\tilde{\boldsymbol{\omega}} \in \mathcal{W}_2$, we have

$$
\begin{aligned}
|\tilde{\Delta}_2| &= \left|\|\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} - \tilde{R}_N(\tilde{\boldsymbol{\omega}})\right| \\
&= \Big|\|\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} + 2\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\} \\
&\quad + \|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} - \tilde{R}_N(\tilde{\boldsymbol{\omega}})\Big| \\
&\leq \|\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} + \left|\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} - \tilde{R}_N(\tilde{\boldsymbol{\omega}})\right| \\
&\quad + 2\left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}\right|.
\end{aligned}
$$

Similarly,

$$
\begin{aligned}
|\tilde{\Delta}_4| &= 2\left|\{\boldsymbol{\mu}_s - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s + f^{-1}\sigma^2 k(\tilde{\boldsymbol{\omega}})\right| \\
&= 2\left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) + \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s + f^{-1}\sigma^2 k(\tilde{\boldsymbol{\omega}})\right| \\
&\leq 2\left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s + f^{-1}\sigma^2 k(\tilde{\boldsymbol{\omega}})\right| + 2\left|\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right|.
\end{aligned}
$$

Therefore, the proof of Equation (16) in Theorem 4 reduces to showing that

$$
\sup_{\breve{\boldsymbol{\omega}}\in\mathcal{W}_1, \tilde{\boldsymbol{\omega}}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}})\|\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} = o_p(1), \tag{S.51}
$$

$$
\sup_{\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}})\|\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|^2_{\boldsymbol{\Phi}_s} = o_p(1), \tag{S.52}
$$

$$\sup_{\breve{\omega}\in\mathcal{W}_1,\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left|\{\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\}' \boldsymbol{\Phi}_s \boldsymbol{e}_s\right| = o_p(1), \tag{S.53}$$

$$\sup_{\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left|\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}' \boldsymbol{\Phi}_s \boldsymbol{e}_s\right| = o_p(1), \tag{S.54}$$

$$\sup_{\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}' \boldsymbol{\Phi}_s \{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}\right| = o_p(1), \tag{S.55}$$

$$\sup_{\breve{\omega}\in\mathcal{W}_1,\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\}' \boldsymbol{\Phi}_s \boldsymbol{e}_s + f^{-1}\sigma^2 k(\breve{\boldsymbol{\omega}})\right| = o_p(1), \tag{S.56}$$

$$\sup_{\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left|\{\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}' \boldsymbol{\Phi}_s \boldsymbol{e}_s + f^{-1}\sigma^2 k(\tilde{\boldsymbol{\omega}})\right| = o_p(1), \tag{S.57}$$

$$\sup_{\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left|\|\boldsymbol{\mu}_s - \tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|_{\boldsymbol{\Phi}_s}^2 - \tilde{R}_N(\tilde{\boldsymbol{\omega}})\right| = o_p(1), \tag{S.58}$$

and

$$\sup_{\breve{\omega}\in\mathcal{W}_1,\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \left\|\tilde{\boldsymbol{P}}_s(\breve{\boldsymbol{\omega}})\boldsymbol{e}_U\right\|_{\boldsymbol{\Phi}_s}^2 = o_p(1). \tag{S.59}$$

Using the same argument as in the proof of Equation (S.35), for any $\delta > 0$, we have

$$\begin{aligned}
&P\left(\sup_{\breve{\omega}\in\mathcal{W}_1,\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \|\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|_{\boldsymbol{\Phi}_s}^2 > \delta\right) \\
&\leq P\left(\sup_{\breve{\omega}\in\mathcal{W}_1} \|\tilde{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\breve{\boldsymbol{\omega}})\|_{\boldsymbol{\Phi}_s}^2 > \delta\tilde{\xi}_N^*\right) \\
&= O\left(\frac{s_0^2 k_M^2}{f\tilde{\xi}_N^*}\right)
\end{aligned}$$

and

$$\begin{aligned}
&P\left(\sup_{\tilde{\omega}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}}) \|\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|_{\boldsymbol{\Phi}_s}^2 > \delta\right) \\
&\leq P\left(\sup_{\tilde{\omega}\in\mathcal{W}_2} \|\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}}) - \hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\|_{\boldsymbol{\Phi}_s}^2 > \delta\tilde{\xi}_N^*\right) \\
&= O\left(\frac{(M-s_0)^2 k_M^2}{f\tilde{\xi}_N^*}\right).
\end{aligned}$$

By Condition 7, both upper bounds are $o(1)$. This establishes Equations (S.51) and (S.52).

Similarly, using the same argument as in the proof of Equation (S.36), for any $\delta > 0$, we have

$$\begin{aligned}
&P\left(\sup_{\check{\boldsymbol{\omega}}\in\mathcal{W}_1,\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}})\left|\{\tilde{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})-\hat{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right|>\delta\right)\\
&\leq P\left(\sup_{\check{\boldsymbol{\omega}}\in\mathcal{W}_1}\left|\{\tilde{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})-\hat{\boldsymbol{\mu}}_s(\check{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right|>\delta\tilde{\xi}_N^*\right)\\
&=O\left(\frac{s_0\sqrt{nk_M}}{f\tilde{\xi}_N^*}\right)
\end{aligned}$$

and

$$\begin{aligned}
&P\left(\sup_{\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}})\left|\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right|>\delta\right)\\
&\leq P\left(\sup_{\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2}\left|\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\boldsymbol{e}_s\right|>\delta\tilde{\xi}_N^*\right)\\
&=O\left(\frac{(M-s_0)\sqrt{nk_M}}{f\tilde{\xi}_N^*}\right).
\end{aligned}$$

By Condition 7, both upper bounds are $o(1)$. This establishes Equations (S.53) and (S.54).

By the argument used to prove Equation (S.37), for any $\delta > 0$, we have

$$\begin{aligned}
&P\left(\sup_{\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2} \tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}})\left|\{\boldsymbol{\mu}_s-\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}\right|>\delta\right)\\
&\leq P\left(\sup_{\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2}\left|\{\boldsymbol{\mu}_s-\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}'\boldsymbol{\Phi}_s\{\tilde{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})-\hat{\boldsymbol{\mu}}_s(\tilde{\boldsymbol{\omega}})\}\right|>\delta\tilde{\xi}_N^*\right)\\
&=O\left(\frac{(M-s_0)^2\sqrt{nk_M}}{f\tilde{\xi}_N^*}\right).
\end{aligned}$$

By Condition 7, the upper bound is $o(1)$, and hence Equation (S.55) holds.

Following the proofs of Equations (S.28) and (S.38), and using Conditions 7 and 8, Equations (S.56) and (S.57) follow by replacing $\tilde{R}_N(\boldsymbol{\omega})$ and $\tilde{\xi}_N$ with $\tilde{R}_N(\tilde{\boldsymbol{\omega}})$ and $\tilde{\xi}_N^*$, respectively. Similarly, Equation (S.58) follows from the same arguments as those used in the proofs of Equations (S.39) and (S.40).

It remains to prove Equation (S.59). By Markov's inequality and the Cauchy–Schwarz inequality, we have

$$
\begin{aligned}
&P\left(\sup_{\check{\boldsymbol{\omega}}\in\mathcal{W}_1,\tilde{\boldsymbol{\omega}}\in\mathcal{W}_2}\tilde{R}_N^{-1}(\tilde{\boldsymbol{\omega}})\left\|\tilde{\boldsymbol{P}}_s(\check{\boldsymbol{\omega}})\boldsymbol{e}_U\right\|_{\boldsymbol{\Phi}_s}^2>\delta\right)\\
&\leq P\left(\sup_{\check{\boldsymbol{\omega}}\in\mathcal{W}_1}\left|\boldsymbol{e}_U'\tilde{\boldsymbol{P}}_s(\check{\boldsymbol{\omega}})'\boldsymbol{\Phi}_s\tilde{\boldsymbol{P}}_s(\check{\boldsymbol{\omega}})\boldsymbol{e}_U\right|>\delta\tilde{\xi}_N^*\right)\\
&\leq \sum_{m=1}^{s_0}\sum_{t=1}^{s_0}P\left(\left|\boldsymbol{e}_U'\boldsymbol{X}_{(m)U}\left(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\boldsymbol{X}_{(t)U}'\boldsymbol{X}_{(t)U}\right)^{-1}\boldsymbol{X}_{(t)U}'\boldsymbol{e}_U\right|>\delta\tilde{\xi}_N^*\right)\\
&\leq \delta^{-1}\tilde{\xi}_N^{*-1}\cdot\sum_{m=1}^{s_0}\sum_{t=1}^{s_0}E\left|\boldsymbol{e}_U'\boldsymbol{X}_{(m)U}\left(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s\boldsymbol{X}_{(t)s}\left(\boldsymbol{X}_{(t)U}'\boldsymbol{X}_{(t)U}\right)^{-1}\boldsymbol{X}_{(t)U}'\boldsymbol{e}_U\right|\\
&\leq \delta^{-1}\tilde{\xi}_N^{*-1}\cdot\left[\sum_{m=1}^{s_0}\left\{E\left\|\boldsymbol{e}_U'\boldsymbol{X}_{(m)U}\left(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}_{(m)s}'\boldsymbol{\Phi}_s^{1/2}\right\|^2\right\}^{1/2}\right]^2\\
&= \delta^{-1}\tilde{\xi}_N^{*-1}\cdot\left[\sum_{m=1}^{s_0}\left\{E_m\left(\boldsymbol{e}_U'\boldsymbol{P}_{(m)U}\boldsymbol{e}_U\right)\right\}^{1/2}\right]^2\\
&= \sigma^2\delta^{-1}\tilde{\xi}_N^{*-1}\cdot\left[\sum_{m=1}^{s_0}\left\{\operatorname{tr}\left(\boldsymbol{P}_{(m)U}\right)\right\}^{1/2}\right]^2\\
&\leq \sigma^2\delta^{-1}s_0^2k_M\tilde{\xi}_N^{*-1},
\end{aligned}
$$

where $\boldsymbol{P}_{(m)U}=\boldsymbol{X}_{(m)U}\left(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}_{(m)U}'$. By Condition 7, the last bound is $o(1)$, and hence Equation (S.59) holds. This completes the proof of Equation (16) in Theorem 4.

We now prove Equation (17) in Theorem 4. By the same argument as that used in the proof of Equation (S.48), together with Conditions 7 and 8, Equation (17) follows after replacing $\tilde{R}_N(\boldsymbol{\omega})$ and $\tilde{\xi}_N$ with $\tilde{R}_N(\tilde{\boldsymbol{\omega}})$ and $\tilde{\xi}_N^*$, respectively. This completes the proof of Theorem 4.

S3.8 Lemmas for the proofs of Theorems 5–7

Lemma 8 *If Conditions 2 and 9 hold, then*

$$
E\left(\frac{1}{N}\sum_{i\in s}\frac{z_i}{\pi_i}-\frac{1}{N}\sum_{i\in U}z_i\right)^4\leq Cn^{-2}\cdot\frac{1}{N}\sum_{i\in U}E_\epsilon(z_i^4).
$$

Proof. By Conditions 2 and 9, Loève's c_r inequality, and the Cauchy–Schwarz inequality,

$$
\begin{aligned}
& E\left(\frac{1}{N}\sum_{i\in s}\frac{z_i}{\pi_i}-\frac{1}{N}\sum_{i\in U}z_i\right)^4 \\
&= \left|E_\epsilon E_d\left[\frac{1}{N}\sum_{i\in U}\frac{z_i(I_i-\pi_i)}{\pi_i}\right\}^4\right| \\
&\le \frac{C}{N^4}E_\epsilon\left\{\sum_{i,j,k,l\in \boldsymbol{D}_4}|z_iz_jz_kz_l|\cdot\left|E_d\left\{(I_i-\pi_i)(I_j-\pi_j)(I_k-\pi_k)(I_l-\pi_l)\right\}\right|\right. \\
&\quad +\sum_{i,j,k\in \boldsymbol{D}_3}|z_i^2z_jz_k|\cdot\left|E_d\left\{(I_i-\pi_i)^2(I_j-\pi_j)(I_k-\pi_k)\right\}\right| \\
&\quad +\sum_{i\neq j\in U}|z_i^2z_j^2|\cdot\left|E_d\left\{(I_i-\pi_i)^2(I_j-\pi_j)^2+(I_i-\pi_i)^3(I_j-\pi_j)\right\}\right| \\
&\quad \left.+\sum_{i\in U}|z_i^4|\cdot E_d\left\{(I_i-\pi_i)^4\right\}\right] \\
&\le Cn^{-2}\cdot\frac{1}{N}\sum_{i\in U}E_\epsilon(z_i^4).
\end{aligned}
$$

This completes the proof of Lemma 8.

Lemma 9 *If Conditions 1–2 and 9 hold, then*

$$E\left\|\hat{\boldsymbol{\beta}}_{(m)}-\boldsymbol{\beta}^*_{(m)}\right\|^4\le Cn^{-2}k_m^2.$$

Proof. By Condition 1, we have

$$
\begin{aligned}
E\left\|\hat{\boldsymbol{\beta}}_{(m)}-\boldsymbol{\beta}^*_{(m)}\right\|^4 =& E\left\|\left(\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}\boldsymbol{x}'_{i(m)}}{\pi_i}\right)^{-1}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^4 \\
\le& E\left\{\lambda_{\min}^{-1}\left(\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}\right)\cdot\left\|\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|\right\}^4 \\
\le& CE\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^4. \qquad \text{(S.60)}
\end{aligned}
$$

Moreover, Condition 1 implies that

$$E_\epsilon(x_{ij(m)}e^*_{i(m)})^4\le C.$$

Applying Lemma 8 yields

$$E\left(\frac{1}{N}\sum_{i\in s}\frac{x_{ij(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}x_{ij(m)}e^*_{i(m)}\right)^4\le Cn^{-2}.$$

Therefore,

$$
\begin{aligned}
& E\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}\right\|^4 \\
&= E\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}+\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^4 \\
&\le CE\left\|\frac{1}{N}\sum_{i\in s}\frac{\boldsymbol{x}_{i(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^4+CE_\epsilon\left\|\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^4 \\
&\le CE\left\{\sum_{j=1}^{k_m}\left(\frac{1}{N}\sum_{i\in s}\frac{x_{ij(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}x_{ij}e^*_{i(m)}\right)^2\right\}^2+CE_\epsilon\left\|\frac{1}{N}\sum_{i\in U}\boldsymbol{x}_{i(m)}e^*_{i(m)}\right\|^4 \\
&\le Ck_m\sum_{j=1}^{k_m}\left[E\left(\frac{1}{N}\sum_{i\in s}\frac{x_{ij(m)}e^*_{i(m)}}{\pi_i}-\frac{1}{N}\sum_{i\in U}x_{ij}e^*_{i(m)}\right)^4+\frac{1}{N^4}\left\{\sum_{i\in U}\sqrt{E_\epsilon\left(x_{ij}e^*_{i(m)}\right)^4}\right\}^2\right] \\
&\le Cn^{-2}k_m^2, \qquad \text{(S.61)}
\end{aligned}
$$

where the third inequality follows from Equation (S.4) with $q = 2$. Combining (S.60) and (S.61), we obtain the desired result.

Lemma 10 *Let* $\bar{\boldsymbol{T}}_{(m)} = N^{-1}\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}$, $\bar{\boldsymbol{t}}_{(m)} = N^{-1}\boldsymbol{X}'_{(m)U}\boldsymbol{y}_U$, $\hat{\bar{\boldsymbol{T}}}_{(m)} = N^{-1}\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}$, *and* $\hat{\bar{\boldsymbol{t}}}_{(m)} = N^{-1}\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{y}_s$. *By a Taylor expansion, we have*

$$\hat{\boldsymbol{\beta}}_{(m)} = \tilde{\boldsymbol{\beta}}_{(m)} + \bar{\boldsymbol{T}}^{-1}_{(m)}\left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right) + \boldsymbol{r}_{(m)},$$

where $\boldsymbol{r}_{(m)}$ *denotes the remainder term. If Conditions 1–2 and 9–10 hold, then*

$$E\left(r^2_{j(m)}\right) = O(n^{-2}),$$

where $r_{j(m)}$ *is the* j*th element of* $\boldsymbol{r}_{(m)}$.

Proof. Let $\bar{t}_{jj'(m)}$, $\bar{t}_{j(m)}$, $\hat{\bar{t}}_{jj'(m)}$, and $\hat{\bar{t}}_{j(m)}$ be the corresponding elements of $\bar{\boldsymbol{T}}_{(m)}$, $\bar{\boldsymbol{t}}_{(m)}$, $\hat{\bar{\boldsymbol{T}}}_{(m)}$, and $\hat{\bar{\boldsymbol{t}}}_{(m)}$, respectively. A Taylor linearization of $\hat{\boldsymbol{\beta}}_{(m)}$ around $(\bar{\boldsymbol{T}}_{(m)}, \bar{\boldsymbol{t}}_{(m)})$ yields

$$\hat{\boldsymbol{\beta}}_{(m)} = \tilde{\boldsymbol{\beta}}_{(m)} + \sum_{j=1}^{k_m}\sum_{j'\le j} a_{jj'(m)}\left(\hat{\bar{t}}_{jj'(m)} - \bar{t}_{jj'(m)}\right) + \sum_{j=1}^{k_m} a_{j(m)}\left(\hat{\bar{t}}_{j(m)} - \bar{t}_{j(m)}\right) + \boldsymbol{r}_{(m)},$$

where $a_{jj'(m)} = \left.\dfrac{\partial\hat{\boldsymbol{\beta}}_{(m)}}{\partial\hat{\bar{t}}_{jj'(m)}}\right|_{\hat{\bar{\boldsymbol{T}}}_{(m)}=\bar{\boldsymbol{T}}_{(m)},\hat{\bar{\boldsymbol{t}}}_{(m)}=\bar{\boldsymbol{t}}_{(m)}}$, $a_{j(m)} = \left.\dfrac{\partial\hat{\boldsymbol{\beta}}_{(m)}}{\partial\hat{\bar{t}}_{j(m)}}\right|_{\hat{\bar{\boldsymbol{T}}}_{(m)}=\bar{\boldsymbol{T}}_{(m)},\hat{\bar{\boldsymbol{t}}}_{(m)}=\bar{\boldsymbol{t}}_{(m)}}$ and $\boldsymbol{r}_{(m)}$ is the remainder term. By standard matrix differentiation results (see, e.g., Graybill 1983), we

have

$$\frac{\partial \hat{\boldsymbol{\beta}}_{(m)}}{\partial \hat{\bar{t}}_{jj'(m)}} = \left(\frac{\partial \hat{\bar{\boldsymbol{T}}}_{(m)}^{-1}}{\partial \hat{\bar{t}}_{jj'(m)}} \right) \hat{\bar{\boldsymbol{t}}}_{(m)} = \left(-\hat{\bar{\boldsymbol{T}}}_{(m)}^{-1} \boldsymbol{\Lambda}_{jj'(m)} \hat{\bar{\boldsymbol{T}}}_{(m)}^{-1} \right) \hat{\bar{\boldsymbol{t}}}_{(m)} = -\hat{\bar{\boldsymbol{T}}}_{(m)}^{-1} \boldsymbol{\Lambda}_{jj'(m)} \hat{\boldsymbol{\beta}}_{(m)},$$

where $\boldsymbol{\Lambda}_{jj'(m)}$ is a $k_m \times k_m$ matrix with one at positions (j, j') and (j', j) and zeros elsewhere. Similarly,

$$\frac{\partial \hat{\boldsymbol{\beta}}_{(m)}}{\partial \hat{\bar{t}}_{j(m)}} = \hat{\bar{\boldsymbol{T}}}_{(m)}^{-1} \left(\frac{\partial \hat{\bar{\boldsymbol{t}}}_{(m)}}{\partial \hat{\bar{t}}_{j(m)}} \right) = \hat{\bar{\boldsymbol{T}}}_{(m)}^{-1} \boldsymbol{\lambda}_{j(m)},$$

where $\boldsymbol{\lambda}_{j(m)}$ is a k_m-vector with the jth component equal to one and zeros elsewhere. Therefore,

$$\begin{aligned}
\hat{\boldsymbol{\beta}}_{(m)} &= \tilde{\boldsymbol{\beta}}_{(m)} - \sum_{j=1}^{k_m} \sum_{j' \leq j} \bar{\boldsymbol{T}}_{(m)}^{-1} \boldsymbol{\Lambda}_{jj'(m)} \tilde{\boldsymbol{\beta}}_{(m)} \left(\hat{\bar{t}}_{jj'(m)} - \bar{t}_{jj'(m)} \right) \\
&\quad + \sum_{j=1}^{k_m} \bar{\boldsymbol{T}}_{(m)}^{-1} \boldsymbol{\lambda}_{j(m)} \left(\hat{\bar{t}}_{j(m)} - \bar{t}_{j(m)} \right) + \boldsymbol{r}_{(m)} \\
&= \tilde{\boldsymbol{\beta}}_{(m)} - \bar{\boldsymbol{T}}_{(m)}^{-1} \left(\hat{\bar{\boldsymbol{T}}}_{(m)} - \bar{\boldsymbol{T}}_{(m)} \right) \tilde{\boldsymbol{\beta}}_{(m)} + \bar{\boldsymbol{T}}_{(m)}^{-1} \left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \bar{\boldsymbol{t}}_{(m)} \right) + \boldsymbol{r}_{(m)} \\
&= \tilde{\boldsymbol{\beta}}_{(m)} + \bar{\boldsymbol{T}}_{(m)}^{-1} \left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)} \tilde{\boldsymbol{\beta}}_{(m)} \right) + \boldsymbol{r}_{(m)}.
\end{aligned}$$

It follows from the above equation that

$$r_{j(m)} = \boldsymbol{\lambda}_{j(m)}' \boldsymbol{r}_{(m)} = \boldsymbol{\lambda}_{j(m)}' \hat{\bar{\boldsymbol{T}}}_{(m)}^{-1} \hat{\bar{\boldsymbol{t}}}_{(m)} - \boldsymbol{\lambda}_{j(m)}' \tilde{\boldsymbol{\beta}}_{(m)} - \boldsymbol{\lambda}_{j(m)}' \bar{\boldsymbol{T}}_{(m)}^{-1} \left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)} \tilde{\boldsymbol{\beta}}_{(m)} \right).$$

Since $\hat{\bar{\boldsymbol{T}}}_{(m)}$ and $\hat{\bar{\boldsymbol{t}}}_{(m)}$ consist of HT estimators, $r_{j(m)}$ is a nonlinear function of these HT estimators. By Lemma 8 and Conditions 1 and 10, the fourth-order moment of each HT estimator is $O(n^{-2})$. Moreover, $r_{j(m)}^2$ and its first three derivatives are equal to zero at the point $(\bar{\boldsymbol{T}}_{(m)}, \bar{\boldsymbol{t}}_{(m)})$. Therefore, applying Theorem 5.4.3 of Fuller (1996) with $\alpha = 1$, $s = 4$, and $a_N^4 = O(n^{-2})$, we conclude that

$$E\left(r_{j(m)}^2 \right) = O(n^{-2}).$$

This completes the proof of Lemma 10.

Lemma 11 *If Conditions 1–2 and 9–10 and $k_M^2 = O(N)$ hold, then*

$$E(b_N^2) = E\left[\frac{1}{N}\sum_{i\in s}\frac{\{\tilde{\mu}_i(\boldsymbol{\omega}^*) - \hat{\mu}_i(\boldsymbol{\omega}^*)\}}{\pi_i} - \frac{1}{N}\sum_{i\in U}\{\tilde{\mu}_i(\boldsymbol{\omega}^*) - \hat{\mu}_i(\boldsymbol{\omega}^*)\}\right]^2 = O(n^{-2}k_M^2),$$

where $\boldsymbol{\omega}^*$ is the optimal weight vector.

Proof. By Lemma 10, we have

$$\begin{aligned}
\hat{\mu}_i(\boldsymbol{\omega}^*) - \tilde{\mu}_i(\boldsymbol{\omega}^*) &= \sum_{m=1}^{M}\omega_m^*\boldsymbol{x}'_{i(m)}\left\{\bar{\boldsymbol{T}}_{(m)}^{-1}\left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right) + \boldsymbol{r}_{(m)}\right\} \\
&= \frac{1}{N}\sum_{j\in U}\frac{I_j - \pi_j}{\pi_j}\sum_{m=1}^{M}\left\{\omega_m^*\boldsymbol{x}'_{i(m)}\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{j(m)}\left(y_j - \boldsymbol{x}'_{j(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)\right\} \\
&\quad + \sum_{m=1}^{M}\left(\omega_m^*\boldsymbol{x}'_{i(m)}\boldsymbol{r}_{(m)}\right) \\
&\triangleq \frac{1}{N}\sum_{j\in U}\frac{I_j - \pi_j}{\pi_j}z_{ij} + r_i(\boldsymbol{\omega}^*).
\end{aligned}$$

Then

$$\begin{aligned}
b_N^2 &= \frac{1}{N^2}\sum_{i,j\in U}\frac{(\pi_i - I_i)(\pi_j - I_j)\{\hat{\mu}_i(\boldsymbol{\omega}^*) - \tilde{\mu}_i(\boldsymbol{\omega}^*)\}\{\hat{\mu}_j(\boldsymbol{\omega}^*) - \tilde{\mu}_j(\boldsymbol{\omega}^*)\}}{\pi_i\pi_j} \\
&= \frac{1}{N^4}\sum_{i,j,k,l\in U}\frac{(\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k)(\pi_l - I_l)z_{ik}z_{jl}}{\pi_i\pi_j\pi_k\pi_l} \\
&\quad + \frac{2}{N^3}\sum_{i,j,k\in U}\frac{(\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k)z_{ik}r_j(\boldsymbol{\omega}^*)}{\pi_i\pi_j\pi_k} \\
&\quad + \frac{1}{N^2}\sum_{i,j\in U}\frac{(\pi_i - I_i)(\pi_j - I_j)r_i(\boldsymbol{\omega}^*)r_j(\boldsymbol{\omega}^*)}{\pi_i\pi_j} \\
&\triangleq b_{1N} + b_{2N} + b_{3N}.
\end{aligned}$$

First, we prove that $|E(b_{1N})| = O(n^{-2}k_M^2)$. Condition 1 and Lemma 1 imply that

$$\left|\boldsymbol{x}'_{i(m)}\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{j(m)}\right| \le \left\|\boldsymbol{x}_{i(m)}\right\|^2 \cdot \lambda_{\min}^{-1}\left(\bar{\boldsymbol{T}}_{(m)}\right) \le Ck_m$$

and

$$\begin{aligned}
&E_\epsilon\left|\left(y_j - \boldsymbol{x}'_{j(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)\left(y_j - \boldsymbol{x}'_{j(l)}\tilde{\boldsymbol{\beta}}_{(l)}\right)\right| \\
&\le \left(E_\epsilon\left|e^*_{j(m)} + \boldsymbol{x}'_{j(m)}\boldsymbol{\beta}^*_{(m)} - \boldsymbol{x}'_{j(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right|^2 \cdot E_\epsilon\left|e^*_{j(l)} + \boldsymbol{x}'_{j(l)}\beta^*_{(l)} - \boldsymbol{x}'_{j(l)}\tilde{\boldsymbol{\beta}}_{(l)}\right|^2\right)^{1/2}
\end{aligned}$$

$$\leq C \left\{ \left(1 + \frac{k_m^2}{N}\right) \left(1 + \frac{k_l^2}{N}\right) \right\}^{1/2}.$$

It follows from the condition $k_M^2 = O(N)$ that

$$E_\epsilon(z_{ij}^2) \leq C \sum_{m=1}^{M} \sum_{l=1}^{M} \omega_m^* \omega_l^* k_m k_l \left\{ \left(1 + \frac{k_m^2}{N}\right) \left(1 + \frac{k_l^2}{N}\right) \right\}^{1/2} = O(k_M^2)$$

uniformly in i and j. Thus, using Conditions 2 and 9, we obtain that

$$\begin{aligned}
|E(b_{1N})| &= \left| E \left\{ \frac{1}{N^4} \sum_{i,j,k,l \in U} \frac{(\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k)(\pi_l - I_l) z_{ik} z_{jl}}{\pi_i \pi_j \pi_k \pi_l} \right\} \right| \\
&= \left| \frac{1}{N^4} \sum_{i,j,k,l \in U} E_\epsilon(z_{ik} z_{jl}) \cdot E_d \left\{ \frac{(\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k)(\pi_l - I_l)}{\pi_i \pi_j \pi_k \pi_l} \right\} \right| \\
&\leq \frac{C k_M^2}{N^4} \sum_{i,j,k,l \in U} \left| E_d \left\{ (\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k)(\pi_l - I_l) \right\} \right| \\
&\leq \frac{C k_M^2}{N^4} \left[\sum_{i,j,k,l \in \boldsymbol{D}_4} \left| E_d \left\{ (I_i - \pi_i)(I_j - \pi_j)(I_k - \pi_k)(I_l - \pi_l) \right\} \right| \right. \\
&\quad + \sum_{i,j,k \in \boldsymbol{D}_3} \left| E_d \left\{ (I_i - \pi_i)^2 (I_j - \pi_j)(I_k - \pi_k) \right\} \right| \\
&\quad + \sum_{i \neq j \in U} \left| E_d \left\{ (I_i - \pi_i)^2 (I_j - \pi_j)^2 + (I_i - \pi_i)(I_j - \pi_j)^3 \right\} \right| \\
&\quad \left. + \sum_{i \in U} \left| E_d \left\{ (I_i - \pi_i)^4 \right\} \right| \right] \\
&= O\left(n^{-2} k_M^2 \right).
\end{aligned}$$

Next, we consider $|E(b_{3N})|$. By Lemma 10, we have

$$E \left\| \boldsymbol{r}_{(m)} \right\|^2 = \sum_{j=1}^{k_m} E\left(r_{j(m)}^2 \right) = O(n^{-2} k_m)$$

uniformly in m. It follows from Condition 1 that

$$\begin{aligned}
\sum_{i \in U} E\left\{ r_i^2(\boldsymbol{\omega}^*) \right\} &= E \left\{ \sum_{m=1}^{M} \sum_{l=1}^{M} \omega_m^* \omega_l^* \boldsymbol{r}'_{(m)} \left(\sum_{i \in U} \boldsymbol{x}_{i(m)} \boldsymbol{x}'_{i(l)} \right) \boldsymbol{r}_{(l)} \right\} \\
&= E \left(\sum_{m=1}^{M} \sum_{l=1}^{M} \omega_m^* \omega_l^* \boldsymbol{r}'_{(m)} \boldsymbol{X}'_{(m)U} \boldsymbol{X}_{(l)U} \boldsymbol{r}_{(l)} \right) \\
&\leq C \sum_{m=1}^{M} \sum_{l=1}^{M} \omega_m^* \omega_l^* \left(E \left\| \boldsymbol{r}_{(m)} \right\|^2 E \left\| \boldsymbol{r}_{(l)} \right\|^2 \right)^{1/2}
\end{aligned}$$

$$= O(Nn^{-2}k_M^2). \tag{S.62}$$

Applying Loève's c_r inequality, we obtain that

$$\begin{aligned}
|E(b_{3N})| &= \left| E\left\{ \frac{1}{N^2} \sum_{i,j\in U} \frac{(\pi_i - I_i)(\pi_j - I_j) r_i(\boldsymbol{\omega}^*) r_j(\boldsymbol{\omega}^*)}{\pi_i \pi_j} \right\} \right| \\
&= E\left\{ \frac{1}{N} \sum_{i\in U} \frac{r_i(\boldsymbol{\omega}^*)(\pi_i - I_i)}{\pi_i} \right\}^2 \\
&\leq \frac{1}{N} \sum_{i\in U} E\left\{ r_i^2(\boldsymbol{\omega}^*) \right\} \\
&= O\left(n^{-2} k_M^2 \right).
\end{aligned}$$

Finally, we prove that $|E(b_{2N})| = O\left(n^{-2}k_M^2\right)$. By the Cauchy–Schwarz inequality,

$$\begin{aligned}
|E(b_{2N})| &= \left| E\left\{ \frac{2}{N^3} \sum_{i,j,k\in U} \frac{(\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k) z_{ik} r_j(\boldsymbol{\omega}^*)}{\pi_i \pi_j \pi_k} \right\} \right| \\
&= \frac{2}{N^3} \left| E\left\{ \sum_{i,k\in U} \frac{(\pi_i - I_i)(\pi_k - I_k) z_{ik}}{\pi_i \pi_k} \cdot \sum_{j\in U} \frac{(\pi_j - I_j) r_j(\boldsymbol{\omega}^*)}{\pi_j} \right\} \right| \\
&\leq \frac{2}{N^3} \left[E\left\{ \sum_{i,k\in U} \frac{(\pi_i - I_i)(\pi_k - I_k) z_{ik}}{\pi_i \pi_k} \right\}^2 \cdot E\left\{ \sum_{j\in U} \frac{(\pi_j - I_j) r_j(\boldsymbol{\omega}^*)}{\pi_j} \right\}^2 \right]^{1/2} \\
&= \frac{2}{N^3} \cdot \sqrt{N^4 |E(b_{1N})| \cdot N^2 |E(b_{3N})|} \\
&= O\left(n^{-2} k_M^2 \right).
\end{aligned}$$

This completes the proof of Lemma 11.

Lemma 12 *If Conditions 1–2 and 10 hold, then*

$$\frac{1}{N} \sum_{i\in U} E\left\{ \hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*) \right\}^2 = O\left(n^{-1} M k_M + M \zeta_n^{1/2} \right),$$

where $\hat{\boldsymbol{\omega}}_0$ *is an estimator of* $\boldsymbol{\omega}^*$ *and* $\zeta_n = E\|\hat{\boldsymbol{\omega}}_0 - \boldsymbol{\omega}^*\|^4$.

Proof. Conditions 1–2 and 10 imply that

$$E\|\boldsymbol{y}_U\|^4 \leq N \sum_{i\in U} E_\epsilon (\mu_i + e_i)^4$$

$$\leq CN \sum_{i \in U} \mu_i^4 + CN \sum_{i \in U} E_\epsilon \left(e_i^4 \right)$$

$$= O(N^2).$$

Then, by Condition 1, Lemma 3, Loève's c_r inequality, and the Cauchy–Schwarz inequality, we have

$$\begin{aligned}
&\sum_{i \in U} E \left\{ \hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*) \right\}^2 \\
&= \sum_{i \in U} E \left\{ \hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\hat{\boldsymbol{\omega}}_0) + \tilde{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*) \right\}^2 \\
&\leq 2 \sum_{i \in U} E \left[\sum_{m=1}^{M} \left\{ \hat{\omega}_{0m} \left(\boldsymbol{x}'_{i(m)} \hat{\boldsymbol{\beta}}_{(m)} - \boldsymbol{x}'_{i(m)} \tilde{\boldsymbol{\beta}}_{(m)} \right) \right\} \right]^2 \\
&\quad + 2 \sum_{i \in U} E \left[\sum_{m=1}^{M} \left\{ (\hat{\omega}_{0m} - \omega_m^*) \boldsymbol{x}'_{i(m)} \tilde{\boldsymbol{\beta}}_{(m)} \right\} \right]^2 \\
&\leq 2E \left\{ \sum_{m=1}^{M} \left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)} \right)' \boldsymbol{X}'_{(m)U} \boldsymbol{X}_{(m)U} \left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)} \right) \right\} \\
&\quad + 2E \left\{ \| \hat{\boldsymbol{\omega}}_0 - \boldsymbol{\omega}^* \|^2 \cdot \sum_{m=1}^{M} \boldsymbol{y}'_U \boldsymbol{X}_{(m)U} \left(\boldsymbol{X}'_{(m)U} \boldsymbol{X}_{(m)U} \right)^{-1} \boldsymbol{X}'_{(m)U} \boldsymbol{y}_U \right\} \\
&\leq C f^{-1} M k_M + CM \left(E \| \boldsymbol{y}_U \|^4 \cdot E \| \hat{\boldsymbol{\omega}}_0 - \boldsymbol{\omega}^* \|^4 \right)^{1/2} \\
&= O \left(f^{-1} M k_M + MN \zeta_n^{1/2} \right),
\end{aligned}$$

where $\hat{\omega}_{0m}$ and ω_m^* denote the mth elements of $\hat{\boldsymbol{\omega}}_0$ and $\boldsymbol{\omega}^*$, respectively. Therefore,

$$\frac{1}{N} \sum_{i \in U} E \left\{ \hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*) \right\}^2 = O \left(n^{-1} M k_M + M \zeta_n^{1/2} \right).$$

This completes the proof of Lemma 12.

S3.9 Proof of Theorem 5

The proof of this theorem relies on Lemmas 1–2 and 8–11.

Proof. Note that

$$\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y = \left(\hat{\bar{t}}_y(\boldsymbol{\omega}^*) - \bar{t}_y \right) + \left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \hat{\bar{t}}_y(\boldsymbol{\omega}^*) \right)$$

$$
\begin{aligned}
&= \frac{1}{N}\left\{\sum_{i\in s}\frac{y_i-\hat{\mu}_i(\boldsymbol{\omega}^*)}{\pi_i}+\sum_{i\in U}\hat{\mu}_i(\boldsymbol{\omega}^*)\right\}-\bar{t}_y \\
&\quad+\frac{1}{N}\sum_{m=1}^{M}\left\{(\hat{\omega}_{0m}-\omega_m^*)\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i}-\sum_{i\in U}\hat{\mu}_{i(m)}\right)\right\} \\
&= \frac{1}{N}\sum_{i\in U}\frac{(I_i-\pi_i)\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}}{\pi_i} \\
&\quad+\frac{1}{N}\sum_{i\in U}\frac{(\pi_i-I_i)\left\{\hat{\mu}_i(\boldsymbol{\omega}^*)-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}}{\pi_i} \\
&\quad+\frac{1}{N}\sum_{m=1}^{M}\left\{(\hat{\omega}_{0m}-\omega_m^*)\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i}-\sum_{i\in U}\hat{\mu}_{i(m)}\right)\right\} \\
&\triangleq a_N+b_N+c_N. \qquad \text{(S.63)}
\end{aligned}
$$

By Condition 1, Lemma 1, Loève's c_r inequality, and the Cauchy–Schwarz inequality,

$$
\begin{aligned}
\sum_{i\in U}E_\epsilon\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 &= \sum_{i\in U}E_\epsilon\left\{y_i-\mu_i^*(\boldsymbol{\omega}^*)+\mu_i^*(\boldsymbol{\omega}^*)-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 \\
&\le 2\sum_{i\in U}E_\epsilon\left(\sum_{m=1}^{M}\omega_m^*e_{i(m)}^*\right)^2+2\sum_{i\in U}E_\epsilon\left\{\sum_{m=1}^{M}\omega_m^*\boldsymbol{x}_{i(m)}'\left(\boldsymbol{\beta}_{(m)}^*-\tilde{\boldsymbol{\beta}}_{(m)}\right)\right\}^2 \\
&\le 2\sum_{i\in U}\sum_{m=1}^{M}\sum_{l=1}^{M}\omega_m^*\omega_l^*E_\epsilon\left(e_{i(m)}^*e_{i(l)}^*\right) \\
&\quad+2\sum_{m=1}^{M}\sum_{l=1}^{M}\omega_m^*\omega_l^*E\left|\left(\boldsymbol{\beta}_{(m)}^*-\tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(l)U}\left(\boldsymbol{\beta}_{(m)}^*-\tilde{\boldsymbol{\beta}}_{(m)}\right)\right| \\
&\le 2\sum_{i\in U}\left[\sum_{m=1}^{M}\omega_m^*\left\{E_\epsilon\left(e_{i(m)}^{*2}\right)\right\}^{1/2}\right]^2 \\
&\quad+2\left\{\sum_{m=1}^{M}\omega_m^*\cdot\lambda_{\max}^{1/2}\left(\boldsymbol{X}_{(m)U}'\boldsymbol{X}_{(m)U}\right)\cdot\left(E\left\|\boldsymbol{\beta}_{(m)}^*-\tilde{\boldsymbol{\beta}}_{(m)}\right\|^2\right)^{1/2}\right\}^2 \\
&= O(N),
\end{aligned}
$$

where $\mu_i^*(\boldsymbol{\omega}^*)=\sum_{m=1}^{M}\omega_m^*\boldsymbol{x}_{i(m)}'\boldsymbol{\beta}_{(m)}^*$. Then, by Lemma 2, we have

$$
\begin{aligned}
E(a_N^2) &= E\left[\frac{1}{N}\sum_{i\in U}\frac{(I_i-\pi_i)\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}}{\pi_i}\right]^2 \\
&= \frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}E_\epsilon\left[\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right] \\
&= O\left(n^{-1}\right). \qquad \text{(S.64)}
\end{aligned}
$$

For b_N, Lemma 11 and Condition 11 imply that

$$E(b_N^2) = O\left(n^{-2}k_M^2\right) = o\left(n^{-1}\right). \tag{S.65}$$

Combining (S.63), (S.64), and (S.65), we obtain that

$$\begin{aligned}\mathrm{MSE}\left(\hat{\bar{t}}_y(\boldsymbol{\omega}^*)\right) &= E(a_N + b_N)^2 \\ &= \frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}E_\epsilon\left[\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right] + O\left(\frac{k_M}{n^{3/2}}\right). \end{aligned} \tag{S.66}$$

Next, we consider c_N. By Conditions 1 and 10,

$$\begin{aligned}\sum_{i\in U}E_\epsilon\left(\mu_{i(m)}^{*4}\right) &= \sum_{i\in U}E_\epsilon\left(\mu_i + e_i - e_{i(m)}^*\right)^4 \\ &\le 27\sum_{i\in U}\mu_i^4 + 27\sum_{i\in U}E_\epsilon\left(e_i^4\right) + 27\sum_{i\in U}E_\epsilon\left(e_{i(m)}^{*4}\right) \\ &= O(N).\end{aligned}$$

Then, Lemmas 8–9 and Loève's c_r inequality yield

$$\begin{aligned}&E\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i} - \sum_{i\in U}\hat{\mu}_{i(m)}\right)^4 \\ &= E\left\{\sum_{i\in U}\frac{(I_i-\pi_i)\left(\hat{\mu}_{i(m)} - \mu_{i(m)}^*\right)}{\pi_i} + \sum_{i\in U}\frac{(I_i-\pi_i)\mu_{i(m)}^*}{\pi_i}\right\}^4 \\ &\le CN^3\sum_{i\in U}E\left\{\boldsymbol{x}_{i(m)}'\left(\hat{\boldsymbol{\beta}}_{(m)} - \boldsymbol{\beta}_{(m)}^*\right)\right\}^4 + Cn^{-2}N^3\sum_{i=1}^N E_\epsilon\left(\mu_{i(m)}^{*4}\right) \\ &\le Cn^{-2}N^4k_M^4.\end{aligned}$$

Hence, using the Cauchy–Schwarz inequality and Condition 11, we have

$$\begin{aligned}E\left(c_N^2\right) &= E\left\{\frac{1}{N}\sum_{m=1}^M(\hat{\omega}_{0m} - \omega_m^*)\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i} - \sum_{i\in U}\hat{\mu}_{i(m)}\right)\right\}^2 \\ &\le \frac{1}{N^2}E\left\{\sum_{m=1}^M(\hat{\omega}_{0m} - \omega_m^*)^2\sum_{m=1}^M\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i} - \sum_{i\in U}\hat{\mu}_{i(m)}\right)^2\right\} \\ &\le \frac{1}{N^2}\left[E\left\{\sum_{m=1}^M(\hat{\omega}_{0m} - \omega_m^*)^2\right\}^2 E\left\{\sum_{m=1}^M\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i} - \sum_{i\in U}\hat{\mu}_{i(m)}\right)^2\right\}^2\right]^{1/2}\end{aligned}$$

$$
\begin{aligned}
&\leq \frac{1}{N^2}\left[E\left\|\hat{\boldsymbol{\omega}}_0-\boldsymbol{\omega}^*\right\|^4\cdot M\left\{\sum_{m=1}^{M}E\left(\sum_{i\in s}\frac{\hat{\mu}_{i(m)}}{\pi_i}-\sum_{i\in U}\hat{\mu}_{i(m)}\right)^4\right\}\right]^{1/2}\\
&=O\left(n^{-1}\zeta_n^{1/2}Mk_M^2\right)\\
&=o\left(n^{-1}\right),
\end{aligned}
\tag{S.67}
$$

where $\hat{\omega}_{0m}$ and ω_m^* are the mth elements of $\hat{\boldsymbol{\omega}}_0$ and $\boldsymbol{\omega}^*$, respectively. Combining (S.63), (S.66), (S.67) and Condition 11, we obtain that

$$
\begin{aligned}
\mathrm{MSE}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)&=\frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}E_\epsilon\left[\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right]\\
&\quad+O\left(\frac{1}{n}\sqrt{\zeta_n^{1/2}Mk_M^2\left(1+\frac{k_M}{\sqrt{n}}\right)}\right)\\
&=\frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}E_\epsilon\left[\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right]+o\left(n^{-1}\right).
\end{aligned}
$$

This completes the proof of Theorem 5.

S3.10 Proof of Theorem 6

The proof of this theorem relies on Lemmas 1–3 and 8–12.

Proof. By Theorem 5, it suffices to show that

$$
\lim_{n\to\infty}nE\left|\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)-\mathrm{AMSE}\left(\hat{\bar{t}}_y(\boldsymbol{\omega}^*)\right)\right|=o\left(1\right),
$$

where

$$
\mathrm{AMSE}\left(\hat{\bar{t}}_y(\boldsymbol{\omega}^*)\right)=\frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}E_\epsilon\left[\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right].
$$

Let

$$
A_N=E\left|\frac{n}{N^2}\sum_{i,j\in U}\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}\frac{I_iI_j-\pi_{ij}}{\pi_{ij}}\right|
$$

and

$$
B_N=E\left|\frac{n}{N^2}\sum_{i,j\in U}\left[2\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\tilde{\mu}_j(\boldsymbol{\omega}^*)-\hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right\}\right.
$$

$$+ \left\{\tilde{\mu}_i(\boldsymbol{\omega}^*) - \hat{\mu}_i(\hat{\boldsymbol{\omega}}_0)\right\}\left\{\tilde{\mu}_j(\boldsymbol{\omega}^*) - \hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right\}\Big] \frac{\pi_{ij} - \pi_i\pi_j}{\pi_i\pi_j}\frac{I_iI_j}{\pi_{ij}}\Bigg|.$$

Then

$$nE\left|\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) - \mathrm{AMSE}\left(\hat{\bar{t}}_y(\boldsymbol{\omega}^*)\right)\right| \leq A_N + B_N.$$

Hence, the proof is completed once we establish that $A_N = o(1)$ and $B_N = o(1)$.

First, we prove that $A_N = o(1)$. Note that

$$\begin{aligned}
&E\left[\frac{n}{N^2}\sum_{i,j\in U}\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}\frac{I_iI_j - \pi_{ij}}{\pi_{ij}}\right]^2 \\
&= n^2E_\epsilon\left[\sum_{i\neq j\in U}\sum_{k\neq l\in U}\frac{(\pi_{ij}-\pi_i\pi_j)(\pi_{kl}-\pi_k\pi_l)}{\pi_i\pi_j\pi_k\pi_l}\frac{E_d\left\{(I_iI_j-\pi_{ij})(I_kI_l-\pi_{kl})\right\}}{\pi_{ij}\pi_{kl}}\right. \\
&\quad\left.\times\frac{\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\left\{y_k - \tilde{\mu}_k(\boldsymbol{\omega}^*)\right\}\left\{y_l - \tilde{\mu}_l(\boldsymbol{\omega}^*)\right\}}{N^4}\right] \\
&\quad + n^2E_\epsilon\left[\sum_{i\in U}\sum_{k\in U}\frac{(1-\pi_i)(1-\pi_k)}{\pi_i\pi_k}\frac{E_d\left\{(I_i-\pi_i)(I_k-\pi_k)\right\}}{\pi_i\pi_k}\right. \\
&\quad\left.\times\frac{\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2\left\{y_k - \tilde{\mu}_k(\boldsymbol{\omega}^*)\right\}^2}{N^4}\right] \\
&\quad + 2n^2E_\epsilon\left[\sum_{i\in U}\sum_{k\neq l\in U}\frac{(1-\pi_i)(\pi_{kl}-\pi_k\pi_l)}{\pi_i\pi_k\pi_l}\frac{E_d\left\{(I_i-\pi_i)(I_kI_l-\pi_{kl})\right\}}{\pi_i\pi_{kl}}\right. \\
&\quad\left.\times\frac{\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2\left\{y_k - \tilde{\mu}_k(\boldsymbol{\omega}^*)\right\}\left\{y_l - \tilde{\mu}_l(\boldsymbol{\omega}^*)\right\}}{N^4}\right] \\
&\triangleq A_{1N} + A_{2N} + A_{3N}.
\end{aligned}$$

By Lemma 1 and Conditions 1 and 11, we have

$$\begin{aligned}
E_\epsilon\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^4 &= E_\epsilon\left\{y_i - \mu_i^*(\boldsymbol{\omega}^*) + \mu_i^*(\boldsymbol{\omega}^*) - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^4 \\
&\leq CE_\epsilon\left(\sum_{m=1}^M \omega_m^* e_{i(m)}^*\right)^4 + CE_\epsilon\left[\sum_{m=1}^M \omega_m^*\left\{\boldsymbol{x}_{i(m)}'\left(\boldsymbol{\beta}_{(m)}^* - \tilde{\boldsymbol{\beta}}_{(m)}\right)\right\}\right]^4 \\
&\leq C\left[\sum_{m=1}^M \omega_m^*\left\{E_\epsilon\left(e_{i(m)}^{*4}\right)\right\}^{1/4}\right]^4 \\
&\quad + C\left[\sum_{m=1}^M \omega_m^*\left\{E_\epsilon\left(\boldsymbol{x}_{i(m)}'\boldsymbol{\beta}_{(m)}^* - \boldsymbol{x}_{i(m)}'\tilde{\boldsymbol{\beta}}_{(m)}\right)^4\right\}^{1/4}\right]^4 \\
&= O(1) \qquad \text{(S.68)}
\end{aligned}$$

uniformly in i. Using Conditions 2 and 9, we obtain that

$$
\begin{aligned}
|A_{1N}| &= n^2 E_\epsilon \left[\sum_{i\neq j\in U} \sum_{k\neq l\in U} \frac{(\pi_{ij}-\pi_i\pi_j)(\pi_{kl}-\pi_k\pi_l)}{\pi_i\pi_j\pi_k\pi_l} \frac{E_d\left\{(I_iI_j-\pi_{ij})(I_kI_l-\pi_{kl})\right\}}{\pi_{ij}\pi_{kl}} \right. \\
&\quad \left. \times \frac{\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\left\{y_k-\tilde{\mu}_k(\boldsymbol{\omega}^*)\right\}\left\{y_l-\tilde{\mu}_l(\boldsymbol{\omega}^*)\right\}}{N^4} \right] \\
&\leq C\left(n\max_{i\neq j\in U}|\pi_{ij}-\pi_i\pi_j|\right)^2 E_\epsilon \left[\sum_{i\neq j\in U} \sum_{k\neq l\in U} \left|E_d\left\{(I_iI_j-\pi_{ij})(I_kI_l-\pi_{kl})\right\}\right| \right. \\
&\quad \left. \times \frac{\left|\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\left\{y_k-\tilde{\mu}_k(\boldsymbol{\omega}^*)\right\}\left\{y_l-\tilde{\mu}_l(\boldsymbol{\omega}^*)\right\}\right|}{N^4} \right] \\
&\leq O\left(\frac{1}{N}\right) + \frac{C}{N}\sum_{i\in U} E_\epsilon\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\}^4 \times \max_{i,j,k,l\in \boldsymbol{D}_4} \left|E_d\left\{(I_iI_j-\pi_{ij})(I_kI_l-\pi_{kl})\right\}\right| \\
&= o(1).
\end{aligned}
$$

For A_{2N}, Conditions 2 and (S.68) imply that

$$
\begin{aligned}
|A_{2N}| &\leq \frac{Cn^2}{N^4}\sum_{i\in U} E_\epsilon\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^4 + \frac{Cn^2}{N}\sum_{i\in U} E_\epsilon\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^4 \times \max_{i\neq k\in U}|\pi_{ik}-\pi_i\pi_k| \\
&= o(1).
\end{aligned}
$$

Similarly, combining Conditions 2 and 9 with (S.68), we obtain $|A_{3N}| = o(1)$. Consequently, $A_N = o(1)$.

Next, we prove that $B_N = o(1)$. By (S.68), Lemma 12, and Condition 11, we have

$$
\begin{aligned}
B_N &= E\left| \frac{n}{N^2}\sum_{i,j\in U} \left[2\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\tilde{\mu}_j(\boldsymbol{\omega}^*)-\hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right\} \right.\right. \\
&\quad \left.\left. + \left\{\tilde{\mu}_i(\boldsymbol{\omega}^*)-\hat{\mu}_i(\hat{\boldsymbol{\omega}}_0)\right\}\left\{\tilde{\mu}_j(\boldsymbol{\omega}^*)-\hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right\}\right] \frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}\frac{I_iI_j}{\pi_{ij}} \right| \\
&\leq \frac{C}{N}\sum_{j\in U} E\left|\tilde{\mu}_j(\boldsymbol{\omega}^*)-\hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right| \times \left(f + n\max_{i\neq j\in U}|\pi_{ij}-\pi_i\pi_j|\right) \\
&\quad + \frac{C}{N}\sum_{j\in U} E\left\{\tilde{\mu}_j(\boldsymbol{\omega}^*)-\hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right\}^2 \times \left(f + n\max_{i\neq j\in U}|\pi_{ij}-\pi_i\pi_j|\right) \\
&= O\left(n^{-1}Mk_M + M\zeta_n^{1/2}\right) \\
&= o(1).
\end{aligned}
$$

This completes the proof of Theorem 6.

S3.11 Proof of Theorem 7

The proof of this theorem relies on Lemmas 1–3 and 8–12.

Proof. It follows from (S.63) that

$$\begin{aligned}
\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y &= a_N + b_N + c_N \\
&= \frac{1}{N}\sum_{i\in U}\frac{(I_i-\pi_i)\left\{y_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}}{\pi_i} + b_N + c_N \\
&= \frac{1}{N}\sum_{i\in U}\frac{(I_i-\pi_i)\left\{y_i-\mu_i+\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}}{\pi_i} + b_N + c_N \\
&= \frac{1}{N}\left(\sum_{i\in s}\frac{y_i-\mu_i}{\pi_i}+\sum_{i\in U}\mu_i\right) - \bar{t}_y \\
&\quad + \frac{1}{N}\left[\sum_{i\in s}\frac{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)}{\pi_i} - \sum_{i\in U}\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\right] + b_N + c_N \\
&= \left(\hat{\bar{t}}_y - \bar{t}_y\right) + a_{2N} + b_N + c_N. \qquad \text{(S.69)}
\end{aligned}$$

Using Conditions 2 and 11, we obtain that

$$\begin{aligned}
E(a_{2N}^2) &= \left|\frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}E_\epsilon\left[\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\mu_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right]\right| \\
&\le E_\epsilon\left|\frac{1}{N^2}\sum_{i\in U}\frac{1-\pi_i}{\pi_i}\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2\right| \\
&\quad + E_\epsilon\left|\frac{1}{N^2}\sum_{i\neq j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}\left[\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\mu_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right]\right| \\
&\le \frac{C}{N^2}E_\epsilon\left[\sum_{i\in U}\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2\right] + \frac{C}{nN^2}E_\epsilon\left|\sum_{i\neq j\in U}\left[\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\mu_j-\tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right]\right| \\
&\le \frac{C}{N^2}\sum_{i\in U}E_\epsilon\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 + \frac{C}{nN}\sum_{i\in U}E_\epsilon\left\{\mu_i-\tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 \\
&= O\left(\frac{\tilde{R}_N(\boldsymbol{\omega}^*)}{nN}\right) \\
&= o\left(n^{-1}\right).
\end{aligned}$$

Lemma 11 and Condition 11 imply that $E(b_N^2) = o\left(n^{-1}\right)$. By (S.67), $E(c_N^2) = o\left(n^{-1}\right)$. Hence, combining (S.69), Markov's inequality, and the Cauchy–Schwarz inequality, we have

$$\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y = \left(\hat{\bar{t}}_y - \bar{t}_y\right) + o_p\left(n^{-1/2}\right) \qquad \text{(S.70)}$$

and

$$\begin{aligned}\mathrm{MSE}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) &= E\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y\right)^2 \\ &= E\left(\hat{\bar{t}}_y - \bar{t}_y\right)^2 + o\left(n^{-1}\right) \\ &= \mathrm{MSE}(\hat{\bar{t}}_y) + o\left(n^{-1}\right). \end{aligned} \tag{S.71}$$

It follows from (S.70) that

$$\begin{aligned}&\frac{\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y}{\sqrt{\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)}} \\ &= \frac{\hat{\bar{t}}_y - \bar{t}_y}{\sqrt{\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)}} + o_p\left(\frac{1}{\sqrt{n\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)}}\right) \\ &= \frac{\hat{\bar{t}}_y - \bar{t}_y}{\sqrt{\mathrm{MSE}(\hat{\bar{t}}_y)}}\sqrt{\frac{\mathrm{MSE}(\hat{\bar{t}}_y)}{\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)}} + o_p\left(\frac{1}{\sqrt{n\mathrm{MSE}(\hat{\bar{t}}_y)}}\sqrt{\frac{\mathrm{MSE}(\hat{\bar{t}}_y)}{\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)}}\right). \end{aligned} \tag{S.72}$$

To complete the proof of this theorem, since $n\mathrm{MSE}(\hat{\bar{t}}_y) > c_9$, it suffices to show that

$$\left|\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) - \mathrm{MSE}(\hat{\bar{t}}_y)\right| = o_p(n^{-1}).$$

By Theorem 6 and Markov's inequality, we have

$$\left|\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) - \mathrm{MSE}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)\right| = o_p(n^{-1}).$$

Together with (S.71), it follows that

$$\left|\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) - \mathrm{MSE}(\hat{\bar{t}}_y)\right| = o_p(n^{-1}).$$

This completes the proof of Theorem 7.

S3.12 Lemmas for the proof of Theorem 8

Note that the following theoretical results are established with respect to the sampling design randomness only.

Lemma 13 *If Conditions 2 and 9 hold, then*

$$E_d\left(\frac{1}{N}\sum_{i\in s}\frac{z_i}{\pi_i}-\frac{1}{N}\sum_{i\in U}z_i\right)^2 \le Cn^{-1}\cdot\frac{1}{N}\sum_{i\in U}z_i^2$$

and

$$E_d\left(\frac{1}{N}\sum_{i\in s}\frac{z_i}{\pi_i}-\frac{1}{N}\sum_{i\in U}z_i\right)^4 \le Cn^{-2}\cdot\frac{1}{N}\sum_{i\in U}z_i^4.$$

Proof. By Condition 2 and the Cauchy–Schwarz inequality, we have

$$\begin{aligned}
E_d\left(\frac{1}{N}\sum_{i\in s}\frac{z_i}{\pi_i}-\frac{1}{N}\sum_{i\in U}z_i\right)^2 &= E_d\left\{\frac{1}{N}\sum_{i\in U}\frac{z_i(I_i-\pi_i)}{\pi_i}\right\}^2 \\
&= \frac{1}{N^2}\left\{\sum_{i\in U}(1-\pi_i)z_i^2+\sum_{i\neq j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}z_iz_j\right\} \\
&\le \frac{C}{N^2}\sum_{i\in U}z_i^2+\frac{C}{nN^2}\left(\sum_{i\in U}z_i\right)^2 \\
&\le Cn^{-1}\cdot\frac{1}{N}\sum_{i\in U}z_i^2.
\end{aligned}$$

Furthermore, Conditions 2 and 9, together with Loève's c_r inequality and the Cauchy–Schwarz inequality, imply that

$$\begin{aligned}
&E_d\left(\frac{1}{N}\sum_{i\in s}\frac{z_i}{\pi_i}-\frac{1}{N}\sum_{i\in U}z_i\right)^4 \\
&\le \frac{C}{N^4}\Bigg[\sum_{i,j,k,l\in\boldsymbol{D}_4}z_iz_jz_kz_lE_d\left\{(I_i-\pi_i)(I_j-\pi_j)(I_k-\pi_k)(I_l-\pi_l)\right\} \\
&\quad+\sum_{i,j,k\in\boldsymbol{D}_3}z_i^2z_jz_kE_d\left\{(I_i-\pi_i)^2(I_j-\pi_j)(I_k-\pi_k)\right\} \\
&\quad+\sum_{i\neq j\in U}z_i^2z_j^2E_d\left\{(I_i-\pi_i)^2(I_j-\pi_j)^2+(I_i-\pi_i)^3(I_j-\pi_j)\right\} \\
&\quad+\sum_{i\in U}z_i^4E_d\left\{(I_i-\pi_i)^4\right\}\Bigg] \\
&\le Cn^{-2}\cdot\frac{1}{N}\sum_{i\in U}z_i^4.
\end{aligned}$$

This completes the proof of Lemma 13.

Lemma 14 *Using a Taylor approximation, we have*

$$\hat{\boldsymbol{\beta}}_{(m)}=\tilde{\boldsymbol{\beta}}_{(m)}+\bar{\boldsymbol{T}}_{(m)}^{-1}\left(\hat{\bar{\boldsymbol{t}}}_{(m)}-\hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)+\boldsymbol{r}_{(m)},$$

where $\boldsymbol{r}_{(m)}$ *is the remainder term of* $\hat{\boldsymbol{\beta}}_{(m)}$*. If Conditions 1, 2, 9, and 13 hold, then*

$$E_d\left(r_{j(m)}^2\right) = O(n^{-2}),$$

where $r_{j(m)}$ *is the* j*th element of* $\boldsymbol{r}_{(m)}$*.*

Proof. Similar to the proof of Lemma 10, by applying Taylor linearization to $\hat{\boldsymbol{\beta}}_{(m)}$ at the point $(\bar{\boldsymbol{T}}_{(m)}, \bar{\boldsymbol{t}}_{(m)})$, we have

$$\hat{\boldsymbol{\beta}}_{(m)} = \tilde{\boldsymbol{\beta}}_{(m)} + \bar{\boldsymbol{T}}_{(m)}^{-1}\left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right) + \boldsymbol{r}_{(m)}.$$

Thus

$$r_{j(m)} = \boldsymbol{\lambda}_{j(m)}'\boldsymbol{r}_{(m)} = \boldsymbol{\lambda}_{j(m)}'\hat{\bar{\boldsymbol{T}}}_{(m)}^{-1}\hat{\bar{\boldsymbol{t}}}_{(m)} - \boldsymbol{\lambda}_{j(m)}'\tilde{\boldsymbol{\beta}}_{(m)} - \boldsymbol{\lambda}_{j(m)}'\bar{\boldsymbol{T}}_{(m)}^{-1}\left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right).$$

Note that the elements of $\hat{\bar{\boldsymbol{T}}}_{(m)}^{-1}$ and $\hat{\bar{\boldsymbol{t}}}_{(m)}$ are HT estimators, so $r_{j(m)}$ is a nonlinear function of the HT estimators. From Lemma 13 and Conditions 1, 2 and 13, we can see that the fourth-order moment of each HT estimator is $O(n^{-2})$. Additionally, $r_{j(m)}^2$ and its first three derivatives are equal to zero at the point $(\bar{\boldsymbol{T}}_{(m)}, \bar{\boldsymbol{t}}_{(m)})$. Therefore, using Theorem 5.4.3 of Fuller (1996), we conclude that

$$E_d\left(r_{j(m)}^2\right) = O(n^{-2}).$$

This completes the proof of Lemma 14.

Lemma 15 *If Conditions 1, 2, 9, and 13 hold, then*

$$E_d(b_N^2) = E_d\left[\frac{1}{N}\sum_{i\in U}\frac{I_i - \pi_i}{\pi_i}\left\{\tilde{\mu}_i(\boldsymbol{\omega}^*) - \hat{\mu}_i(\boldsymbol{\omega}^*)\right\}\right]^2 = O(n^{-2}k_M^2)$$

and

$$E_d\left[\frac{1}{N}\sum_{i\in U}\frac{I_i - \pi_i}{\pi_i}\left\{\tilde{\mu}_{i(m)} - \hat{\mu}_{i(m)}\right\}\right]^2 \leq Cn^{-2}k_m^2.$$

Proof. By Lemma 14, we have

$$\hat{\mu}_i(\boldsymbol{\omega}^*) - \tilde{\mu}_i(\boldsymbol{\omega}^*) = \sum_{m=1}^{M}\omega_m^*\boldsymbol{x}_{i(m)}'\left\{\bar{\boldsymbol{T}}_{(m)}^{-1}\left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right) + \boldsymbol{r}_{(m)}\right\}$$

$$
\begin{aligned}
&= \frac{1}{N}\sum_{j\in U}\frac{I_j-\pi_j}{\pi_j}\sum_{m=1}^{M}\left\{\omega_m^*\boldsymbol{x}'_{i(m)}\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{j(m)}\left(y_j-\boldsymbol{x}'_{j(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)\right\}\\
&\quad+\sum_{m=1}^{M}\left(\omega_m^*\boldsymbol{x}'_{i(m)}\boldsymbol{r}_{(m)}\right)\\
&\triangleq \frac{1}{N}\sum_{j\in U}\frac{I_j-\pi_j}{\pi_j}z_{ij}+r_i(\boldsymbol{\omega}^*).
\end{aligned}
$$

Thus

$$
\begin{aligned}
b_N^2 &= \frac{1}{N^4}\sum_{i,j,k,l\in U}\frac{(\pi_i-I_i)(\pi_j-I_j)(\pi_k-I_k)(\pi_l-I_l)z_{ik}z_{jl}}{\pi_i\pi_j\pi_k\pi_l}\\
&\quad+\frac{2}{N^3}\sum_{i,j,k\in U}\frac{(\pi_i-I_i)(\pi_j-I_j)(\pi_k-I_k)z_{ik}r_j(\boldsymbol{\omega}^*)}{\pi_i\pi_j\pi_k}\\
&\quad+\frac{1}{N^2}\sum_{i,j\in U}\frac{(\pi_i-I_i)(\pi_j-I_j)r_i(\boldsymbol{\omega}^*)r_j(\boldsymbol{\omega}^*)}{\pi_i\pi_j}\\
&\triangleq b_{1N}+b_{2N}+b_{3N}.
\end{aligned}
$$

We first consider $|E_d(b_{1N})|$. Conditions 1 and 13 imply that

$$
\begin{aligned}
|z_{ij}| &= \left|\sum_{m=1}^{M}\left\{\omega_m^*\boldsymbol{x}'_{i(m)}\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{j(m)}\left(y_j-\boldsymbol{x}'_{j(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)\right\}\right|\\
&\le \sum_{m=1}^{M}\left|\omega_m^*\boldsymbol{x}'_{i(m)}\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{j(m)}\left(y_j-\tilde{\mu}_{j(m)}\right)\right|\\
&\le \sum_{m=1}^{M}\omega_m^*\|\boldsymbol{x}'_{i(m)}\|\cdot\|\bar{\boldsymbol{T}}_{(m)}^{-1}\|\cdot\|x_{j(m)}\|\cdot\left|y_j-\tilde{\mu}_{j(m)}\right|\\
&= O(k_M)
\end{aligned}
$$

uniformly in i and j. Then, by Conditions 2 and 9, we have

$$
\begin{aligned}
|E_d(b_{1N})| &= \left|\frac{1}{N^4}\sum_{i,j,k,l\in U}z_{ik}z_{jl}E_d\left\{\frac{(\pi_i-I_i)(\pi_j-I_j)(\pi_k-I_k)(\pi_l-I_l)}{\pi_i\pi_j\pi_k\pi_l}\right\}\right|\\
&\le \frac{Ck_M^2}{N^4}\sum_{i,j,k,l\in U}\left|E_d\left\{(\pi_i-I_i)(\pi_j-I_j)(\pi_k-I_k)(\pi_l-I_l)\right\}\right|\\
&\le \frac{Ck_M^2}{N^4}\left[\sum_{i,j,k,l\in\boldsymbol{D}_4}\left|E_d\left\{(I_i-\pi_i)(I_j-\pi_j)(I_k-\pi_k)(I_l-\pi_l)\right\}\right|\right.\\
&\quad\left.+\sum_{i,j,k\in\boldsymbol{D}_3}\left|E_d\left\{(I_i-\pi_i)^2(I_j-\pi_j)(I_k-\pi_k)\right\}\right|\right.
\end{aligned}
$$

$$
\begin{aligned}
&+ \sum_{i \neq j \in U} \left| E_d \left\{ (I_i - \pi_i)^2 (I_j - \pi_j)^2 + (I_i - \pi_i)(I_j - \pi_j)^3 \right\} \right| \\
&+ \sum_{i \in U} \left| E_d \left\{ (I_i - \pi_i)^4 \right\} \right| \Bigg] \\
&= O\left(n^{-2} k_M^2 \right).
\end{aligned}
$$

Next, we consider $|E_d(b_{3N})|$. Condition 1 and Lemma 14 imply that

$$
\begin{aligned}
\sum_{i \in U} E_d \left\{ r_i^2(\boldsymbol{\omega}^*) \right\} &= E_d \left(\sum_{m=1}^{M} \sum_{l=1}^{M} \omega_m^* \omega_l^* \boldsymbol{r}'_{(m)} \boldsymbol{X}'_{(m)U} \boldsymbol{X}_{(l)U} \boldsymbol{r}_{(l)} \right) \\
&\leq C \sum_{m=1}^{M} \sum_{l=1}^{M} \omega_m^* \omega_l^* \left(E_d \left\| \boldsymbol{r}_{(m)} \right\|^2 E_d \left\| \boldsymbol{r}_{(l)} \right\|^2 \right)^{1/2} \\
&= O(N n^{-2} k_M^2). \qquad \text{(S.73)}
\end{aligned}
$$

Then, from Condition 2 and Loève's c_r inequality, we have

$$
\begin{aligned}
|E_d(b_{3N})| &= \left| \frac{1}{N^2} \sum_{i,j \in U} E_d \left\{ \frac{(\pi_i - I_i)(\pi_j - I_j) r_i(\boldsymbol{\omega}^*) r_j(\boldsymbol{\omega}^*)}{\pi_i \pi_j} \right\} \right| \\
&\leq E_d \left\{ \frac{1}{N} \sum_{i \in U} \frac{r_i(\boldsymbol{\omega}^*)(\pi_i - I_i)}{\pi_i} \right\}^2 \\
&\leq \frac{1}{N} \sum_{i \in U} E_d \left\{ r_i^2(\boldsymbol{\omega}^*) \right\} \\
&= O\left(n^{-2} k_M^2 \right).
\end{aligned}
$$

It remains to show that $|E_d(b_{2N})| = O(n^{-2} k_M^2)$. By the Cauchy–Schwarz inequality,

$$
\begin{aligned}
|E_d(b_{2N})| &= \left| E_d \left\{ \frac{2}{N^3} \sum_{i,j,k \in U} \frac{(\pi_i - I_i)(\pi_j - I_j)(\pi_k - I_k) z_{ik} r_j(\boldsymbol{\omega}^*)}{\pi_i \pi_j \pi_k} \right\} \right| \\
&= \frac{2}{N^3} \left| E_d \left\{ \sum_{i,k \in U} \frac{(\pi_i - I_i)(\pi_k - I_k) z_{ik}}{\pi_i \pi_k} \cdot \sum_{j \in U} \frac{(\pi_j - I_j) r_j(\boldsymbol{\omega}^*)}{\pi_j} \right\} \right| \\
&= \frac{2}{N^3} \left| E_d \left\{ \sum_{i,k \in U} \frac{(\pi_i - I_i)(\pi_k - I_k) z_{ik}}{\pi_i \pi_k} \right\}^2 \cdot E_d \left\{ \sum_{j \in U} \frac{(\pi_j - I_j) r_j(\boldsymbol{\omega}^*)}{\pi_j} \right\}^2 \right|^{1/2} \\
&= \frac{2}{N^3} \cdot \sqrt{N^4 |E_d(b_{1N})| \cdot N^2 |E_d(b_{3N})|} \\
&= O\left(n^{-2} k_M^2 \right).
\end{aligned}
$$

Therefore, $E_d(b_N^2) \leq |E_d(b_{1N})| + |E_d(b_{2N})| + |E_d(b_{3N})| = O\left(n^{-2}k_M^2\right)$.

Note that $\tilde{\mu}_{i(m)} = \tilde{\mu}_i(\boldsymbol{\omega}_m^0)$, where $\boldsymbol{\omega}_m^0$ is an $M \times 1$ vector with one in the mth position and zeros elsewhere. Similar to the proof of $E_d(b_N^2) = O\left(n^{-2}k_M^2\right)$, we have

$$E_d\left\{\frac{1}{N}\sum_{i\in U}\frac{I_i - \pi_i}{\pi_i}\left(\tilde{\mu}_{i(m)} - \hat{\mu}_{i(m)}\right)\right\}^2 \leq Cn^{-2}k_m^2.$$

This completes the proof of Lemma 15.

S3.13 Proof of Theorem 8

The proof of this theorem relies on Lemmas 13–15.

Proof. By Equation (S.69), we have

$$\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y = a_N + b_N + c_N = \left(\hat{\bar{t}}_y - \bar{t}_y\right) + a_{2N} + b_N + c_N. \tag{S.74}$$

For a_N, Conditions 2 and 13 imply that

$$\begin{aligned} E_d(a_N^2) &= E_d\left[\frac{1}{N}\sum_{i\in U}\frac{(I_i - \pi_i)\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}}{\pi_i}\right]^2 \\ &= \frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij} - \pi_i\pi_j}{\pi_i\pi_j}\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\} \\ &= O\left(n^{-1}\right). \end{aligned} \tag{S.75}$$

For a_{2N}, it follows from Conditions 2 and 15 that

$$\begin{aligned} E_d(a_{2N}^2) &= E_d\left[\frac{1}{N}\sum_{i\in s}\frac{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)}{\pi_i} - \frac{1}{N}\sum_{i\in U}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\right]^2 \\ &= \left|\frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij} - \pi_i\pi_j}{\pi_i\pi_j}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\mu_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right| \\ &\leq \frac{C}{N^2}\sum_{i\in U}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 + \frac{C}{nN^2}\left|\sum_{i\neq j\in U}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{\mu_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\}\right| \\ &\leq \frac{C}{N^2}\sum_{i\in U}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 + \frac{C}{nN}\sum_{i\in U}\left\{\mu_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 \end{aligned}$$

$$= O\left(\frac{\tilde{L}_N(\boldsymbol{\omega}^*)}{nN}\right)$$
$$= o\left(n^{-1}\right). \tag{S.76}$$

Next, we consider c_N. Using Lemmas 13 and 15, Conditions 13 and 14, Loève's c_r inequality, and the Cauchy–Schwarz inequality, we have

$$\begin{aligned}
E_d(c_N^2) &= E_d\left\{\frac{1}{N}\sum_{m=1}^{M}(\hat{\omega}_{0m}-\omega_m^*)\left(\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\hat{\mu}_{i(m)}\right)\right\}^2 \\
&= E_d\left\{\sum_{m=1}^{M}(\hat{\omega}_{0m}-\omega_m^*)\cdot\frac{1}{N}\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\left(\hat{\mu}_{i(m)}-\tilde{\mu}_{i(m)}\right)\right. \\
&\quad \left.+\sum_{m=1}^{M}(\hat{\omega}_{0m}-\omega_m^*)\cdot\frac{1}{N}\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\tilde{\mu}_{i(m)}\right\}^2 \\
&\le 2M\sum_{m=1}^{M}\left[E_d\left\{\frac{1}{N}\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\left(\hat{\mu}_{i(m)}-\tilde{\mu}_{i(m)}\right)\right\}^2\right] \\
&\quad +2E_d\left\{\sum_{m=1}^{M}(\hat{\omega}_{0m}-\omega_m^*)^2\cdot\sum_{m=1}^{M}\left(\frac{1}{N}\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\tilde{\mu}_{i(m)}\right)^2\right\} \\
&\le 2M\sum_{m=1}^{M}\left[E_d\left\{\frac{1}{N}\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\left(\hat{\mu}_{i(m)}-\tilde{\mu}_{i(m)}\right)\right\}^2\right] \\
&\quad +C\left\{M\sum_{m=1}^{M}E_d\left(\frac{1}{N}\sum_{i\in U}\frac{I_i-\pi_i}{\pi_i}\tilde{\mu}_{i(m)}\right)^4\cdot E_d\|\hat{\boldsymbol{\omega}}_0-\boldsymbol{\omega}^*\|^4\right\}^{1/2} \\
&\le Cn^{-2}M^2k_M^2+Cn^{-1}M\tilde{\zeta}_n^{1/2} \\
&= o\left(n^{-1}\right).
\end{aligned} \tag{S.77}$$

Condition 14 and Lemma 15 imply that $E_d(b_N^2) = o\left(n^{-1}\right)$. Combining this result with (S.74), (S.76), and (S.77), we obtain that

$$\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)-\bar{t}_y = \left(\hat{\bar{t}}_y-\bar{t}_y\right)+o_{\tilde{p}}\left(n^{-1/2}\right)$$

and

$$\begin{aligned}
\mathrm{MSE}_d\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) &= E_d\left\{\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)-\bar{t}_y\right\}^2 \\
&= E_d\left(\hat{\bar{t}}_y-\bar{t}_y\right)^2+o\left(n^{-1}\right)
\end{aligned}$$

$$= \mathrm{MSE}_d(\hat{\bar{t}}_y) + o\left(n^{-1}\right). \tag{S.78}$$

Theorem 2 of Berger (1996), together with the conditions $D(P\|R) \to 0$, $e_R \to 0$, and $d_R \to \infty$, implies that

$$\frac{\sqrt{n}(\hat{\bar{t}}_y - \bar{t}_y)}{\sqrt{\mathrm{MSE}_d(\hat{\bar{t}}_y)}} \xrightarrow{\tilde{d}} N(0,1).$$

Therefore, to complete the proof of this theorem, from the condition $n\mathrm{MSE}_d(\hat{\bar{t}}_y) > c_{10}$ and Equations (S.72), (S.75) and (S.78), it suffices to show that

$$\left|\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) - \mathrm{MSE}_d\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)\right| = o_{\tilde{p}}\left(n^{-1}\right),$$

where

$$\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) = \frac{1}{N^2}\sum_{i,j\in s}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_{ij}\pi_i\pi_j}\left\{y_i - \hat{\mu}_i(\hat{\boldsymbol{\omega}}_0)\right\}\left\{y_j - \hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right\}$$

and

$$\mathrm{MSE}_d\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) = \frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}\left\{y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}\left\{y_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right\} + o\left(n^{-1}\right).$$

Similar to the proof of Theorem 6, we need to verify that $|y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)| = O(1)$ and

$$\frac{1}{N}\sum_{i\in U} E_d\left\{\hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 = o(1).$$

The first equation follows directly from Condition 13. Moreover, by Lemmas 13–14 and Conditions 1, 2 and 13, we have

$$\begin{aligned}
E_d\left\|\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right\|^2 &= E_d\left\|\bar{\boldsymbol{T}}_{(m)}^{-1}\left(\hat{\bar{\boldsymbol{t}}}_{(m)} - \hat{\bar{\boldsymbol{T}}}_{(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right) + \boldsymbol{r}_{(m)}\right\|^2 \\
&\le \frac{2}{N}\sum_{i,j\in U}\frac{|E_d(I_i-\pi_i)(I_j-\pi_j)|}{\pi_i\pi_j}\left|\left\{\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{i(m)}\left(y_i - \boldsymbol{x}_{i(m)}'\tilde{\boldsymbol{\beta}}_{(m)}\right)\right\}'\right. \\
&\quad \left.\left\{\bar{\boldsymbol{T}}_{(m)}^{-1}\boldsymbol{x}_{j(m)}\left(y_j - \boldsymbol{x}_{j(m)}'\tilde{\boldsymbol{\beta}}_{(m)}\right)\right\}\right| + 2E_d\left\|\boldsymbol{r}_{(m)}\right\|^2 \\
&= O(n^{-1}k_m).
\end{aligned}$$

Then, Conditions 2, 13 and 14 imply that

$$\frac{1}{N}\sum_{i\in U} E_d\left\{\hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2$$

$$
\begin{aligned}
&= \frac{1}{N}\sum_{i\in U} E_d\left\{\hat{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\hat{\boldsymbol{\omega}}_0) + \tilde{\mu}_i(\hat{\boldsymbol{\omega}}_0) - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right\}^2 \\
&\le \frac{2}{N}\sum_{i\in U} E_d\left\{\sum_{m=1}^{M}\hat{\omega}_{0m}^2\sum_{m=1}^{M}\left(\boldsymbol{x}'_{i(m)}\hat{\boldsymbol{\beta}}_{(m)} - \boldsymbol{x}'_{i(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)^2\right\} \\
&\quad + \frac{2}{N}\sum_{i\in U} E_d\left\{\sum_{m=1}^{M}(\hat{\omega}_{0m} - \omega_m^*)^2\sum_{m=1}^{M}\left(\boldsymbol{x}'_{i(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right)^2\right\} \\
&\le \frac{2}{N}E_d\left\{\sum_{m=1}^{M}\left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)'\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\left(\hat{\boldsymbol{\beta}}_{(m)} - \tilde{\boldsymbol{\beta}}_{(m)}\right)\right\} \\
&\quad + \frac{2}{N}E_d\left\{\|\hat{\boldsymbol{\omega}}_0 - \boldsymbol{\omega}^*\|^2\cdot\sum_{m=1}^{M}\boldsymbol{y}'_U\boldsymbol{X}_{(m)U}\left(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}\right)^{-1}\boldsymbol{X}'_{(m)U}\boldsymbol{y}_U\right\} \\
&\le Cn^{-1}Mk_M + CN^{-1}M\|\boldsymbol{y}_U\|^2\left(E_d\|\hat{\boldsymbol{\omega}}_0 - \boldsymbol{\omega}^*\|^4\right)^{1/2} \\
&= O\left(n^{-1}Mk_M + \tilde{\zeta}_n^{1/2}M\right) \\
&= o(1).
\end{aligned}
$$

This completes the proof of Theorem 8.

S4 Relationship with High-Dimensional Regression

In this section, we explain how the number and dimensions of the candidate models relate to the number of covariates and discuss the relationship between the proposed framework and high-dimensional regression in survey sampling.

The number of candidate models M and their dimensions $\{k_m\}$ depend on how the candidate models are constructed, so their relationship with the number of covariates p is not fixed. For example, under the nested candidate models, where the mth model contains the first m covariates (treating the intercept as a covariate), we have $p = M = k_M$. In this case, Conditions 3, 4, 6, and 11, which restrict the number and dimensions of candidate models, can be expressed in terms of the number of covariates by replacing both M and k_M with p. More generally, if the mth candidate model contains the first k_m covariates, then $p = k_M$, which may be much larger than M. If the sets of covariates included in

the candidate models are mutually disjoint as in Ando & Li (2014), then $p = \sum_{m=1}^{M} k_m$. When the candidate models correspond to all nonempty subsets of the p covariates, we have $p = \log_2(M+1)$. Consequently, we express our conditions in terms of the number and dimensions of candidate models in this paper.

Although our framework allows both M and k_M to diverge (so that p also diverges), their growth rates are constrained by Conditions 3, 4, 6, and 11. Detailed discussions of these conditions are provided in the main manuscript. Since the theoretical properties studied in this paper are different from those in the survey sampling literature on high-dimensional regression (Ta et al. 2020, Chauvet & Goga 2022), the divergence rate of p is also different. For instance, in the discussion of Condition 3, we require $p = O(n^v)$ with $0 < v < 3/28$ to establish the asymptotic optimality of the proposed model averaging estimators. By comparison, to establish asymptotic equivalence between estimators, Ta et al. (2020) require $p/n \to 0$, whereas Chauvet & Goga (2022) require $p = O(n^v)$ with $0 < v < 1/2$.

Moreover, when $p > n$, Ta et al. (2020) showed that, under some sparsity and regularity conditions, the generalized regression (GREG) estimator based on the ℓ_1-penalized weighted least squares estimator is asymptotically equivalent to the generalized difference estimator. In contrast, the proposed model averaging method may not be directly applicable without additional dimensionality-reduction techniques. A possible solution is to incorporate a preliminary feature screening or grouping step, as is commonly adopted in high-dimensional model averaging (Ando & Li 2014, Xie et al. 2021), before constructing the candidate models. This discussion has been added to the Conclusion section in the main manuscript.

S5 Additional Numerical Analyses

S5.1 Simulation results for $n = 50, 150, 200$

Table S1: Values of MSE_μ for various methods in the simulation ($n = 50$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.1708	0.2046	0.1115	0.1814	0.1561	0.1207	0.1057	**0.0864**	0.1457	*0.1039*	0.1035
	0.2	0.2540	0.2747	0.1627	0.3751	0.2176	0.1699	0.1815	**0.1455**	0.2038	*0.1519*	0.1516
	0.3	0.3188	0.3195	0.2113	0.5318	0.2687	0.2181	0.2529	*0.2059*	0.2614	**0.1980**	0.1978
	0.4	0.3859	0.3769	*0.2643*	0.6026	0.3266	0.2740	0.3342	0.2737	0.3216	**0.2502**	0.2501
	0.5	0.4257	0.4193	0.3303	0.6459	0.3753	*0.3168*	0.4021	0.3406	0.3721	**0.3004**	0.3005
	0.6	0.4933	0.4947	0.4164	0.6430	0.4563	*0.4013*	0.5001	0.4273	0.4588	**0.3786**	0.3787
	0.7	0.5798	0.5944	0.5521	0.7339	0.5719	*0.5143*	0.6501	0.5617	0.5755	**0.4909**	0.4911
	0.8	0.7641	0.7969	0.7963	0.9523	0.7685	*0.7170*	0.8907	0.7888	0.7817	**0.6989**	0.6991
	0.9	1.3004	1.3489	1.4902	1.6650	1.3354	*1.2906*	1.5287	1.4116	1.3555	**1.2823**	1.2829
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.1479	0.1966	0.0960	0.2570	0.1332	0.1020	*0.0763*	**0.0600**	0.1247	0.0862	0.0857
	0.2	0.1901	0.2171	0.1213	0.4521	0.1553	0.1185	0.1184	**0.0895**	0.1477	*0.1053*	0.1048
	0.3	0.2335	0.2404	0.1438	0.4625	0.1750	0.1376	0.1509	**0.1158**	0.1654	*0.1241*	0.1240
	0.4	0.2641	0.2664	0.1650	0.4133	0.2029	0.1585	0.1789	**0.1390**	0.1939	*0.1445*	0.1443
	0.5	0.2834	0.2762	0.1875	0.3619	0.2214	0.1780	0.2018	*0.1636*	0.2143	**0.1628**	0.1627
	0.6	0.2966	0.2911	0.2207	0.3674	0.2393	0.1909	0.2312	*0.1873*	0.2294	**0.1807**	0.1805
	0.7	0.3208	0.3192	0.2600	0.3598	0.2713	*0.2254*	0.2746	0.2279	0.2664	**0.2137**	0.2136
	0.8	0.3419	0.3448	0.3123	0.3907	0.3036	*0.2565*	0.3298	0.2733	0.3027	**0.2474**	0.2474
	0.9	0.3966	0.4123	0.4314	0.4973	0.3859	*0.3433*	0.4425	0.3799	0.3913	**0.3369**	0.3370
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.1793	0.2175	0.1125	0.1798	0.1658	0.1292	*0.1081*	**0.0895**	0.1573	0.1116	0.1109
	0.2	0.2578	0.2790	0.1627	0.3670	0.2220	0.1762	0.1835	**0.1482**	0.2150	*0.1567*	0.1561
	0.3	0.3184	0.3216	0.2081	0.5028	0.2681	0.2176	0.2519	*0.2040*	0.2619	**0.1973**	0.1971
	0.4	0.3801	0.3760	*0.2590*	0.5578	0.3257	0.2698	0.3255	0.2686	0.3181	**0.2473**	0.2471
	0.5	0.4328	0.4267	0.3292	0.5707	0.3811	*0.3234*	0.4116	0.3434	0.3777	**0.3029**	0.3028
	0.6	0.4916	0.4943	0.4168	0.6250	0.4622	*0.4058*	0.5128	0.4336	0.4616	**0.3834**	0.3834
	0.7	0.5769	0.5952	0.5445	0.6914	0.5680	*0.5134*	0.6468	0.5616	0.5732	**0.4945**	0.4946
	0.8	0.7662	0.7994	0.7914	0.9429	0.7823	*0.7261*	0.8958	0.7997	0.7910	**0.7098**	0.7103
	0.9	1.2979	1.3591	1.4641	1.6266	1.3338	*1.2908*	1.5332	1.4107	1.3648	**1.2818**	1.2821
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.1412	0.1877	0.0931	0.2521	0.1288	0.0983	*0.0754*	**0.0596**	0.1205	0.0844	0.0838
	0.2	0.1944	0.2237	0.1187	0.4139	0.1580	0.1211	0.1201	**0.0909**	0.1481	*0.1087*	0.1080
	0.3	0.2288	0.2410	0.1416	0.4110	0.1756	0.1385	0.1481	**0.1156**	0.1682	*0.1257*	0.1252
	0.4	0.2655	0.2632	0.1647	0.3779	0.2026	0.1592	0.1832	**0.1429**	0.1923	*0.1456*	0.1452
	0.5	0.2822	0.2754	0.1843	0.3325	0.2133	0.1742	0.2019	*0.1636*	0.2072	**0.1619**	0.1617
	0.6	0.3019	0.2991	0.2134	0.3324	0.2446	0.1974	0.2337	*0.1894*	0.2374	**0.1856**	0.1853
	0.7	0.3164	0.3161	0.2553	0.3408	0.2696	*0.2213*	0.2725	0.2233	0.2625	**0.2121**	0.2121
	0.8	0.3542	0.3522	0.2987	0.3674	0.3109	*0.2635*	0.3348	0.2766	0.3080	**0.2534**	0.2535
	0.9	0.4099	0.4175	0.4332	0.4885	0.3919	*0.3471*	0.4508	0.3862	0.3963	**0.3420**	0.3422

Table S2: Values of MSE_μ for various methods in the simulation ($n = 150$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.1002	0.1064	*0.0587*	0.1826	0.0743	0.0592	0.0705	0.0592	0.0732	**0.0544**	0.0543
	0.2	0.1378	0.1375	*0.0886*	0.3591	0.1078	0.0889	0.1129	0.0980	0.1066	**0.0818**	0.0817
	0.3	0.1537	0.1534	0.1136	0.3611	0.1314	*0.1118*	0.1515	0.1320	0.1304	**0.1046**	0.1046
	0.4	0.1706	0.1745	0.1425	0.3178	0.1578	*0.1373*	0.1877	0.1664	0.1576	**0.1285**	0.1286
	0.5	0.1956	0.2032	0.1810	0.3092	0.1914	*0.1714*	0.2331	0.2097	0.1914	**0.1616**	0.1617
	0.6	0.2246	0.2368	0.2279	0.3182	0.2313	*0.2118*	0.2917	0.2638	0.2320	**0.2037**	0.2038
	0.7	0.2779	0.2917	0.2975	0.3660	0.2875	*0.2722*	0.3688	0.3359	0.2908	**0.2656**	0.2658
	0.8	*0.3857*	0.4014	0.4309	0.4848	0.4003	0.3890	0.4965	0.4590	0.4027	**0.3835**	0.3837
	0.9	**0.7028**	0.7157	0.7917	0.8374	0.7180	0.7141	0.8289	0.7880	0.7204	*0.7141*	0.7144
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.0780	0.0897	0.0438	0.2568	0.0513	0.0408	0.0454	**0.0358**	0.0505	*0.0382*	0.0380
	0.2	0.0963	0.0985	0.0537	0.2992	0.0605	*0.0499*	0.0614	0.0502	0.0597	**0.0469**	0.0468
	0.3	0.1079	0.1078	0.0656	0.2204	0.0723	*0.0591*	0.0729	0.0605	0.0709	**0.0557**	0.0556
	0.4	0.1132	0.1129	0.0737	0.1834	0.0798	*0.0663*	0.0854	0.0717	0.0782	**0.0629**	0.0628
	0.5	0.1178	0.1180	0.0850	0.1575	0.0877	*0.0732*	0.0976	0.0833	0.0866	**0.0705**	0.0704
	0.6	0.1246	0.1251	0.0967	0.1537	0.0991	*0.0832*	0.1101	0.0942	0.0980	**0.0786**	0.0786
	0.7	0.1340	0.1352	0.1124	0.1529	0.1130	*0.0964*	0.1294	0.1120	0.1122	**0.0916**	0.0916
	0.8	0.1459	0.1504	0.1349	0.1620	0.1323	*0.1155*	0.1581	0.1387	0.1323	**0.1104**	0.1105
	0.9	0.1602	0.1687	0.1762	0.1988	0.1620	*0.1464*	0.2054	0.1818	0.1632	**0.1427**	0.1428
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.1036	0.1113	*0.0578*	0.1814	0.0774	0.0616	0.0711	0.0595	0.0756	**0.0562**	0.0560
	0.2	0.1387	0.1371	*0.0850*	0.3304	0.1066	0.0892	0.1149	0.0988	0.1055	**0.0815**	0.0814
	0.3	0.1560	0.1557	*0.1113*	0.3177	0.1323	0.1127	0.1491	0.1307	0.1310	**0.1049**	0.1049
	0.4	0.1759	0.1799	0.1424	0.2822	0.1625	*0.1417*	0.1899	0.1686	0.1618	**0.1316**	0.1316
	0.5	0.1930	0.2018	0.1772	0.2781	0.1891	*0.1695*	0.2316	0.2067	0.1894	**0.1607**	0.1608
	0.6	0.2300	0.2411	0.2258	0.2987	0.2343	*0.2158*	0.2886	0.2615	0.2357	**0.2058**	0.2059
	0.7	0.2782	0.2946	0.2968	0.3524	0.2896	*0.2750*	0.3743	0.3418	0.2921	**0.2680**	0.2682
	0.8	*0.3845*	0.3995	0.4256	0.4673	0.3998	0.3890	0.4958	0.4594	0.4025	**0.3827**	0.3830
	0.9	**0.7072**	0.7213	0.7842	0.8290	0.7228	0.7180	0.8313	0.7925	0.7255	*0.7176*	0.7179
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.0744	0.0872	0.0411	0.2478	0.0492	0.0396	0.0444	**0.0353**	0.0483	*0.0373*	0.0370
	0.2	0.0959	0.0970	0.0520	0.2505	0.0598	*0.0490*	0.0613	0.0504	0.0590	**0.0471**	0.0470
	0.3	0.1090	0.1086	0.0631	0.1810	0.0729	*0.0596*	0.0715	0.0601	0.0718	**0.0564**	0.0563
	0.4	0.1135	0.1136	0.0715	0.1492	0.0809	*0.0659*	0.0852	0.0708	0.0794	**0.0625**	0.0624
	0.5	0.1189	0.1188	0.0806	0.1388	0.0893	*0.0740*	0.0969	0.0822	0.0883	**0.0704**	0.0704
	0.6	0.1285	0.1276	0.0947	0.1352	0.1020	*0.0855*	0.1119	0.0961	0.1007	**0.0805**	0.0805
	0.7	0.1349	0.1356	0.1092	0.1399	0.1137	*0.0970*	0.1304	0.1125	0.1126	**0.0917**	0.0917
	0.8	0.1421	0.1455	0.1317	0.1564	0.1294	*0.1128*	0.1559	0.1347	0.1289	**0.1080**	0.1081
	0.9	0.1606	0.1692	0.1773	0.1982	0.1633	*0.1480*	0.2079	0.1843	0.1645	**0.1440**	0.1441

Table S3: Values of MSE_μ for various methods in the simulation ($n = 200$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.0904	0.0933	*0.0499*	0.1821	0.0635	0.0517	0.0646	0.0544	0.0630	**0.0469**	0.0468
	0.2	0.1158	0.1142	*0.0740*	0.3347	0.0897	0.0755	0.0998	0.0873	0.0887	**0.0688**	0.0688
	0.3	0.1284	0.1296	0.0971	0.2983	0.1113	*0.0956*	0.1302	0.1159	0.1103	**0.0889**	0.0890
	0.4	0.1426	0.1473	0.1213	0.2631	0.1331	*0.1177*	0.1638	0.1475	0.1330	**0.1101**	0.1102
	0.5	0.1622	0.1691	0.1546	0.2596	0.1621	*0.1468*	0.2027	0.1836	0.1625	**0.1376**	0.1377
	0.6	0.1864	0.1961	0.1936	0.2649	0.1923	*0.1782*	0.2515	0.2295	0.1937	**0.1720**	0.1721
	0.7	0.2336	0.2458	0.2521	0.3045	0.2439	*0.2331*	0.3192	0.2941	0.2459	**0.2271**	0.2272
	0.8	**0.3258**	0.3372	0.3650	0.4064	0.3374	0.3300	0.4306	0.4003	0.3398	*0.3272*	0.3274
	0.9	**0.6028**	0.6109	0.6692	0.7037	0.6120	*0.6106*	0.7048	0.6754	0.6141	0.6110	0.6112
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.0662	0.0753	0.0359	0.2524	0.0407	0.0323	0.0391	*0.0315*	0.0398	**0.0307**	0.0306
	0.2	0.0809	0.0830	0.0455	0.2514	0.0507	*0.0409*	0.0495	0.0421	0.0497	**0.0390**	0.0389
	0.3	0.0876	0.0879	0.0523	0.1810	0.0575	*0.0467*	0.0604	0.0507	0.0565	**0.0449**	0.0448
	0.4	0.0935	0.0940	0.0620	0.1465	0.0657	*0.0540*	0.0715	0.0612	0.0650	**0.0518**	0.0518
	0.5	0.0991	0.0989	0.0681	0.1248	0.0728	*0.0600*	0.0807	0.0695	0.0716	**0.0569**	0.0569
	0.6	0.1043	0.1035	0.0798	0.1208	0.0815	*0.0688*	0.0922	0.0801	0.0808	**0.0646**	0.0646
	0.7	0.1073	0.1078	0.0885	0.1183	0.0908	*0.0773*	0.1075	0.0936	0.0902	**0.0729**	0.0729
	0.8	0.1179	0.1209	0.1079	0.1288	0.1071	*0.0932*	0.1292	0.1141	0.1067	**0.0883**	0.0883
	0.9	0.1324	0.1389	0.1467	0.1636	0.1335	*0.1211*	0.1731	0.1558	0.1343	**0.1173**	0.1174
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.0902	0.0953	*0.0491*	0.1817	0.0643	0.0512	0.0647	0.0547	0.0631	**0.0474**	0.0472
	0.2	0.1148	0.1130	*0.0718*	0.2951	0.0881	0.0748	0.0969	0.0850	0.0876	**0.0687**	0.0687
	0.3	0.1298	0.1303	*0.0950*	0.2556	0.1127	0.0973	0.1313	0.1161	0.1122	**0.0895**	0.0895
	0.4	0.1443	0.1485	0.1210	0.2318	0.1350	*0.1197*	0.1655	0.1486	0.1349	**0.1119**	0.1119
	0.5	0.1604	0.1674	0.1502	0.2239	0.1599	*0.1449*	0.2015	0.1831	0.1604	**0.1368**	0.1369
	0.6	0.1907	0.2019	0.1937	0.2466	0.1958	*0.1819*	0.2543	0.2323	0.1972	**0.1749**	0.1751
	0.7	0.2340	0.2455	0.2526	0.2929	0.2457	*0.2337*	0.3184	0.2942	0.2472	**0.2283**	0.2285
	0.8	**0.3258**	0.3376	0.3622	0.3946	0.3376	0.3307	0.4303	0.4011	0.3398	*0.3274*	0.3276
	0.9	**0.6085**	0.6174	0.6679	0.7007	0.6177	0.6155	0.6990	0.6731	0.6192	*0.6148*	0.6150
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.0662	0.0752	0.0340	0.2398	0.0406	0.0323	0.0388	**0.0309**	0.0387	*0.0309*	0.0307
	0.2	0.0799	0.0814	0.0419	0.1987	0.0501	*0.0405*	0.0501	0.0421	0.0495	**0.0389**	0.0388
	0.3	0.0884	0.0889	0.0504	0.1401	0.0578	*0.0476*	0.0607	0.0509	0.0570	**0.0451**	0.0450
	0.4	0.0945	0.0946	0.0590	0.1187	0.0660	*0.0548*	0.0705	0.0608	0.0652	**0.0520**	0.0519
	0.5	0.0984	0.0982	0.0684	0.1091	0.0726	*0.0607*	0.0809	0.0699	0.0715	**0.0580**	0.0580
	0.6	0.1031	0.1030	0.0753	0.1068	0.0798	*0.0675*	0.0924	0.0801	0.0789	**0.0644**	0.0643
	0.7	0.1118	0.1126	0.0895	0.1137	0.0936	*0.0807*	0.1095	0.0958	0.0931	**0.0761**	0.0761
	0.8	0.1182	0.1207	0.1073	0.1255	0.1077	*0.0941*	0.1312	0.1154	0.1074	**0.0890**	0.0891
	0.9	0.1335	0.1398	0.1426	0.1580	0.1341	*0.1219*	0.1715	0.1555	0.1346	**0.1178**	0.1179

Table S4: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 50$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	2.3356	2.3532	2.1954	**2.1707**	2.2721	2.2149	2.2375	*2.1804*	2.2479	2.1916	2.1918
	0.2	2.3712	2.4489	2.2446	2.2381	2.3254	*2.2322*	2.3256	2.2324	2.3185	**2.2082**	2.2086
	0.3	2.6507	2.6019	2.4599	2.5657	2.5194	*2.4403*	2.5120	2.4579	2.5232	**2.4072**	2.4067
	0.4	2.6446	2.5707	**2.4153**	2.5575	2.5494	2.4620	2.5729	2.4771	2.5504	*2.4181*	2.4185
	0.5	2.5697	2.5846	2.3571	2.5118	2.4431	*2.3565*	2.5990	2.4810	2.4826	**2.3404**	2.3413
	0.6	2.9517	2.9397	**2.7370**	2.8836	2.8611	2.7769	2.9866	2.8411	2.8872	*2.7723*	2.7724
	0.7	2.7368	2.7214	2.6480	2.7258	2.7775	*2.6199*	2.8481	2.6749	2.7481	**2.5827**	2.5832
	0.8	3.3757	3.4217	3.3083	3.3530	3.3752	*3.2988*	3.5618	3.4139	3.4280	**3.2556**	3.2567
	0.9	*4.3581*	4.4543	4.5263	4.6566	4.3918	4.3608	4.8711	4.6240	4.4136	**4.3113**	4.3125
$\rho = 0.1$ $\alpha = 1.0$	0.1	2.2134	2.2984	2.0239	**1.9568**	2.1465	2.0855	2.0056	*1.9822*	2.1306	2.0488	2.0478
	0.2	2.3335	2.3693	2.1188	*2.0711*	2.2238	2.1730	2.1392	**2.0612**	2.2287	2.1326	2.1310
	0.3	2.2824	2.3281	2.1578	**2.1311**	2.2272	2.1832	2.2001	2.1492	2.1984	*2.1483*	2.1474
	0.4	2.3816	2.3967	**2.1401**	2.1907	2.2909	2.1957	2.2195	*2.1434*	2.2738	2.1867	2.1869
	0.5	2.5975	2.5862	2.3989	*2.3573*	2.4864	2.3776	2.4212	**2.3474**	2.4672	2.3720	2.3720
	0.6	2.6270	2.6171	2.3951	2.3932	2.4695	*2.3736*	2.4490	2.3742	2.4314	**2.3539**	2.3536
	0.7	2.5488	2.5713	2.4537	*2.4193*	2.5221	2.4205	2.5119	2.4453	2.5199	**2.4097**	2.4098
	0.8	2.7294	2.7407	2.5855	2.5847	2.6551	*2.5630*	2.7130	2.5733	2.6243	**2.5513**	2.5513
	0.9	2.9151	2.9411	2.8385	2.7997	2.8269	*2.7855*	3.0396	2.8711	2.8116	**2.7715**	2.7704
$\rho = 0.6$ $\alpha = 0.5$	0.1	2.3833	2.3918	2.1990	**2.1049**	2.3115	2.2465	2.1979	*2.1636*	2.2964	2.2212	2.2216
	0.2	2.5356	2.6353	2.3474	**2.2212**	2.5115	2.4172	2.4098	*2.3232*	2.4959	2.3649	2.3630
	0.3	2.4602	2.4875	*2.2483*	**2.2088**	2.4193	2.3217	2.3250	2.2493	2.4326	2.2573	2.2593
	0.4	2.5704	2.5375	**2.3145**	2.4267	2.4662	2.3542	2.5022	2.4197	2.4728	*2.3263*	2.3269
	0.5	2.5419	2.4899	*2.2293*	2.2915	2.3734	2.2504	2.3916	2.2918	2.3413	**2.2166**	2.2180
	0.6	2.7453	2.7203	2.5849	2.6989	2.6643	*2.5467*	2.8334	2.6538	2.6196	**2.5309**	2.5315
	0.7	3.0402	3.0819	3.0455	3.0805	3.1348	*2.9740*	3.3010	3.0651	3.0989	**2.9335**	2.9346
	0.8	3.4678	3.5087	*3.3416*	3.3809	3.5068	3.3614	3.5512	3.4113	3.4919	**3.2858**	3.2842
	0.9	4.7501	4.8761	4.8555	4.9126	4.8014	*4.7026*	5.2296	4.9295	4.9178	**4.6568**	4.6551
$\rho = 0.6$ $\alpha = 1.0$	0.1	2.2071	2.3165	2.0424	**1.8993**	2.1742	2.1066	2.0418	*2.0100*	2.1662	2.0754	2.0764
	0.2	2.3414	2.4485	2.1667	**2.0267**	2.2659	2.1953	2.2108	*2.1331*	2.2343	2.1774	2.1772
	0.3	2.3544	2.4114	2.1411	**2.0346**	2.2429	2.1917	2.1869	*2.1254*	2.2476	2.1499	2.1499
	0.4	2.4672	2.4497	*2.2347*	**2.1625**	2.3822	2.2834	2.3404	2.2548	2.3782	2.2442	2.2450
	0.5	2.6687	2.6389	*2.3174*	**2.2978**	2.4845	2.4151	2.4429	2.3610	2.4720	2.3951	2.3960
	0.6	2.4832	2.4521	*2.2534*	**2.2068**	2.3916	2.3146	2.3542	2.2960	2.3990	2.2801	2.2809
	0.7	2.6330	2.6138	2.4685	2.4551	2.5186	2.4423	2.5124	*2.4224*	2.5249	**2.4116**	2.4124
	0.8	2.7659	2.7328	2.5022	**2.4749**	2.6019	2.5266	2.6024	2.5160	2.5786	*2.4976*	2.4978
	0.9	2.9262	2.9745	2.8876	2.8790	2.8731	*2.8094*	3.0375	2.8878	2.8851	**2.7923**	2.7924

Table S5: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 150$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.6580	0.6529	*0.6340*	0.6364	0.6496	0.6356	0.6409	0.6348	0.6451	**0.6313**	0.6313
	0.2	0.7231	0.7234	**0.6964**	0.7374	0.7031	0.6991	0.7271	0.7139	0.7020	*0.6967*	0.6968
	0.3	0.7125	0.7188	0.7011	0.7435	0.7110	*0.6932*	0.7392	0.7244	0.7097	**0.6892**	0.6894
	0.4	0.6976	0.7029	0.6710	0.7204	0.6758	*0.6653*	0.7072	0.6928	0.6753	**0.6642**	0.6643
	0.5	0.7147	0.7189	0.7264	0.7637	0.7232	*0.7088*	0.7576	0.7384	0.7220	**0.7039**	0.7040
	0.6	0.7989	0.8035	0.8092	0.8297	0.8006	**0.7896**	0.8681	0.8448	0.7987	*0.7922*	0.7923
	0.7	0.7965	0.7995	0.7822	0.7934	0.7930	*0.7793*	0.8247	0.8080	0.7885	**0.7750**	0.7750
	0.8	**0.9080**	0.9245	0.9472	0.9606	0.9264	*0.9161*	1.0068	0.9911	0.9269	0.9168	0.9169
	0.9	**1.0788**	1.0909	1.0953	1.0929	1.0891	1.0834	1.1476	1.1283	1.0895	*1.0797*	1.0798
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.6726	0.6777	0.6589	0.6666	0.6691	**0.6586**	0.6719	0.6643	0.6678	*0.6587*	0.6587
	0.2	0.6506	0.6508	*0.6346*	0.6391	0.6389	0.6361	0.6479	0.6364	0.6389	**0.6303**	0.6303
	0.3	0.7113	0.7167	*0.6843*	0.6879	0.7004	0.6855	0.6974	0.6899	0.6986	**0.6842**	0.6843
	0.4	0.6774	0.6780	0.6625	0.6602	0.6570	*0.6505*	0.6609	0.6528	0.6574	**0.6490**	0.6488
	0.5	0.6772	0.6852	0.6637	0.6612	0.6683	*0.6597*	0.6815	0.6729	0.6688	**0.6565**	0.6563
	0.6	0.6530	0.6585	0.6506	0.6496	0.6530	*0.6450*	0.6659	0.6606	0.6524	**0.6427**	0.6427
	0.7	0.7218	0.7229	0.7063	0.7099	0.7164	**0.7023**	0.7314	0.7220	0.7210	*0.7032*	0.7033
	0.8	0.6701	0.6717	0.6531	0.6538	0.6655	*0.6473*	0.6771	0.6626	0.6582	**0.6428**	0.6427
	0.9	0.6989	0.7031	0.7154	0.7190	0.6975	**0.6935**	0.7411	0.7304	0.6989	*0.6963*	0.6964
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.6421	0.6586	0.6276	**0.6035**	0.6376	0.6312	0.6270	*0.6201*	0.6357	0.6258	0.6259
	0.2	0.7059	0.6916	*0.6658*	0.6726	0.6823	0.6666	0.6911	0.6821	0.6793	**0.6619**	0.6621
	0.3	0.6888	0.6859	0.6603	0.6685	0.6713	*0.6541*	0.6812	0.6670	0.6684	**0.6525**	0.6525
	0.4	0.7462	0.7445	*0.7014*	**0.6961**	0.7470	0.7275	0.7436	0.7331	0.7471	0.7126	0.7128
	0.5	0.7049	0.7197	0.6963	0.7082	0.7111	*0.6949*	0.7162	0.7139	0.7065	**0.6895**	0.6897
	0.6	0.8876	0.8975	*0.8713*	0.8757	0.8856	0.8731	0.9157	0.8920	0.8849	**0.8652**	0.8654
	0.7	0.8135	0.8334	**0.7981**	0.8015	0.8115	0.8091	0.8607	0.8448	0.8079	*0.7983*	0.7984
	0.8	**0.8700**	0.8750	0.8968	0.9056	0.8810	*0.8724*	0.9351	0.9199	0.8841	0.8732	0.8735
	0.9	**1.0798**	*1.0846*	1.1514	1.1652	1.0957	1.0954	1.1649	1.1537	1.0973	1.0984	1.0988
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.6852	0.6931	0.6665	**0.6269**	0.6768	0.6683	0.6661	*0.6632*	0.6766	0.6654	0.6657
	0.2	0.6956	0.6918	0.6501	**0.6267**	0.6596	0.6498	0.6570	*0.6477*	0.6599	0.6485	0.6487
	0.3	0.6885	0.6870	0.6654	**0.6521**	0.6707	0.6659	0.6714	0.6645	0.6709	*0.6599*	0.6600
	0.4	0.6911	0.6904	0.6452	**0.6317**	0.6555	0.6473	0.6573	0.6483	0.6544	*0.6449*	0.6450
	0.5	0.6793	0.6776	*0.6455*	**0.6312**	0.6666	0.6561	0.6784	0.6639	0.6652	0.6520	0.6519
	0.6	0.7625	0.7531	0.7345	**0.7272**	0.7510	0.7403	0.7566	0.7426	0.7494	*0.7322*	0.7325
	0.7	0.7361	0.7334	0.7129	**0.7083**	0.7150	0.7106	0.7280	0.7203	0.7143	*0.7094*	0.7096
	0.8	0.6568	0.6576	0.6306	0.6288	0.6392	0.6281	0.6363	**0.6207**	0.6330	*0.6222*	0.6223
	0.9	*0.7632*	0.7670	0.7720	0.7797	0.7656	**0.7577**	0.8235	0.8069	0.7695	0.7641	0.7642

Table S6: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 200$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.4872	0.4785	**0.4614**	0.4804	0.4681	0.4635	0.4779	0.4728	0.4689	*0.4630*	0.4631
	0.2	0.4712	0.4717	**0.4483**	0.4750	0.4652	0.4547	0.4692	0.4604	0.4635	*0.4523*	0.4523
	0.3	0.5461	0.5508	**0.5266**	0.5501	0.5376	0.5321	0.5588	0.5473	0.5354	*0.5317*	0.5317
	0.4	0.5394	0.5463	**0.5178**	0.5264	0.5309	0.5259	0.5424	0.5379	0.5321	*0.5235*	0.5236
	0.5	0.5358	0.5366	0.5393	0.5579	0.5389	*0.5321*	0.5607	0.5564	0.5382	**0.5291**	0.5292
	0.6	**0.5181**	0.5300	0.5218	0.5347	0.5274	0.5209	0.5638	0.5486	0.5310	*0.5198*	0.5199
	0.7	**0.5368**	0.5412	0.5552	0.5632	0.5429	*0.5391*	0.5797	0.5666	0.5414	0.5416	0.5417
	0.8	0.6038	0.6082	0.6255	0.6354	0.6034	**0.6014**	0.6440	0.6357	0.6070	*0.6029*	0.6030
	0.9	**0.6540**	*0.6567*	0.6936	0.6916	0.6576	0.6592	0.7089	0.7014	0.6603	0.6639	0.6641
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.5073	0.5152	0.4942	**0.4893**	0.5027	0.4993	0.4937	*0.4910*	0.5018	0.4945	0.4944
	0.2	0.4916	0.4930	0.4805	0.4835	0.4817	*0.4766*	0.4831	0.4816	0.4824	**0.4743**	0.4743
	0.3	0.4945	0.4945	0.4759	0.4815	0.4749	**0.4717**	0.4880	0.4822	*0.4742*	0.4743	0.4742
	0.4	0.5534	0.5539	0.5393	0.5458	0.5412	**0.5364**	0.5477	0.5414	0.5403	*0.5388*	0.5387
	0.5	0.4903	0.4894	0.4737	0.4719	0.4782	*0.4702*	0.4727	0.4712	0.4778	**0.4686**	0.4686
	0.6	0.4743	0.4755	0.4728	0.4706	0.4766	*0.4703*	0.4744	0.4725	0.4763	**0.4660**	0.4661
	0.7	0.5042	0.5072	0.4975	0.4977	0.5018	*0.4936*	0.5028	0.4966	0.5001	**0.4922**	0.4923
	0.8	0.4861	0.4873	0.4769	0.4783	0.4854	*0.4769*	0.4974	0.4913	0.4845	**0.4750**	0.4750
	0.9	0.5360	0.5420	0.5293	0.5304	0.5391	**0.5276**	0.5452	0.5379	0.5391	*0.5277*	0.5278
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.4632	0.4787	0.4598	**0.4556**	0.4620	0.4582	0.4698	0.4653	0.4614	*0.4567*	0.4568
	0.2	0.5130	0.5228	**0.4900**	0.4940	0.5018	0.4949	0.5032	0.4985	0.5011	*0.4927*	0.4927
	0.3	0.5491	0.5471	**0.5128**	0.5202	0.5218	0.5144	0.5325	0.5255	0.5190	*0.5144*	0.5144
	0.4	0.5456	0.5441	0.5374	0.5392	0.5393	**0.5338**	0.5620	0.5563	0.5380	*0.5344*	0.5345
	0.5	0.5087	0.5129	**0.4953**	0.4994	0.5069	0.5016	0.5191	0.5148	0.5062	*0.4972*	0.4972
	0.6	**0.5591**	0.5669	0.5752	0.5825	0.5742	*0.5650*	0.6081	0.5972	0.5749	0.5650	0.5652
	0.7	*0.5948*	0.6012	0.6132	0.6191	0.6024	**0.5942**	0.6363	0.6258	0.6023	0.5966	0.5968
	0.8	*0.5937*	0.5991	0.6071	0.6062	0.6020	0.5969	0.6391	0.6221	0.6050	**0.5910**	0.5910
	0.9	**0.7606**	0.7658	0.7724	0.7794	0.7681	0.7668	0.8026	0.7900	0.7695	*0.7633*	0.7632
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.4864	0.4969	0.4736	**0.4514**	0.4768	0.4707	0.4731	*0.4698*	0.4743	0.4732	0.4731
	0.2	0.4840	0.4849	0.4611	**0.4391**	0.4642	0.4616	0.4571	*0.4550*	0.4624	0.4600	0.4599
	0.3	0.4848	0.4853	0.4761	**0.4682**	0.4749	0.4721	0.4833	0.4789	0.4753	*0.4691*	0.4694
	0.4	0.5107	0.5112	0.4944	**0.4862**	0.4989	0.4951	0.5022	0.5002	0.5002	*0.4918*	0.4918
	0.5	0.5383	0.5404	0.5160	**0.5067**	0.5232	0.5138	0.5210	0.5175	0.5204	*0.5130*	0.5131
	0.6	0.5170	0.5196	0.5060	*0.5008*	0.5062	0.5020	0.5125	0.5072	0.5052	**0.4983**	0.4983
	0.7	0.5073	0.5087	**0.4874**	*0.4890*	0.5033	0.4965	0.5103	0.5019	0.5037	0.4924	0.4925
	0.8	0.5340	0.5365	0.5327	0.5337	0.5350	*0.5322*	0.5539	0.5457	0.5383	**0.5292**	0.5292
	0.9	0.4881	**0.4845**	0.4967	0.4958	0.4879	*0.4858*	0.5238	0.5153	0.4882	0.4866	0.4867

S5.2 Simulation results for $M = [n/3]$

Table S7: Values of MSE_μ for various methods in the simulation ($n = 50$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.2118	0.3226	0.1211	0.1755	0.2040	0.1611	*0.1115*	**0.0905**	0.1791	0.1270	0.1263
	0.2	0.2865	0.3770	*0.1638*	0.3459	0.2563	0.2009	0.1805	**0.1455**	0.2321	0.1660	0.1655
	0.3	0.3806	0.4414	*0.2188*	0.4845	0.3259	0.2627	0.2592	**0.2088**	0.2969	0.2191	0.2186
	0.4	0.4646	0.4994	*0.2650*	0.5374	0.3823	0.3101	0.3278	0.2687	0.3508	**0.2604**	0.2598
	0.5	0.5378	0.5416	*0.3332*	0.5954	0.4423	0.3649	0.4081	0.3430	0.4132	**0.3174**	0.3171
	0.6	0.6403	0.6298	*0.4117*	0.6117	0.5358	0.4478	0.5110	0.4349	0.5082	**0.3891**	0.3890
	0.7	0.7386	0.7304	*0.5372*	0.7040	0.6446	0.5532	0.6522	0.5615	0.6230	**0.4891**	0.4890
	0.8	0.9041	0.8958	0.7513	0.9091	0.8306	*0.7353*	0.8989	0.7830	0.8266	**0.6678**	0.6679
	0.9	1.3666	1.3886	1.3426	1.5030	1.3323	*1.2357*	1.5064	1.3355	1.3490	**1.1492**	1.1496
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.1766	0.2888	0.1042	0.2420	0.1735	0.1316	*0.0750*	**0.0599**	0.1402	0.1017	0.1012
	0.2	0.2314	0.3332	0.1267	0.3915	0.2111	0.1595	*0.1173*	**0.0895**	0.1731	0.1267	0.1259
	0.3	0.2792	0.3533	0.1579	0.4272	0.2319	0.1753	0.1496	**0.1161**	0.1932	*0.1414*	0.1409
	0.4	0.3090	0.3669	0.1708	0.3918	0.2390	0.1867	0.1709	**0.1381**	0.2089	*0.1563*	0.1558
	0.5	0.3661	0.4086	0.2084	0.3822	0.2744	0.2204	0.2000	**0.1628**	0.2423	*0.1844*	0.1838
	0.6	0.3890	0.4162	0.2330	0.3597	0.3006	0.2393	0.2299	**0.1857**	0.2675	*0.2007*	0.2003
	0.7	0.4214	0.4450	0.2616	0.3726	0.3249	0.2608	0.2742	**0.2237**	0.2903	*0.2238*	0.2236
	0.8	0.4767	0.4869	0.3113	0.3914	0.3757	0.3088	0.3316	*0.2769*	0.3443	**0.2697**	0.2695
	0.9	0.5629	0.5662	0.4276	0.4886	0.4731	0.4000	0.4549	*0.3813*	0.4493	**0.3507**	0.3507
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.1954	0.3109	0.1219	0.1742	0.1991	0.1563	*0.1093*	**0.0901**	0.1720	0.1255	0.1245
	0.2	0.2868	0.3751	*0.1653*	0.3379	0.2662	0.2101	0.1797	**0.1464**	0.2367	0.1721	0.1714
	0.3	0.3816	0.4409	0.2178	0.4569	0.3238	0.2614	0.2550	**0.2079**	0.2967	*0.2167*	0.2160
	0.4	0.4691	0.5004	**0.2594**	0.4874	0.3895	0.3135	0.3257	0.2678	0.3584	*0.2609*	0.2605
	0.5	0.5500	0.5622	*0.3312*	0.5555	0.4578	0.3769	0.4110	0.3457	0.4249	**0.3240**	0.3235
	0.6	0.6369	0.6326	*0.4087*	0.5857	0.5427	0.4538	0.5105	0.4357	0.5111	**0.3941**	0.3940
	0.7	0.7482	0.7339	*0.5408*	0.6891	0.6580	0.5700	0.6670	0.5720	0.6363	**0.5053**	0.5054
	0.8	0.9342	0.9226	0.7552	0.8866	0.8466	*0.7537*	0.9187	0.8043	0.8380	**0.6857**	0.6858
	0.9	1.3727	1.3996	1.3304	1.4753	1.3302	*1.2293*	1.5161	1.3417	1.3505	**1.1543**	1.1545
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.1902	0.2997	0.1069	0.2306	0.1782	0.1374	*0.0811*	**0.0641**	0.1488	0.1060	0.1051
	0.2	0.2310	0.3307	0.1260	0.3586	0.2027	0.1522	*0.1168*	**0.0894**	0.1677	0.1230	0.1219
	0.3	0.2810	0.3483	0.1437	0.3691	0.2212	0.1731	0.1482	**0.1145**	0.1959	*0.1420*	0.1413
	0.4	0.3131	0.3617	0.1639	0.3353	0.2280	0.1802	0.1773	**0.1401**	0.1985	*0.1552*	0.1545
	0.5	0.3507	0.3935	0.1941	0.3325	0.2626	0.2070	0.2027	**0.1639**	0.2296	*0.1783*	0.1776
	0.6	0.3945	0.4186	0.2229	0.3381	0.3007	0.2441	0.2338	**0.1904**	0.2735	*0.2041*	0.2035
	0.7	0.4174	0.4426	0.2524	0.3400	0.3264	0.2602	0.2721	**0.2241**	0.2880	*0.2250*	0.2245
	0.8	0.4773	0.4907	0.3187	0.3921	0.3869	0.3159	0.3345	*0.2785*	0.3528	**0.2724**	0.2720
	0.9	0.5716	0.5651	0.4258	0.4898	0.4834	0.4066	0.4591	*0.3872*	0.4545	**0.3567**	0.3566

Table S8: Values of MSE_μ for various methods in the simulation ($n = 100$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.1593	0.2725	*0.0795*	0.1746	0.1181	0.0928	0.0821	**0.0678**	0.1046	0.0830	0.0825
	0.2	0.2322	0.3237	**0.1104**	0.3165	0.1599	0.1294	0.1341	0.1131	0.1429	*0.1129*	0.1124
	0.3	0.3036	0.3791	**0.1467**	0.3593	0.2132	0.1752	0.1825	0.1566	0.1906	*0.1474*	0.1470
	0.4	0.3611	0.4136	*0.1812*	0.3602	0.2528	0.2110	0.2325	0.2003	0.2314	**0.1759**	0.1756
	0.5	0.4015	0.4411	*0.2207*	0.3639	0.3062	0.2558	0.2847	0.2513	0.2757	**0.2122**	0.2118
	0.6	0.4647	0.4951	*0.2775*	0.3945	0.3696	0.3162	0.3544	0.3154	0.3397	**0.2601**	0.2598
	0.7	0.5486	0.5650	*0.3503*	0.4423	0.4504	0.3923	0.4515	0.4076	0.4198	**0.3235**	0.3234
	0.8	0.6636	0.6619	*0.4752*	0.5467	0.5726	0.5140	0.6144	0.5539	0.5480	**0.4291**	0.4290
	0.9	0.9137	0.9210	0.7994	0.8672	0.8561	*0.7913*	1.0057	0.9198	0.8467	**0.6925**	0.6926
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.1257	0.2460	0.0645	0.2304	0.0899	0.0713	*0.0544*	**0.0424**	0.0781	0.0653	0.0647
	0.2	0.1718	0.2712	0.0821	0.2982	0.1004	0.0803	0.0788	**0.0633**	0.0891	*0.0759*	0.0754
	0.3	0.2090	0.2963	0.0944	0.2648	0.1206	0.0960	0.0948	**0.0779**	0.1076	*0.0889*	0.0884
	0.4	0.2363	0.3139	0.1068	0.2315	0.1357	0.1084	0.1087	**0.0900**	0.1201	*0.0983*	0.0978
	0.5	0.2561	0.3326	0.1182	0.2100	0.1532	0.1234	0.1263	**0.1048**	0.1322	*0.1096*	0.1091
	0.6	0.2764	0.3409	0.1341	0.2051	0.1694	0.1382	0.1460	*0.1221*	0.1508	**0.1210**	0.1206
	0.7	0.3022	0.3632	0.1625	0.2205	0.1870	0.1536	0.1687	*0.1439*	0.1645	**0.1385**	0.1381
	0.8	0.3285	0.3829	0.1856	0.2306	0.2190	0.1822	0.2016	*0.1728*	0.1951	**0.1569**	0.1566
	0.9	0.3968	0.4291	0.2423	0.2748	0.2903	0.2462	0.2715	*0.2366*	0.2620	**0.2057**	0.2055
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.1603	0.2706	*0.0791*	0.1696	0.1191	0.0947	0.0827	**0.0678**	0.1028	0.0851	0.0842
	0.2	0.2400	0.3369	**0.1111**	0.2874	0.1670	0.1352	0.1379	*0.1148*	0.1509	0.1182	0.1174
	0.3	0.3057	0.3782	**0.1423**	0.3184	0.2120	0.1719	0.1841	0.1574	0.1896	*0.1466*	0.1460
	0.4	0.3534	0.4055	**0.1771**	0.3161	0.2500	0.2104	0.2279	0.1994	0.2297	*0.1780*	0.1775
	0.5	0.4059	0.4481	*0.2196*	0.3270	0.2956	0.2497	0.2838	0.2502	0.2710	**0.2113**	0.2109
	0.6	0.4638	0.4931	*0.2663*	0.3520	0.3623	0.3104	0.3545	0.3141	0.3332	**0.2564**	0.2561
	0.7	0.5492	0.5610	*0.3460*	0.4227	0.4492	0.3894	0.4543	0.4050	0.4151	**0.3199**	0.3197
	0.8	0.6578	0.6610	*0.4674*	0.5299	0.5755	0.5113	0.6157	0.5538	0.5477	**0.4277**	0.4276
	0.9	0.9145	0.9229	*0.7854*	0.8468	0.8520	0.7886	1.0115	0.9272	0.8445	**0.6937**	0.6939
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.1349	0.2584	0.0640	0.2153	0.0883	0.0724	*0.0549*	**0.0434**	0.0799	0.0669	0.0660
	0.2	0.1778	0.2764	*0.0790*	0.2525	0.1092	0.0872	0.0808	**0.0640**	0.0961	0.0794	0.0786
	0.3	0.2048	0.2920	0.0888	0.2195	0.1176	0.0925	0.0914	**0.0753**	0.1020	*0.0865*	0.0857
	0.4	0.2363	0.3088	0.1010	0.1921	0.1383	0.1109	0.1099	**0.0911**	0.1208	*0.1004*	0.0998
	0.5	0.2587	0.3262	0.1144	0.1792	0.1482	0.1199	0.1236	**0.1031**	0.1305	*0.1093*	0.1087
	0.6	0.2782	0.3343	0.1337	0.1878	0.1705	0.1417	0.1466	**0.1238**	0.1548	*0.1239*	0.1234
	0.7	0.2979	0.3604	0.1522	0.1938	0.1904	0.1593	0.1686	*0.1435*	0.1714	**0.1385**	0.1380
	0.8	0.3387	0.3902	0.1858	0.2181	0.2272	0.1898	0.2050	*0.1767*	0.2038	**0.1650**	0.1644
	0.9	0.3996	0.4332	0.2455	0.2786	0.2871	0.2420	0.2747	*0.2407*	0.2560	**0.2071**	0.2068

Table S9: Values of MSE_μ for various methods in the simulation ($n = 150$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.1480	0.2713	*0.0653*	0.1718	0.0890	0.0729	0.0722	**0.0605**	0.0838	0.0690	0.0685
	0.2	0.2169	0.3209	**0.0920**	0.2730	0.1249	0.1013	0.1144	0.0988	0.1137	*0.0948*	0.0942
	0.3	0.2665	0.3489	*0.1169*	0.2802	0.1561	0.1317	0.1499	0.1307	0.1446	**0.1161**	0.1157
	0.4	0.3075	0.3818	*0.1459*	0.2762	0.1971	0.1661	0.1896	0.1686	0.1800	**0.1415**	0.1412
	0.5	0.3437	0.4101	*0.1761*	0.2837	0.2314	0.1983	0.2331	0.2088	0.2128	**0.1684**	0.1681
	0.6	0.3952	0.4469	*0.2170*	0.2990	0.2889	0.2499	0.2928	0.2639	0.2638	**0.2041**	0.2039
	0.7	0.4597	0.5009	*0.2721*	0.3358	0.3604	0.3126	0.3682	0.3338	0.3260	**0.2527**	0.2525
	0.8	0.5615	0.5767	*0.3668*	0.4187	0.4579	0.4095	0.4964	0.4564	0.4233	**0.3319**	0.3318
	0.9	0.7595	0.7596	*0.5869*	0.6279	0.6833	0.6313	0.7950	0.7383	0.6581	**0.5192**	0.5192
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.1174	0.2332	0.0494	0.2135	0.0572	0.0459	*0.0449*	**0.0357**	0.0523	0.0495	0.0490
	0.2	0.1563	0.2627	0.0602	0.2198	0.0755	0.0600	0.0599	**0.0496**	0.0683	*0.0594*	0.0589
	0.3	0.1909	0.2844	0.0715	0.1954	0.0894	0.0712	0.0718	**0.0600**	0.0782	*0.0686*	0.0682
	0.4	0.2024	0.2871	0.0812	0.1748	0.0910	0.0746	0.0844	**0.0714**	0.0827	*0.0742*	0.0737
	0.5	0.2132	0.2918	0.0899	0.1588	0.1068	0.0870	0.0969	**0.0820**	0.0962	*0.0826*	0.0822
	0.6	0.2391	0.3191	0.1039	0.1604	0.1221	0.1017	0.1120	*0.0966*	0.1097	**0.0946**	0.0942
	0.7	0.2560	0.3335	0.1170	0.1571	0.1393	0.1161	0.1293	*0.1110*	0.1256	**0.1044**	0.1040
	0.8	0.2866	0.3615	0.1379	0.1662	0.1602	*0.1362*	0.1582	0.1385	0.1470	**0.1234**	0.1230
	0.9	0.3313	0.3909	0.1798	0.2030	0.2105	*0.1795*	0.2059	0.1839	0.1869	**0.1546**	0.1543
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.1475	0.2705	*0.0633*	0.1637	0.0866	0.0690	0.0706	**0.0592**	0.0796	0.0676	0.0668
	0.2	0.2184	0.3192	**0.0902**	0.2346	0.1266	0.1032	0.1153	0.0985	0.1141	*0.0948*	0.0941
	0.3	0.2675	0.3489	**0.1129**	0.2376	0.1541	0.1296	0.1513	0.1327	0.1414	*0.1163*	0.1157
	0.4	0.3143	0.3882	**0.1417**	0.2388	0.1947	0.1656	0.1894	0.1671	0.1794	*0.1421*	0.1416
	0.5	0.3484	0.4125	*0.1734*	0.2494	0.2300	0.1965	0.2314	0.2077	0.2096	**0.1688**	0.1683
	0.6	0.3957	0.4477	*0.2160*	0.2736	0.2826	0.2455	0.2908	0.2614	0.2586	**0.2044**	0.2041
	0.7	0.4730	0.5097	*0.2727*	0.3241	0.3667	0.3215	0.3670	0.3347	0.3328	**0.2579**	0.2575
	0.8	0.5553	0.5717	*0.3656*	0.4108	0.4556	0.4036	0.4977	0.4578	0.4219	**0.3314**	0.3313
	0.9	0.7576	0.7531	*0.5885*	0.6307	0.6801	0.6270	0.7899	0.7369	0.6544	**0.5191**	0.5191
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.1186	0.2378	0.0484	0.1866	0.0584	0.0461	*0.0454*	**0.0359**	0.0533	0.0503	0.0495
	0.2	0.1587	0.2707	*0.0598*	0.1850	0.0756	0.0601	0.0610	**0.0500**	0.0671	0.0606	0.0598
	0.3	0.1818	0.2722	0.0685	0.1595	0.0796	*0.0663*	0.0715	**0.0603**	0.0751	0.0680	0.0673
	0.4	0.2065	0.2910	*0.0776*	0.1360	0.0978	0.0804	0.0852	**0.0721**	0.0882	0.0777	0.0771
	0.5	0.2187	0.3059	0.0864	0.1324	0.1044	0.0851	0.0968	**0.0824**	0.0939	*0.0831*	0.0825
	0.6	0.2387	0.3167	0.1018	0.1403	0.1229	0.1001	0.1113	*0.0957*	0.1074	**0.0933**	0.0927
	0.7	0.2542	0.3238	0.1150	0.1450	0.1369	0.1136	0.1314	*0.1132*	0.1234	**0.1046**	0.1042
	0.8	0.2832	0.3503	0.1338	0.1562	0.1563	*0.1323*	0.1533	0.1340	0.1416	**0.1211**	0.1207
	0.9	0.3314	0.3902	0.1826	0.2009	0.2070	*0.1765*	0.2085	0.1858	0.1859	**0.1561**	0.1558

Table S10: Values of MSE_μ for various methods in the simulation ($n = 200$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.1397	0.2638	*0.0540*	0.1667	0.0723	0.0590	0.0642	**0.0540**	0.0684	0.0573	0.0569
	0.2	0.2056	0.3066	**0.0783**	0.2353	0.0992	0.0827	0.0998	0.0874	0.0942	*0.0798*	0.0794
	0.3	0.2459	0.3360	**0.0991**	0.2261	0.1299	0.1087	0.1311	0.1170	0.1196	*0.0995*	0.0992
	0.4	0.2782	0.3621	*0.1198*	0.2198	0.1578	0.1330	0.1621	0.1452	0.1438	**0.1179**	0.1175
	0.5	0.3165	0.3910	*0.1471*	0.2280	0.1906	0.1636	0.2028	0.1840	0.1759	**0.1427**	0.1424
	0.6	0.3576	0.4243	*0.1828*	0.2482	0.2339	0.2040	0.2519	0.2295	0.2133	**0.1730**	0.1728
	0.7	0.4161	0.4721	*0.2260*	0.2730	0.2986	0.2615	0.3140	0.2889	0.2678	**0.2111**	0.2109
	0.8	0.4939	0.5260	*0.3035*	0.3420	0.3822	0.3416	0.4263	0.3967	0.3522	**0.2757**	0.2756
	0.9	0.6704	0.6705	*0.4844*	0.5150	0.5778	0.5332	0.6781	0.6343	0.5461	**0.4271**	0.4270
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.1114	0.2336	0.0408	0.1941	0.0439	*0.0354*	0.0393	**0.0316**	0.0414	0.0414	0.0408
	0.2	0.1428	0.2404	0.0491	0.1800	0.0521	*0.0424*	0.0503	**0.0422**	0.0496	0.0470	0.0466
	0.3	0.1688	0.2645	0.0586	0.1519	0.0651	*0.0529*	0.0614	**0.0517**	0.0608	0.0558	0.0553
	0.4	0.1821	0.2752	0.0665	0.1372	0.0704	**0.0583**	0.0708	*0.0603*	0.0665	0.0620	0.0616
	0.5	0.1946	0.2849	0.0752	0.1274	0.0781	**0.0655**	0.0814	0.0704	0.0761	*0.0680*	0.0676
	0.6	0.2097	0.2987	0.0828	0.1242	0.0889	*0.0750*	0.0916	0.0802	0.0843	**0.0748**	0.0743
	0.7	0.2265	0.3081	0.0956	0.1288	0.1002	*0.0849*	0.1064	0.0930	0.0955	**0.0835**	0.0831
	0.8	0.2551	0.3320	0.1137	0.1365	0.1234	*0.1050*	0.1294	0.1147	0.1134	**0.0993**	0.0989
	0.9	0.2907	0.3611	0.1465	0.1628	0.1547	*0.1355*	0.1719	0.1548	0.1473	**0.1246**	0.1244
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.1394	0.2648	**0.0533**	0.1567	0.0696	0.0564	0.0637	*0.0539*	0.0658	0.0575	0.0568
	0.2	0.2033	0.3107	**0.0757**	0.1979	0.1018	0.0840	0.1004	0.0876	0.0938	*0.0800*	0.0793
	0.3	0.2476	0.3362	**0.0965**	0.1899	0.1250	0.1058	0.1304	0.1159	0.1182	*0.0986*	0.0980
	0.4	0.2815	0.3663	*0.1198*	0.1922	0.1568	0.1344	0.1634	0.1470	0.1469	**0.1197**	0.1192
	0.5	0.3171	0.3911	*0.1465*	0.2033	0.1958	0.1692	0.2020	0.1830	0.1796	**0.1440**	0.1436
	0.6	0.3618	0.4285	*0.1830*	0.2324	0.2407	0.2115	0.2524	0.2312	0.2209	**0.1753**	0.1749
	0.7	0.4211	0.4686	*0.2275*	0.2650	0.2980	0.2628	0.3186	0.2942	0.2707	**0.2134**	0.2130
	0.8	0.4954	0.5267	*0.3041*	0.3355	0.3845	0.3453	0.4264	0.3958	0.3532	**0.2780**	0.2778
	0.9	0.6659	0.6639	*0.4790*	0.5084	0.5737	0.5257	0.6730	0.6323	0.5358	**0.4232**	0.4231
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.1089	0.2313	0.0391	0.1630	0.0452	*0.0375*	0.0394	**0.0321**	0.0434	0.0413	0.0407
	0.2	0.1417	0.2490	0.0468	0.1376	0.0518	*0.0427*	0.0492	**0.0413**	0.0492	0.0479	0.0472
	0.3	0.1719	0.2687	0.0563	0.1217	0.0657	*0.0540*	0.0610	**0.0516**	0.0622	0.0565	0.0558
	0.4	0.1832	0.2675	0.0634	0.1144	0.0697	**0.0577**	0.0697	*0.0592*	0.0652	0.0614	0.0608
	0.5	0.1929	0.2801	0.0695	0.1060	0.0784	**0.0649**	0.0801	0.0694	0.0734	*0.0670*	0.0665
	0.6	0.2136	0.3007	0.0816	0.1080	0.0911	*0.0766*	0.0926	0.0803	0.0851	**0.0760**	0.0755
	0.7	0.2300	0.3117	0.0925	0.1129	0.1040	*0.0861*	0.1088	0.0951	0.0969	**0.0856**	0.0852
	0.8	0.2504	0.3285	0.1092	0.1277	0.1201	*0.1040*	0.1299	0.1147	0.1148	**0.0988**	0.0984
	0.9	0.2928	0.3606	0.1440	0.1604	0.1603	*0.1384*	0.1719	0.1533	0.1483	**0.1253**	0.1250

Table S11: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 50$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	2.4335	2.6698	2.2872	*2.2529*	2.4883	2.3826	2.2562	**2.2215**	2.4473	2.3093	2.3066
	0.2	2.5462	2.7367	*2.2280*	2.2324	2.4956	2.3416	2.2761	**2.1931**	2.3939	2.2766	2.2749
	0.3	2.7850	2.9158	**2.4568**	2.5798	2.7053	2.6003	2.6502	2.5397	2.6956	*2.5163*	2.5157
	0.4	2.8109	2.8980	**2.3587**	2.4326	2.6075	2.4991	2.4833	2.3868	2.5770	*2.3817*	2.3785
	0.5	3.1337	3.1756	2.7745	2.8770	3.0406	2.8365	2.8972	**2.7497**	2.8613	*2.7604*	2.7603
	0.6	3.0808	3.0271	*2.7801*	2.8871	2.9715	2.7908	3.0086	2.8380	2.9290	**2.7090**	2.7093
	0.7	3.2647	3.2200	*2.8178*	2.8943	3.1346	2.9061	3.1669	2.9946	3.0606	**2.7753**	2.7768
	0.8	3.5069	3.4795	3.3299	3.4625	3.4576	*3.2683*	3.6524	3.4328	3.5165	**3.1462**	3.1495
	0.9	4.5089	4.6381	4.3010	4.3834	4.4075	*4.1759*	4.7571	4.3817	4.3751	**4.0078**	4.0076
$\rho = 0.1$ $\alpha = 1.0$	0.1	2.2717	2.4755	2.0929	**2.0175**	2.2255	2.1799	2.0376	*2.0196*	2.1691	2.1167	2.1166
	0.2	2.1411	2.3163	**1.9566**	*1.9630*	2.1769	2.0560	2.0343	1.9740	2.1344	2.0081	2.0073
	0.3	2.4549	2.5737	2.2355	*2.2288*	2.4234	2.3095	2.3137	**2.2249**	2.3545	2.2544	2.2539
	0.4	2.3531	2.5393	2.1304	*2.1210*	2.3863	2.2351	2.1554	**2.1011**	2.2898	2.1703	2.1693
	0.5	2.8121	2.8316	**2.4098**	2.4611	2.6258	2.5113	2.5259	*2.4402*	2.5954	2.4444	2.4454
	0.6	2.5731	2.5782	2.3298	2.3158	2.4105	*2.3020*	2.3955	2.3374	2.3844	**2.2755**	2.2756
	0.7	2.8484	2.9165	*2.4373*	2.4497	2.6163	2.4829	2.5951	2.4746	2.5796	**2.4307**	2.4306
	0.8	3.0285	3.0600	2.6190	*2.6167*	2.8403	2.6839	2.7659	2.6272	2.7258	**2.6109**	2.6108
	0.9	2.8477	2.9770	2.5190	2.5390	2.7025	2.5491	2.6752	*2.5030*	2.6529	**2.4666**	2.4673
$\rho = 0.6$ $\alpha = 0.5$	0.1	2.3915	2.7468	2.1752	**2.0003**	2.5194	2.3964	2.1726	*2.1245*	2.4618	2.2663	2.2695
	0.2	2.6483	2.8265	2.3662	**2.2163**	2.5846	2.4821	2.3923	*2.3053*	2.4834	2.3930	2.3942
	0.3	2.5746	2.6589	**2.1500**	2.2033	2.3811	2.2557	2.3014	*2.1883*	2.3297	2.1914	2.1893
	0.4	2.6165	2.7429	2.2873	**2.2646**	2.5805	2.4101	2.3743	*2.2729*	2.4860	2.3115	2.3098
	0.5	3.0528	3.1191	2.6769	2.7578	2.8583	2.7153	2.7741	**2.6373**	2.7766	*2.6466*	2.6462
	0.6	3.2916	3.2010	*2.6825*	2.6999	2.9900	2.8066	2.8824	2.7381	2.9402	**2.6675**	2.6697
	0.7	3.5306	3.4177	*3.0254*	3.0348	3.3227	3.1476	3.4166	3.2129	3.3193	**3.0136**	3.0150
	0.8	4.1697	4.1563	*3.7728*	3.8221	3.9589	3.8055	4.1484	3.9393	4.0017	**3.6674**	3.6656
	0.9	4.8924	4.9501	4.7133	4.8579	4.8560	*4.6316*	5.2348	4.8921	4.8839	**4.5143**	4.5148
$\rho = 0.6$ $\alpha = 1.0$	0.1	2.5971	2.8473	2.3655	**2.1549**	2.5384	2.4800	2.3442	*2.3190*	2.5090	2.3989	2.4012
	0.2	2.3051	2.4699	2.0473	**1.9967**	2.2119	2.1026	2.1028	*2.0429*	2.1347	2.0589	2.0616
	0.3	2.4713	2.7001	*2.1174*	**2.0364**	2.4100	2.2722	2.1914	2.1241	2.3570	2.2103	2.2104
	0.4	2.5871	2.6565	2.2429	**2.1631**	2.4033	2.3059	2.3032	*2.2325*	2.3623	2.2574	2.2551
	0.5	2.7205	2.7829	*2.2233*	**2.1594**	2.4618	2.3513	2.3028	2.2453	2.4060	2.3076	2.3067
	0.6	2.7210	2.8243	**2.2686**	*2.2765*	2.5189	2.4042	2.3905	2.3226	2.4634	2.3379	2.3380
	0.7	2.6130	2.7251	2.3460	2.3361	2.4467	2.3705	2.3761	**2.3026**	2.3825	*2.3042*	2.3039
	0.8	2.5794	2.7216	2.4288	2.3991	2.5618	2.4384	2.4856	*2.3600*	2.5366	**2.3493**	2.3500
	0.9	2.8513	2.8443	2.5372	*2.5237*	2.7227	2.5484	2.7370	2.5873	2.7062	**2.4689**	2.4688

Table S12: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 100$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	1.1503	1.2286	**1.0401**	1.0535	1.0557	*1.0443*	1.0671	1.0543	1.0597	1.0455	1.0458
	0.2	1.1663	1.2341	**1.0238**	1.0561	1.0984	1.0676	1.0767	*1.0517*	1.0787	1.0578	1.0574
	0.3	1.2952	1.3857	*1.1501*	1.1697	1.2335	1.1925	1.1704	1.1514	1.1904	**1.1456**	1.1452
	0.4	1.2769	1.3042	*1.0862*	1.1117	1.1772	1.1243	1.1434	1.0990	1.1319	**1.0753**	1.0755
	0.5	1.2736	1.3206	**1.0852**	1.1212	1.1820	1.1286	1.1929	1.1430	1.1528	*1.0870*	1.0870
	0.6	1.3150	1.3639	**1.1723**	1.2172	1.2959	1.2307	1.2749	1.2440	1.2731	*1.1836*	1.1837
	0.7	1.4016	1.4723	*1.2106*	1.2270	1.3302	1.2814	1.3098	1.2560	1.3115	**1.2014**	1.2006
	0.8	1.5240	1.5060	1.3355	1.3640	1.3833	*1.3307*	1.4530	1.4056	1.3627	**1.2707**	1.2706
	0.9	1.8465	1.8643	1.7636	1.7632	1.8144	*1.7475*	1.9104	1.8426	1.8086	**1.6475**	1.6474
$\rho = 0.1$ $\alpha = 1.0$	0.1	1.0433	1.2084	1.0113	*1.0018*	1.0368	1.0231	1.0042	**0.9895**	1.0281	1.0202	1.0193
	0.2	1.1088	1.1492	0.9932	1.0038	1.0112	0.9988	1.0099	**0.9892**	1.0027	*0.9927*	0.9922
	0.3	1.1625	1.2161	*1.0463*	**1.0389**	1.1025	1.0764	1.0796	1.0581	1.0942	1.0510	1.0507
	0.4	1.1691	1.2236	**1.0262**	1.0318	1.0594	1.0343	1.0549	*1.0313*	1.0334	1.0399	1.0392
	0.5	1.2006	1.2565	**1.0432**	1.0497	1.0955	1.0683	1.0688	*1.0464*	1.0650	1.0508	1.0503
	0.6	1.2217	1.3084	*1.0671*	**1.0567**	1.1155	1.0978	1.1051	1.0864	1.0960	1.0830	1.0826
	0.7	1.2261	1.3515	1.1022	1.0941	1.1461	1.1076	*1.0753*	**1.0588**	1.1020	1.0854	1.0847
	0.8	1.2019	1.2517	1.0374	**1.0251**	1.0726	1.0567	1.0794	1.0484	1.0880	*1.0321*	1.0321
	0.9	1.2460	1.2632	*1.1143*	1.1209	1.1986	1.1475	1.1543	1.1229	1.1804	**1.1000**	1.0998
$\rho = 0.6$ $\alpha = 0.5$	0.1	1.1666	1.2565	1.0842	**1.0410**	1.1466	1.1109	1.1011	1.0803	1.1286	*1.0787*	1.0802
	0.2	1.2106	1.3123	1.0418	**1.0237**	1.1090	1.0931	1.0508	*1.0324*	1.0969	1.0659	1.0667
	0.3	1.2996	1.3339	**1.0908**	1.1166	1.1935	1.1484	1.1591	1.1467	1.1821	*1.1152*	1.1166
	0.4	1.4131	1.4888	*1.1718*	**1.1503**	1.2647	1.2327	1.2109	1.1898	1.2508	1.1998	1.1998
	0.5	1.3471	1.3981	**1.1509**	*1.1538*	1.2440	1.1993	1.2012	1.1864	1.1975	1.1638	1.1645
	0.6	1.4055	1.4426	**1.1699**	*1.1709*	1.2816	1.2292	1.2820	1.2443	1.2541	1.1826	1.1822
	0.7	1.4719	1.4760	**1.1579**	*1.1599*	1.3390	1.2881	1.2715	1.2317	1.3088	1.2021	1.2019
	0.8	1.5626	1.5659	*1.4194*	1.4373	1.4986	1.4256	1.5566	1.5029	1.4644	**1.3732**	1.3736
	0.9	1.9126	1.9239	1.8319	1.8329	1.8808	*1.7986*	2.1261	1.9962	1.8433	**1.7092**	1.7088
$\rho = 0.6$ $\alpha = 1.0$	0.1	1.2033	1.3168	1.0721	**1.0053**	1.0927	1.0851	1.0687	*1.0622*	1.0920	1.0860	1.0858
	0.2	1.1221	1.2565	1.0349	**0.9896**	1.0841	1.0648	1.0392	*1.0279*	1.0592	1.0503	1.0504
	0.3	1.2143	1.2894	1.0244	**0.9909**	1.0588	1.0319	1.0205	*1.0065*	1.0424	1.0331	1.0332
	0.4	1.3490	1.4410	1.1794	**1.1415**	1.2368	1.2062	1.1687	*1.1561*	1.2239	1.1934	1.1932
	0.5	1.2044	1.2667	1.0242	**0.9938**	1.0912	1.0541	1.0242	*1.0173*	1.0602	1.0338	1.0334
	0.6	1.2468	1.2843	*1.0818*	**1.0710**	1.1082	1.0865	1.1035	1.0849	1.0940	1.0854	1.0856
	0.7	1.2422	1.3343	*1.1069*	**1.0956**	1.1786	1.1449	1.1604	1.1254	1.1678	1.1181	1.1184
	0.8	1.3047	1.3528	1.1122	**1.1026**	1.1880	1.1403	1.1364	1.1157	1.1547	*1.1099*	1.1101
	0.9	1.3890	1.3790	*1.1977*	1.2078	1.2524	1.2062	1.2353	1.2116	1.2282	**1.1855**	1.1856

Table S13: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 150$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.7126	0.7911	*0.6431*	**0.6369**	0.6677	0.6591	0.6543	0.6452	0.6657	0.6488	0.6486
	0.2	0.7631	0.8120	*0.7283*	0.7481	0.7471	0.7361	0.7525	0.7410	0.7454	**0.7232**	0.7231
	0.3	0.7906	0.8286	0.7027	0.7207	0.7254	0.7173	*0.6996*	**0.6951**	0.7227	0.7012	0.7010
	0.4	0.8756	0.9505	**0.7780**	0.8076	0.8064	0.7890	0.8236	0.8034	0.7921	*0.7832*	0.7830
	0.5	0.8218	0.8269	**0.7172**	0.7414	0.7520	0.7437	0.7755	0.7601	0.7465	*0.7174*	0.7174
	0.6	0.8479	0.8583	*0.7332*	0.7466	0.7767	0.7488	0.7533	0.7410	0.7603	**0.7129**	0.7126
	0.7	0.9543	0.9810	*0.8254*	0.8376	0.8679	0.8378	0.8598	0.8535	0.8373	**0.8010**	0.8006
	0.8	1.0274	1.0483	*0.8910*	0.9025	0.9535	0.9140	0.9920	0.9573	0.9220	**0.8732**	0.8732
	0.9	1.0842	1.0888	*0.9519*	0.9572	1.0161	0.9876	1.1176	1.0706	1.0128	**0.9181**	0.9178
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.6702	0.7389	**0.6344**	0.6435	0.6450	0.6405	0.6421	*0.6359*	0.6435	0.6390	0.6387
	0.2	0.7025	0.7512	0.6285	0.6368	0.6332	*0.6260*	0.6293	**0.6221**	0.6277	0.6285	0.6282
	0.3	0.7565	0.8503	0.6705	0.6794	0.6765	0.6676	*0.6646*	**0.6569**	0.6805	0.6714	0.6710
	0.4	0.8063	0.8304	**0.7088**	*0.7150*	0.7233	0.7193	0.7340	0.7214	0.7226	0.7201	0.7197
	0.5	0.7214	0.7550	*0.6075*	**0.6040**	0.6293	0.6192	0.6206	0.6103	0.6260	0.6169	0.6165
	0.6	0.7718	0.8040	**0.6825**	*0.6825*	0.7158	0.6994	0.7060	0.7005	0.6997	0.6918	0.6918
	0.7	0.7463	0.8012	*0.6837*	0.6837	0.7144	0.6936	0.7021	0.6937	0.7031	**0.6815**	0.6815
	0.8	0.8044	0.8684	0.7185	*0.7155*	0.7286	0.7180	0.7317	0.7183	0.7221	**0.7112**	0.7108
	0.9	0.8369	0.8771	**0.7500**	*0.7530*	0.7827	0.7692	0.8085	0.7901	0.7645	0.7567	0.7568
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.7156	0.7957	0.6448	**0.6310**	0.6586	0.6540	0.6575	*0.6446*	0.6577	0.6486	0.6495
	0.2	0.7860	0.8692	*0.6795*	**0.6727**	0.7180	0.7041	0.7005	0.6864	0.7102	0.6929	0.6931
	0.3	0.7841	0.8566	**0.6742**	0.6766	0.6940	0.6877	0.7349	0.7163	0.7005	*0.6765*	0.6769
	0.4	0.8497	0.9167	**0.7000**	0.7101	0.7426	0.7190	0.7396	0.7255	0.7221	*0.7075*	0.7078
	0.5	0.8547	0.9146	0.7339	**0.7306**	0.7648	0.7416	0.7717	0.7523	0.7435	*0.7316*	0.7310
	0.6	0.8357	0.8618	0.7445	*0.7444*	0.7898	0.7713	0.7877	0.7737	0.7876	**0.7376**	0.7379
	0.7	0.9575	0.9768	*0.8115*	0.8158	0.8674	0.8434	0.8501	0.8285	0.8404	**0.8057**	0.8056
	0.8	0.9735	0.9651	0.8483	*0.8469*	0.9254	0.8830	0.9167	0.8945	0.8973	**0.8303**	0.8303
	0.9	1.0859	1.0980	*0.9783*	0.9877	1.0321	1.0039	1.0900	1.0658	1.0465	**0.9402**	0.9403
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.7284	0.7795	0.6821	**0.6466**	0.6959	0.6891	0.6744	*0.6697*	0.6902	0.6839	0.6848
	0.2	0.6988	0.8081	0.6431	**0.6275**	0.6731	0.6580	*0.6414*	0.6417	0.6602	0.6524	0.6532
	0.3	0.8009	0.8583	0.6883	**0.6657**	0.7056	0.7009	0.6885	*0.6814*	0.7013	0.6984	0.6983
	0.4	0.7307	0.7942	*0.6571*	**0.6482**	0.6689	0.6618	0.6631	0.6596	0.6695	0.6606	0.6608
	0.5	0.7506	0.8288	*0.6607*	**0.6528**	0.6833	0.6660	0.6722	0.6665	0.6733	0.6616	0.6612
	0.6	0.7336	0.7966	0.6571	**0.6479**	0.6677	0.6574	0.6795	0.6652	*0.6550*	0.6633	0.6632
	0.7	0.8401	0.8887	0.7215	0.7093	0.7467	0.7302	*0.7033*	**0.6949**	0.7263	0.7131	0.7128
	0.8	0.8308	0.8775	*0.7116*	**0.7019**	0.7368	0.7185	0.7319	0.7213	0.7325	0.7136	0.7138
	0.9	0.7991	0.8223	0.6738	0.6753	0.6904	0.6733	0.6864	0.6771	**0.6646**	*0.6665*	0.6664

Table S14: Values of $\mathrm{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 200$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.4882	0.5472	0.4526	0.4644	0.4642	0.4530	0.4543	*0.4522*	0.4585	**0.4513**	0.4512
	0.2	0.5389	0.5798	*0.4774*	0.4906	0.4854	0.4784	0.4865	0.4802	0.4828	**0.4765**	0.4762
	0.3	0.5851	0.6105	**0.5140**	0.5289	0.5466	0.5272	0.5292	0.5234	0.5367	*0.5197*	0.5196
	0.4	0.5197	0.5822	**0.4470**	0.4593	0.4802	0.4649	0.4599	*0.4539*	0.4671	0.4561	0.4556
	0.5	0.6224	0.6426	**0.5585**	*0.5647*	0.5936	0.5807	0.5813	0.5740	0.5802	0.5649	0.5650
	0.6	0.6400	0.6768	**0.5595**	0.5676	0.6000	0.5806	0.6035	0.5902	0.5815	*0.5596*	0.5595
	0.7	0.6064	0.6371	*0.4925*	0.5002	0.5444	0.5260	0.5071	0.4977	0.5213	**0.4920**	0.4919
	0.8	0.6620	0.6778	*0.5797*	0.5847	0.6120	0.5910	0.6358	0.6238	0.5935	**0.5676**	0.5676
	0.9	0.7398	0.7437	*0.6909*	0.6978	0.7157	0.6997	0.7833	0.7617	0.7165	**0.6496**	0.6497
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.4892	0.5524	0.4611	0.4661	0.4618	**0.4584**	0.4611	*0.4589*	0.4598	0.4607	0.4605
	0.2	0.5438	0.5692	0.4844	0.4850	0.4833	**0.4795**	0.4908	0.4873	*0.4819*	0.4821	0.4820
	0.3	0.5327	0.5719	**0.4915**	0.4944	0.4978	*0.4938*	0.5050	0.5002	0.4973	0.4957	0.4954
	0.4	0.5184	0.5745	0.4817	0.4824	0.4812	*0.4759*	0.4844	0.4791	**0.4734**	0.4818	0.4813
	0.5	0.5721	0.6206	0.5182	**0.5108**	0.5294	0.5231	0.5266	*0.5180*	0.5291	0.5202	0.5201
	0.6	0.5215	0.5624	0.4612	**0.4605**	0.4696	0.4611	0.4814	0.4754	0.4632	*0.4605*	0.4606
	0.7	0.5267	0.5693	0.4778	0.4760	0.4743	**0.4656**	0.4743	*0.4683*	0.4730	0.4688	0.4686
	0.8	0.5977	0.6413	0.5190	**0.5157**	0.5376	0.5245	0.5197	*0.5173*	0.5227	0.5180	0.5177
	0.9	0.5844	0.6105	0.5151	0.5128	**0.5036**	*0.5045*	0.5348	0.5288	0.5086	0.5072	0.5071
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.4808	0.5451	0.4411	**0.4267**	0.4525	0.4484	0.4433	*0.4394*	0.4482	0.4480	0.4481
	0.2	0.5351	0.6023	*0.4718*	**0.4702**	0.4899	0.4764	0.4849	0.4783	0.4809	0.4751	0.4755
	0.3	0.5567	0.5957	*0.4827*	**0.4825**	0.4987	0.4888	0.4975	0.4916	0.4933	0.4870	0.4873
	0.4	0.6026	0.6387	*0.5155*	**0.5106**	0.5256	0.5166	0.5353	0.5306	0.5225	0.5184	0.5184
	0.5	0.5791	0.6045	**0.5029**	*0.5075*	0.5189	0.5112	0.5326	0.5267	0.5093	0.5079	0.5080
	0.6	0.6844	0.7046	**0.5696**	*0.5738*	0.6162	0.5997	0.5907	0.5823	0.6102	0.5751	0.5751
	0.7	0.6650	0.6849	**0.5417**	*0.5428*	0.5901	0.5727	0.5915	0.5792	0.5813	0.5474	0.5474
	0.8	0.7305	0.7341	*0.6061*	0.6084	0.6473	0.6328	0.6691	0.6563	0.6345	**0.6021**	0.6019
	0.9	0.7556	0.7520	*0.6461*	0.6520	0.7087	0.6848	0.7295	0.7109	0.6923	**0.6396**	0.6394
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.5408	0.6048	0.5066	**0.4835**	0.5118	0.5056	0.5015	*0.4997*	0.5068	0.5062	0.5068
	0.2	0.4905	0.5548	0.4455	**0.4248**	0.4574	0.4531	0.4443	*0.4407*	0.4515	0.4488	0.4491
	0.3	0.5334	0.5714	0.4871	**0.4786**	0.4914	0.4875	0.4875	*0.4840*	0.4919	0.4844	0.4847
	0.4	0.6008	0.6310	*0.5101*	**0.4985**	0.5262	0.5136	0.5231	0.5173	0.5203	0.5150	0.5151
	0.5	0.5770	0.6271	0.5013	**0.4906**	0.5035	*0.4965*	0.5056	0.4988	0.4977	0.5024	0.5024
	0.6	0.5823	0.6122	*0.5182*	**0.5122**	0.5253	0.5219	0.5325	0.5266	0.5186	0.5239	0.5242
	0.7	0.5608	0.5960	0.5055	**0.5017**	0.5171	0.5055	0.5197	0.5116	0.5208	*0.5048*	0.5051
	0.8	0.5569	0.6051	0.4862	*0.4847*	0.4955	**0.4843**	0.4934	0.4898	0.4929	0.4873	0.4869
	0.9	0.5874	0.6251	*0.5251*	0.5254	0.5430	0.5319	0.5386	0.5308	0.5401	**0.5221**	0.5223

S5.3 Simulation results for $\sigma^2 = 1$

Table S15: Values of MSE_μ for various methods in the simulation ($n = 50$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.1797	0.2224	0.1130	0.1801	0.1686	0.1289	0.1076	**0.0880**	0.1587	*0.1074*	0.1070
	0.2	0.2530	0.2734	0.1629	0.3857	0.2222	0.1760	0.1807	**0.1480**	0.2149	*0.1527*	0.1525
	0.3	0.3240	0.3236	0.2092	0.5471	0.2662	0.2173	0.2537	*0.2052*	0.2578	**0.1959**	0.1957
	0.4	0.3769	0.3684	*0.2664*	0.6058	0.3223	0.2666	0.3268	0.2708	0.3158	**0.2437**	0.2437
	0.5	0.4243	0.4153	0.3317	0.6081	0.3736	*0.3150*	0.4001	0.3384	0.3702	**0.2979**	0.2979
	0.6	0.4886	0.4917	0.4227	0.6297	0.4564	*0.4017*	0.5064	0.4362	0.4575	**0.3767**	0.3767
	0.7	0.5806	0.5937	0.5570	0.7131	0.5698	*0.5112*	0.6542	0.5683	0.5740	**0.4888**	0.4887
	0.8	0.7693	0.8021	0.8056	0.9370	0.7715	*0.7211*	0.9043	0.7991	0.7845	**0.6953**	0.6953
	0.9	1.3112	1.3711	1.4672	1.6090	1.3388	*1.2965*	1.5358	1.4167	1.3633	**1.2613**	1.2614
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.1431	0.1896	0.0929	0.2576	0.1233	0.0967	*0.0764*	**0.0614**	0.1134	0.0814	0.0810
	0.2	0.1863	0.2156	0.1208	0.4636	0.1525	0.1166	0.1154	**0.0889**	0.1412	*0.1006*	0.1003
	0.3	0.2348	0.2470	0.1475	0.4768	0.1815	0.1433	0.1517	**0.1180**	0.1735	*0.1251*	0.1248
	0.4	0.2590	0.2568	0.1644	0.4125	0.1909	0.1515	0.1721	**0.1380**	0.1853	*0.1382*	0.1381
	0.5	0.2778	0.2719	0.1852	0.3587	0.2111	0.1691	0.1995	*0.1630*	0.2029	**0.1558**	0.1557
	0.6	0.2924	0.2890	0.2149	0.3457	0.2405	0.1950	0.2269	*0.1862*	0.2337	**0.1791**	0.1789
	0.7	0.3149	0.3127	0.2492	0.3484	0.2643	*0.2177*	0.2667	0.2183	0.2564	**0.2047**	0.2046
	0.8	0.3411	0.3429	0.3144	0.3907	0.3102	*0.2620*	0.3309	0.2783	0.3090	**0.2511**	0.2511
	0.9	0.3943	0.4066	0.4318	0.4970	0.3833	*0.3399*	0.4390	0.3718	0.3865	**0.3279**	0.3280
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.1831	0.2194	0.1147	0.1811	0.1627	0.1273	0.1081	**0.0870**	0.1515	*0.1072*	0.1065
	0.2	0.2487	0.2703	0.1616	0.3753	0.2163	0.1711	0.1836	**0.1500**	0.2097	*0.1527*	0.1522
	0.3	0.3283	0.3325	*0.2046*	0.5017	0.2796	0.2218	0.2537	0.2050	0.2696	**0.1987**	0.1982
	0.4	0.3775	0.3715	*0.2557*	0.5344	0.3233	0.2655	0.3251	0.2688	0.3181	**0.2439**	0.2436
	0.5	0.4286	0.4213	0.3354	0.5495	0.3820	*0.3240*	0.4091	0.3440	0.3781	**0.3029**	0.3027
	0.6	0.4951	0.4940	0.4197	0.5880	0.4556	*0.4005*	0.5106	0.4377	0.4561	**0.3801**	0.3800
	0.7	0.5901	0.6053	0.5567	0.6852	0.5767	*0.5188*	0.6610	0.5784	0.5811	**0.4976**	0.4976
	0.8	0.7762	0.8056	0.8019	0.9181	0.7819	*0.7283*	0.9109	0.8074	0.7949	**0.6999**	0.6999
	0.9	1.3052	1.3616	1.4714	1.6188	1.3395	*1.2962*	1.5494	1.4255	1.3667	**1.2601**	1.2602
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.1419	0.1886	0.0920	0.2571	0.1240	0.0949	*0.0752*	**0.0590**	0.1152	0.0797	0.0790
	0.2	0.1921	0.2273	0.1170	0.4247	0.1569	0.1220	0.1193	**0.0919**	0.1490	*0.1057*	0.1051
	0.3	0.2268	0.2372	0.1387	0.4159	0.1724	0.1361	0.1508	**0.1193**	0.1633	*0.1232*	0.1228
	0.4	0.2567	0.2579	0.1523	0.3488	0.1884	0.1490	0.1733	*0.1352*	0.1819	**0.1350**	0.1347
	0.5	0.2796	0.2786	0.1812	0.3246	0.2139	0.1696	0.1979	*0.1630*	0.2067	**0.1590**	0.1586
	0.6	0.2880	0.2858	0.2114	0.3287	0.2339	0.1892	0.2288	*0.1861*	0.2267	**0.1770**	0.1768
	0.7	0.3108	0.3096	0.2462	0.3272	0.2642	*0.2164*	0.2706	0.2213	0.2582	**0.2046**	0.2044
	0.8	0.3419	0.3450	0.3082	0.3830	0.3045	*0.2592*	0.3276	0.2726	0.3027	**0.2465**	0.2465
	0.9	0.4098	0.4196	0.4287	0.4921	0.3992	*0.3561*	0.4549	0.3909	0.4030	**0.3442**	0.3443

Table S16: Values of MSE_μ for various methods in the simulation ($n = 100$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.1204	0.1384	0.0750	0.1823	0.0966	0.0757	0.0802	**0.0664**	0.0937	*0.0680*	0.0678
	0.2	0.1769	0.1759	*0.1074*	0.3759	0.1360	0.1103	0.1363	0.1143	0.1325	**0.1015**	0.1014
	0.3	0.2054	0.2010	0.1420	0.4403	0.1671	*0.1409*	0.1818	0.1543	0.1657	**0.1293**	0.1292
	0.4	0.2320	0.2334	0.1813	0.4188	0.2055	*0.1764*	0.2284	0.1982	0.2041	**0.1630**	0.1630
	0.5	0.2556	0.2615	0.2215	0.3864	0.2433	*0.2132*	0.2894	0.2529	0.2436	**0.2010**	0.2010
	0.6	0.2939	0.3087	0.2837	0.3985	0.2914	*0.2631*	0.3550	0.3148	0.2933	**0.2508**	0.2508
	0.7	0.3572	0.3765	0.3787	0.4710	0.3681	*0.3419*	0.4543	0.4080	0.3713	**0.3294**	0.3295
	0.8	0.4829	0.5037	0.5349	0.6043	0.5025	*0.4826*	0.6161	0.5615	0.5070	**0.4693**	0.4694
	0.9	*0.8687*	0.8933	0.9846	1.0514	0.8902	0.8810	1.0309	0.9744	0.8993	**0.8649**	0.8650
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.0984	0.1184	0.0588	0.2608	0.0711	0.0550	0.0548	**0.0423**	0.0681	*0.0491*	0.0488
	0.2	0.1289	0.1350	0.0740	0.3850	0.0875	0.0688	0.0777	**0.0625**	0.0845	*0.0629*	0.0627
	0.3	0.1451	0.1453	0.0866	0.2992	0.0985	0.0776	0.0923	*0.0753*	0.0947	**0.0727**	0.0725
	0.4	0.1536	0.1514	0.0998	0.2435	0.1112	*0.0894*	0.1088	0.0901	0.1086	**0.0838**	0.0837
	0.5	0.1662	0.1651	0.1139	0.2158	0.1251	*0.1016*	0.1261	0.1046	0.1227	**0.0937**	0.0936
	0.6	0.1690	0.1696	0.1311	0.2079	0.1334	*0.1113*	0.1452	0.1221	0.1323	**0.1062**	0.1062
	0.7	0.1838	0.1837	0.1507	0.2057	0.1546	*0.1307*	0.1674	0.1437	0.1523	**0.1235**	0.1235
	0.8	0.1970	0.2009	0.1843	0.2270	0.1777	*0.1531*	0.2034	0.1768	0.1772	**0.1474**	0.1475
	0.9	0.2263	0.2375	0.2446	0.2790	0.2254	*0.2024*	0.2737	0.2402	0.2272	**0.1962**	0.1963
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.1211	0.1389	0.0727	0.1815	0.0971	0.0769	0.0811	**0.0669**	0.0945	*0.0691*	0.0688
	0.2	0.1741	0.1741	*0.1056*	0.3587	0.1341	0.1105	0.1367	0.1150	0.1316	**0.1014**	0.1012
	0.3	0.2052	0.2026	0.1395	0.3950	0.1662	*0.1394*	0.1842	0.1569	0.1646	**0.1307**	0.1306
	0.4	0.2303	0.2319	*0.1763*	0.3563	0.2055	0.1768	0.2290	0.2003	0.2045	**0.1645**	0.1644
	0.5	0.2568	0.2631	0.2191	0.3459	0.2438	*0.2131*	0.2876	0.2520	0.2437	**0.2013**	0.2014
	0.6	0.2972	0.3131	0.2818	0.3764	0.2970	*0.2683*	0.3568	0.3182	0.2989	**0.2549**	0.2550
	0.7	0.3624	0.3798	0.3711	0.4392	0.3730	*0.3469*	0.4588	0.4107	0.3762	**0.3331**	0.3331
	0.8	0.4809	0.5041	0.5345	0.5927	0.4990	*0.4788*	0.6165	0.5617	0.5033	**0.4661**	0.4662
	0.9	*0.8677*	0.8957	0.9808	1.0456	0.8936	0.8820	1.0334	0.9720	0.9009	**0.8659**	0.8661
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.0964	0.1186	0.0560	0.2547	0.0717	0.0550	0.0549	**0.0424**	0.0684	*0.0498*	0.0494
	0.2	0.1288	0.1364	0.0709	0.3321	0.0879	0.0689	0.0801	**0.0636**	0.0845	*0.0641*	0.0638
	0.3	0.1418	0.1418	0.0842	0.2567	0.0956	0.0769	0.0926	*0.0759*	0.0932	**0.0724**	0.0722
	0.4	0.1519	0.1504	0.0948	0.1990	0.1073	*0.0875*	0.1066	0.0895	0.1054	**0.0825**	0.0824
	0.5	0.1619	0.1631	0.1107	0.1876	0.1239	*0.1013*	0.1241	0.1044	0.1211	**0.0952**	0.0950
	0.6	0.1721	0.1733	0.1275	0.1853	0.1389	*0.1141*	0.1473	0.1228	0.1361	**0.1081**	0.1081
	0.7	0.1827	0.1818	0.1490	0.1915	0.1546	*0.1297*	0.1689	0.1451	0.1533	**0.1233**	0.1233
	0.8	0.1931	0.1969	0.1761	0.2154	0.1773	*0.1523*	0.2011	0.1727	0.1765	**0.1448**	0.1448
	0.9	0.2263	0.2366	0.2429	0.2690	0.2271	*0.2041*	0.2758	0.2418	0.2292	**0.1972**	0.1973

Table S17: Values of MSE_μ for various methods in the simulation ($n = 150$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.1054	0.1139	*0.0599*	0.1824	0.0788	0.0632	0.0713	0.0600	0.0775	**0.0568**	0.0567
	0.2	0.1371	0.1359	*0.0871*	0.3586	0.1055	0.0878	0.1128	0.0972	0.1041	**0.0800**	0.0799
	0.3	0.1543	0.1535	0.1151	0.3666	0.1310	*0.1113*	0.1502	0.1313	0.1300	**0.1034**	0.1035
	0.4	0.1734	0.1781	0.1448	0.3149	0.1597	*0.1399*	0.1897	0.1678	0.1591	**0.1302**	0.1302
	0.5	0.1924	0.2006	0.1804	0.2999	0.1893	*0.1704*	0.2341	0.2085	0.1898	**0.1600**	0.1601
	0.6	0.2274	0.2403	0.2298	0.3172	0.2328	*0.2144*	0.2929	0.2645	0.2346	**0.2039**	0.2040
	0.7	0.2763	0.2901	0.2928	0.3518	0.2881	*0.2723*	0.3664	0.3345	0.2905	**0.2626**	0.2627
	0.8	*0.3822*	0.3974	0.4275	0.4721	0.3969	0.3860	0.4961	0.4609	0.3993	**0.3776**	0.3777
	0.9	**0.7059**	0.7196	0.7854	0.8260	0.7197	0.7163	0.8307	0.7914	0.7234	*0.7077*	0.7078
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.0780	0.0904	0.0440	0.2595	0.0504	0.0404	0.0447	**0.0351**	0.0501	*0.0375*	0.0373
	0.2	0.0975	0.0996	0.0547	0.3035	0.0631	0.0504	0.0604	*0.0498*	0.0616	**0.0470**	0.0469
	0.3	0.1067	0.1071	0.0650	0.2201	0.0706	*0.0574*	0.0728	0.0608	0.0692	**0.0547**	0.0546
	0.4	0.1133	0.1127	0.0749	0.1818	0.0801	*0.0661*	0.0847	0.0711	0.0792	**0.0623**	0.0622
	0.5	0.1189	0.1191	0.0861	0.1577	0.0880	*0.0735*	0.0978	0.0831	0.0872	**0.0705**	0.0705
	0.6	0.1272	0.1273	0.0974	0.1525	0.1010	*0.0849*	0.1115	0.0962	0.0994	**0.0805**	0.0805
	0.7	0.1332	0.1335	0.1092	0.1471	0.1118	*0.0944*	0.1277	0.1102	0.1104	**0.0891**	0.0890
	0.8	0.1398	0.1439	0.1342	0.1619	0.1286	*0.1112*	0.1545	0.1355	0.1284	**0.1062**	0.1063
	0.9	0.1592	0.1676	0.1774	0.1980	0.1603	*0.1449*	0.2040	0.1802	0.1616	**0.1401**	0.1402
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.1029	0.1115	*0.0584*	0.1819	0.0754	0.0611	0.0708	0.0592	0.0741	**0.0557**	0.0554
	0.2	0.1378	0.1350	*0.0867*	0.3273	0.1051	0.0876	0.1148	0.0985	0.1040	**0.0808**	0.0807
	0.3	0.1558	0.1550	*0.1124*	0.3077	0.1319	0.1126	0.1508	0.1325	0.1308	**0.1044**	0.1044
	0.4	0.1743	0.1786	0.1418	0.2708	0.1608	*0.1405*	0.1892	0.1678	0.1605	**0.1301**	0.1301
	0.5	0.1912	0.2010	0.1747	0.2643	0.1895	*0.1694*	0.2303	0.2066	0.1897	**0.1590**	0.1591
	0.6	0.2234	0.2358	0.2252	0.2892	0.2276	*0.2094*	0.2918	0.2634	0.2292	**0.2006**	0.2007
	0.7	0.2776	0.2926	0.2959	0.3461	0.2885	*0.2737*	0.3670	0.3364	0.2912	**0.2641**	0.2643
	0.8	*0.3818*	0.3966	0.4259	0.4634	0.3967	0.3856	0.5019	0.4641	0.4000	**0.3770**	0.3771
	0.9	**0.7072**	0.7193	0.7831	0.8257	0.7214	0.7178	0.8262	0.7881	0.7254	*0.7091*	0.7092
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.0756	0.0882	0.0418	0.2485	0.0511	0.0403	0.0447	**0.0357**	0.0498	*0.0382*	0.0379
	0.2	0.1008	0.1035	0.0540	0.2511	0.0637	*0.0507*	0.0626	0.0513	0.0621	**0.0483**	0.0481
	0.3	0.1089	0.1080	0.0626	0.1841	0.0732	*0.0591*	0.0727	0.0613	0.0717	**0.0557**	0.0555
	0.4	0.1157	0.1157	0.0727	0.1502	0.0822	*0.0669*	0.0853	0.0718	0.0808	**0.0637**	0.0635
	0.5	0.1181	0.1181	0.0802	0.1321	0.0879	*0.0724*	0.0969	0.0816	0.0867	**0.0690**	0.0689
	0.6	0.1272	0.1269	0.0937	0.1341	0.1003	*0.0838*	0.1104	0.0947	0.0995	**0.0795**	0.0795
	0.7	0.1326	0.1327	0.1081	0.1388	0.1120	*0.0943*	0.1304	0.1135	0.1113	**0.0908**	0.0908
	0.8	0.1436	0.1461	0.1319	0.1568	0.1316	*0.1146*	0.1568	0.1369	0.1315	**0.1088**	0.1089
	0.9	0.1640	0.1722	0.1788	0.2006	0.1656	*0.1501*	0.2070	0.1846	0.1667	**0.1445**	0.1446

Table S18: Values of MSE_μ for various methods in the simulation ($n = 200$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.0899	0.0946	*0.0495*	0.1822	0.0639	0.0521	0.0639	0.0537	0.0630	**0.0469**	0.0468
	0.2	0.1139	0.1116	0.0742	0.3383	0.0880	*0.0741*	0.0984	0.0859	0.0874	**0.0681**	0.0681
	0.3	0.1295	0.1315	0.0979	0.2982	0.1124	*0.0971*	0.1295	0.1151	0.1119	**0.0894**	0.0893
	0.4	0.1420	0.1472	0.1218	0.2596	0.1339	*0.1178*	0.1630	0.1465	0.1336	**0.1095**	0.1095
	0.5	0.1608	0.1680	0.1542	0.2485	0.1607	*0.1462*	0.2024	0.1824	0.1611	**0.1373**	0.1373
	0.6	0.1895	0.2003	0.1928	0.2604	0.1947	*0.1805*	0.2524	0.2309	0.1964	**0.1723**	0.1724
	0.7	0.2347	0.2472	0.2567	0.3071	0.2461	*0.2344*	0.3176	0.2931	0.2479	**0.2264**	0.2265
	0.8	*0.3265*	0.3375	0.3663	0.4040	0.3379	0.3312	0.4314	0.4021	0.3402	**0.3249**	0.3250
	0.9	**0.6074**	0.6154	0.6704	0.6994	0.6160	0.6150	0.6989	0.6724	0.6186	*0.6091*	0.6091
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.0661	0.0752	0.0361	0.2541	0.0401	0.0323	0.0389	*0.0311*	0.0394	**0.0306**	0.0305
	0.2	0.0835	0.0853	0.0452	0.2429	0.0517	*0.0419*	0.0513	0.0429	0.0508	**0.0394**	0.0393
	0.3	0.0885	0.0884	0.0533	0.1809	0.0579	*0.0469*	0.0599	0.0506	0.0567	**0.0446**	0.0445
	0.4	0.0900	0.0907	0.0591	0.1440	0.0624	*0.0518*	0.0707	0.0602	0.0617	**0.0497**	0.0497
	0.5	0.0994	0.0991	0.0687	0.1235	0.0729	*0.0617*	0.0814	0.0699	0.0720	**0.0576**	0.0576
	0.6	0.1042	0.1040	0.0785	0.1202	0.0807	*0.0678*	0.0913	0.0793	0.0794	**0.0641**	0.0641
	0.7	0.1096	0.1099	0.0922	0.1237	0.0922	*0.0784*	0.1082	0.0945	0.0914	**0.0738**	0.0738
	0.8	0.1172	0.1206	0.1098	0.1332	0.1074	*0.0934*	0.1302	0.1152	0.1072	**0.0886**	0.0886
	0.9	0.1304	0.1380	0.1435	0.1592	0.1322	*0.1195*	0.1704	0.1529	0.1329	**0.1150**	0.1151
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.0906	0.0943	*0.0495*	0.1815	0.0644	0.0525	0.0650	0.0547	0.0637	**0.0477**	0.0476
	0.2	0.1169	0.1142	*0.0730*	0.3003	0.0901	0.0751	0.1008	0.0879	0.0891	**0.0689**	0.0688
	0.3	0.1296	0.1306	*0.0962*	0.2468	0.1124	0.0974	0.1310	0.1166	0.1120	**0.0889**	0.0889
	0.4	0.1433	0.1465	0.1207	0.2183	0.1343	*0.1186*	0.1649	0.1481	0.1340	**0.1106**	0.1106
	0.5	0.1637	0.1712	0.1533	0.2266	0.1628	*0.1478*	0.2026	0.1838	0.1632	**0.1383**	0.1384
	0.6	0.1903	0.2011	0.1928	0.2450	0.1962	*0.1822*	0.2510	0.2301	0.1972	**0.1738**	0.1739
	0.7	0.2339	0.2460	0.2501	0.2850	0.2438	*0.2327*	0.3157	0.2904	0.2456	**0.2251**	0.2253
	0.8	*0.3265*	0.3368	0.3593	0.3894	0.3375	0.3302	0.4290	0.4007	0.3396	**0.3242**	0.3243
	0.9	**0.6049**	0.6126	0.6651	0.6945	0.6145	0.6127	0.7025	0.6769	0.6161	*0.6077*	0.6078
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.0680	0.0752	0.0350	0.2429	0.0406	0.0327	0.0398	*0.0322*	0.0399	**0.0315**	0.0313
	0.2	0.0824	0.0838	0.0436	0.2025	0.0505	*0.0412*	0.0514	0.0428	0.0496	**0.0394**	0.0392
	0.3	0.0880	0.0881	0.0498	0.1413	0.0576	*0.0466*	0.0610	0.0516	0.0563	**0.0452**	0.0451
	0.4	0.0924	0.0929	0.0585	0.1204	0.0651	*0.0538*	0.0703	0.0597	0.0644	**0.0507**	0.0506
	0.5	0.0978	0.0971	0.0660	0.1093	0.0706	*0.0595*	0.0795	0.0690	0.0696	**0.0565**	0.0564
	0.6	0.1041	0.1031	0.0772	0.1066	0.0815	*0.0692*	0.0931	0.0803	0.0807	**0.0648**	0.0647
	0.7	0.1090	0.1097	0.0876	0.1069	0.0906	*0.0772*	0.1077	0.0946	0.0900	**0.0737**	0.0736
	0.8	0.1176	0.1205	0.1045	0.1212	0.1067	*0.0937*	0.1306	0.1151	0.1063	**0.0885**	0.0886
	0.9	0.1316	0.1386	0.1424	0.1599	0.1332	*0.1206*	0.1707	0.1535	0.1336	**0.1156**	0.1157

Table S19: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 50$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	2.3463	2.4610	2.2018	*2.1788*	2.2888	2.2336	2.1974	**2.1643**	2.2935	2.2120	2.2114
	0.2	2.4075	2.3861	*2.2233*	2.3180	2.3382	2.2433	2.2787	**2.2229**	2.3303	2.2344	2.2347
	0.3	2.5834	2.5546	**2.2903**	2.4632	2.4963	2.3481	2.4657	2.3516	2.4435	*2.3132*	2.3122
	0.4	2.7951	2.8021	**2.5596**	2.6984	2.7322	2.6352	2.6766	2.5703	2.7446	*2.5633*	2.5645
	0.5	2.5418	2.5485	2.3489	2.4797	2.4923	2.3318	2.4233	*2.3205*	2.4438	**2.3056**	2.3058
	0.6	2.7379	2.7849	2.6708	2.7353	2.6875	*2.6393*	2.9173	2.7766	2.7206	**2.5685**	2.5661
	0.7	3.1503	3.1009	3.0130	3.1035	3.0902	*2.9883*	3.2207	3.0610	3.1026	**2.9271**	2.9273
	0.8	3.3510	3.4394	3.5328	3.6437	3.3425	*3.2757*	3.7971	3.5518	3.3690	**3.2254**	3.2259
	0.9	4.7502	4.8426	4.7626	4.7972	4.8175	*4.7076*	5.0328	4.8091	4.8446	**4.6144**	4.6146
$\rho = 0.1$ $\alpha = 1.0$	0.1	2.2736	2.4017	2.1263	*2.0819*	2.2304	2.1606	2.0932	**2.0647**	2.1834	2.1407	2.1401
	0.2	2.2244	2.2873	*2.0493*	**2.0405**	2.1541	2.0778	2.1125	2.0507	2.1262	2.0540	2.0543
	0.3	2.3795	2.4302	*2.1814*	**2.1790**	2.2879	2.2230	2.2675	2.2030	2.3074	2.1850	2.1842
	0.4	2.4431	2.4556	2.1867	2.2203	2.2753	2.1928	2.2210	**2.1601**	2.2812	*2.1800*	2.1804
	0.5	2.6505	2.6520	*2.4077*	2.4657	2.4984	2.4110	2.4890	2.4130	2.4695	**2.3912**	2.3910
	0.6	2.5058	2.5452	2.3702	2.3411	2.4117	*2.3327*	2.4398	2.3412	2.3909	**2.3240**	2.3245
	0.7	2.5256	2.4971	2.3792	2.3904	2.3997	*2.3337*	2.4572	2.3633	2.4006	**2.2890**	2.2899
	0.8	2.5817	2.6132	*2.4187*	**2.3849**	2.5510	2.4776	2.6016	2.4990	2.5481	2.4316	2.4306
	0.9	2.6891	2.7155	2.7261	2.6992	2.7300	*2.6177*	2.8287	2.6837	2.7332	**2.6006**	2.6011
$\rho = 0.6$ $\alpha = 0.5$	0.1	2.2377	2.2454	2.0700	**2.0060**	2.1102	2.0603	2.0905	2.0519	2.0974	*2.0305*	2.0326
	0.2	2.5192	2.5855	2.3126	**2.2208**	2.4359	2.3552	2.3642	*2.3036*	2.4119	2.3070	2.3074
	0.3	2.5584	2.5939	*2.3258*	**2.3038**	2.4823	2.3915	2.5026	2.3892	2.4602	2.3558	2.3558
	0.4	2.6686	2.6349	*2.3490*	**2.2927**	2.5354	2.4177	2.4833	2.3785	2.5142	2.3550	2.3551
	0.5	2.7819	2.7973	2.5378	**2.5259**	2.7398	2.5932	2.7155	2.6232	2.6956	*2.5370*	2.5374
	0.6	2.9476	2.9681	2.8076	*2.8031*	2.8860	2.8063	3.0177	2.9019	2.8942	**2.7588**	2.7584
	0.7	3.1788	3.2082	3.1347	3.1962	3.1604	*3.0636*	3.3902	3.2205	3.1879	**3.0263**	3.0269
	0.8	3.5468	3.5933	3.4926	3.5135	3.4814	*3.4276*	3.8639	3.5852	3.5571	**3.3667**	3.3656
	0.9	4.4869	4.5859	4.6678	4.7624	4.5195	*4.4597*	4.9739	4.7192	4.6163	**4.3910**	4.3902
$\rho = 0.6$ $\alpha = 1.0$	0.1	2.1098	2.1914	1.9922	**1.8354**	2.0997	2.0341	2.0123	*1.9633*	2.0790	2.0055	2.0044
	0.2	2.4161	2.4544	*2.1944*	**2.1073**	2.3015	2.2334	2.2796	2.2092	2.2854	2.1994	2.2003
	0.3	2.4777	2.4803	2.2394	**2.1311**	2.2935	2.2378	2.3366	2.2498	2.2722	*2.2259*	2.2252
	0.4	2.2441	2.2749	*1.9828*	**1.9238**	2.1124	2.0512	2.1137	2.0426	2.1140	2.0165	2.0187
	0.5	2.4083	2.4222	2.2004	**2.1572**	2.2846	*2.1924*	2.2601	2.2116	2.2711	2.1985	2.1966
	0.6	2.3980	2.4236	2.2699	**2.2227**	2.3723	2.2898	2.3513	2.2848	2.3716	*2.2454*	2.2461
	0.7	2.6576	2.6556	2.4506	**2.4129**	2.5668	2.4847	2.5668	2.4667	2.5613	*2.4449*	2.4463
	0.8	2.7509	2.7635	2.5796	2.5802	2.6272	2.5539	2.5754	**2.5132**	2.6198	*2.5172*	2.5176
	0.9	2.8173	2.8408	2.7409	2.7563	2.8260	*2.6961*	2.9525	2.7825	2.7849	**2.6721**	2.6722

Table S20: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 100$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	1.0198	1.0444	0.9903	*0.9871*	1.0322	1.0039	0.9928	**0.9826**	1.0242	0.9922	0.9917
	0.2	1.1157	1.1002	*1.0713*	1.1273	1.0914	1.0763	1.1112	1.0922	1.0847	**1.0698**	1.0701
	0.3	1.1077	1.0901	**1.0168**	1.0907	1.0687	1.0339	1.0850	1.0589	1.0633	*1.0276*	1.0278
	0.4	1.1996	1.1913	**1.1313**	1.1942	1.1591	1.1362	1.2006	1.1701	1.1575	*1.1324*	1.1324
	0.5	1.2138	1.2082	*1.1495*	1.1670	1.1829	1.1563	1.2332	1.1966	1.1817	**1.1401**	1.1403
	0.6	1.2765	1.2892	1.2445	1.2662	1.2734	*1.2340*	1.3023	1.2725	1.2795	**1.2182**	1.2180
	0.7	1.3674	1.3955	1.3414	1.3629	1.3591	*1.3322*	1.4372	1.3898	1.3663	**1.3252**	1.3252
	0.8	1.5895	1.6103	1.5882	1.5939	1.5772	*1.5649*	1.6476	1.5979	1.5845	**1.5567**	1.5567
	0.9	**1.6718**	1.7046	1.8144	1.8567	1.6940	1.6891	1.8689	1.8265	1.6982	*1.6847*	1.6850
$\rho = 0.1$ $\alpha = 1.0$	0.1	1.0158	1.0436	0.9797	**0.9485**	1.0078	0.9897	0.9765	*0.9630*	1.0046	0.9770	0.9769
	0.2	1.0623	1.0714	**0.9925**	1.0298	1.0239	*0.9979*	1.0251	1.0115	1.0233	1.0058	1.0058
	0.3	1.0053	1.0075	0.9690	0.9664	0.9663	**0.9468**	0.9789	0.9590	0.9695	*0.9487*	0.9484
	0.4	1.1623	1.1596	*1.1071*	1.1112	1.1398	1.1188	1.1226	**1.1069**	1.1353	1.1084	1.1084
	0.5	1.1756	1.1654	1.1219	1.1185	1.1236	*1.1115*	1.1439	1.1178	1.1304	**1.0988**	1.0986
	0.6	1.0670	1.0782	1.0253	1.0207	1.0366	**1.0108**	1.0517	1.0333	1.0338	*1.0144*	1.0144
	0.7	1.1486	1.1438	1.0936	1.0975	1.1236	1.0912	1.0990	**1.0701**	1.1242	*1.0759*	1.0758
	0.8	1.1610	1.1514	1.1672	1.1806	1.1460	*1.1191*	1.1815	1.1543	1.1347	**1.1191**	1.1193
	0.9	1.0956	1.1111	1.0865	1.0925	1.1012	*1.0793*	1.1493	1.1078	1.1030	**1.0679**	1.0678
$\rho = 0.6$ $\alpha = 0.5$	0.1	1.0572	1.0666	0.9958	**0.9454**	1.0299	1.0101	1.0029	*0.9918*	1.0301	0.9977	0.9975
	0.2	1.1100	1.1076	*1.0139*	**1.0028**	1.0576	1.0359	1.0628	1.0421	1.0502	1.0201	1.0202
	0.3	1.1865	1.1812	**1.1209**	1.1745	1.1600	1.1345	1.1891	1.1707	1.1598	*1.1287*	1.1293
	0.4	1.1525	1.1591	*1.1024*	1.1265	1.1189	1.1048	1.1436	1.1193	1.1136	**1.0893**	1.0894
	0.5	1.2241	1.2227	**1.1423**	*1.1502*	1.1919	1.1636	1.2318	1.1912	1.2007	1.1525	1.1522
	0.6	1.2627	1.2885	1.2610	1.2670	1.2405	*1.2217*	1.2801	1.2512	1.2541	**1.2076**	1.2082
	0.7	1.2928	1.3231	1.2992	1.3184	1.3206	*1.2862*	1.4167	1.3675	1.3219	**1.2764**	1.2760
	0.8	1.3070	1.3328	1.3660	1.3807	1.3315	*1.3040*	1.4341	1.3819	1.3314	**1.2861**	1.2868
	0.9	**1.8133**	1.8473	1.9681	1.9726	1.8362	1.8331	2.0049	1.9462	1.8417	*1.8188*	1.8194
$\rho = 0.6$ $\alpha = 1.0$	0.1	1.0934	1.0958	1.0339	**0.9736**	1.0775	1.0505	1.0320	*1.0222*	1.0710	1.0402	1.0405
	0.2	1.0808	1.0815	1.0056	**0.9629**	1.0208	1.0071	1.0154	*1.0039*	1.0142	1.0116	1.0118
	0.3	1.1395	1.1468	1.0734	**1.0304**	1.0892	1.0654	1.0885	*1.0636*	1.0817	1.0668	1.0667
	0.4	1.0636	1.0782	1.0087	**0.9783**	1.0189	*1.0049*	1.0246	1.0122	1.0176	1.0062	1.0064
	0.5	1.1618	1.1598	1.0920	**1.0656**	1.1088	1.0893	1.1053	1.0838	1.1078	*1.0835*	1.0841
	0.6	1.2247	1.2242	1.1619	**1.1419**	1.2046	1.1628	1.1660	1.1492	1.2012	*1.1461*	1.1461
	0.7	1.1785	1.1786	*1.1052*	**1.0959**	1.1412	1.1188	1.1352	1.1186	1.1479	1.1077	1.1073
	0.8	1.1511	1.1639	1.1003	**1.0884**	1.1481	1.1122	1.1370	1.1241	1.1488	*1.0985*	1.0989
	0.9	1.1900	1.1904	*1.1421*	**1.1412**	1.1788	1.1577	1.2208	1.1812	1.1872	1.1496	1.1495

Table S21: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 150$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.6899	0.6849	**0.6530**	0.6648	0.6622	0.6562	0.6687	0.6611	0.6600	*0.6543*	0.6542
	0.2	0.7044	0.6983	0.6845	0.7045	0.6915	*0.6800*	0.6935	0.6873	0.6888	**0.6750**	0.6752
	0.3	0.7573	0.7635	*0.7274*	0.7625	0.7339	0.7277	0.7511	0.7412	0.7353	**0.7233**	0.7233
	0.4	0.7760	0.7748	**0.7431**	0.7638	0.7715	0.7547	0.7655	0.7576	0.7678	*0.7463*	0.7462
	0.5	0.7240	0.7235	0.7147	0.7361	0.7263	0.7119	0.7237	*0.7101*	0.7271	**0.7002**	0.7002
	0.6	**0.7538**	0.7633	0.7734	0.7959	0.7727	0.7574	0.8191	0.8011	0.7803	*0.7562*	0.7564
	0.7	0.7898	0.7990	*0.7741*	0.7832	0.7908	0.7748	0.8038	0.7931	0.7907	**0.7685**	0.7685
	0.8	0.8483	0.8480	0.8386	0.8434	0.8415	*0.8372*	0.8894	0.8663	0.8439	**0.8317**	0.8316
	0.9	**1.1486**	1.1649	1.2037	1.2285	*1.1559*	1.1594	1.2398	1.2154	1.1580	1.1562	1.1563
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.6600	0.6701	*0.6382*	0.6446	0.6532	0.6446	0.6443	**0.6370**	0.6532	0.6403	0.6403
	0.2	0.7145	0.7114	*0.6752*	0.6772	0.6891	0.6861	0.6814	**0.6742**	0.6854	0.6809	0.6806
	0.3	0.7067	0.6995	*0.6757*	0.6813	0.6846	0.6762	0.6904	0.6794	0.6864	**0.6743**	0.6744
	0.4	0.7332	0.7285	0.6991	0.7042	0.6988	**0.6936**	0.7031	0.6970	0.6964	*0.6940*	0.6938
	0.5	0.6920	0.6956	0.6842	0.6787	0.6732	**0.6697**	0.6971	0.6933	0.6744	*0.6717*	0.6717
	0.6	0.7114	0.7090	**0.6712**	0.6779	0.6856	0.6735	0.6992	0.6870	0.6810	*0.6731*	0.6731
	0.7	0.6640	0.6603	*0.6261*	0.6267	0.6442	0.6318	0.6336	0.6274	0.6407	**0.6242**	0.6241
	0.8	0.6937	0.6935	0.6960	0.6963	0.6869	**0.6757**	0.7164	0.7019	0.6904	*0.6774*	0.6774
	0.9	0.6250	0.6290	0.6430	0.6448	*0.6205*	**0.6149**	0.6702	0.6563	0.6242	0.6230	0.6231
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.7068	0.7085	0.6823	**0.6464**	0.6864	0.6761	0.6863	0.6771	0.6860	*0.6711*	0.6710
	0.2	0.7515	0.7400	*0.7236*	0.7321	0.7445	0.7275	0.7327	0.7253	0.7438	**0.7216**	0.7219
	0.3	0.7334	0.7285	**0.7041**	0.7131	0.7250	0.7088	0.7321	0.7284	0.7239	*0.7047*	0.7050
	0.4	0.7304	0.7328	*0.6991*	0.7009	0.7152	0.7050	0.7104	0.7033	0.7167	**0.6902**	0.6904
	0.5	0.6565	0.6640	0.6528	0.6615	0.6531	*0.6406*	0.6981	0.6801	0.6558	**0.6395**	0.6396
	0.6	0.7892	0.7976	0.7759	0.7803	0.7807	**0.7691**	0.8381	0.8144	0.7770	*0.7694*	0.7696
	0.7	*0.7392*	0.7475	0.7733	0.7809	0.7454	**0.7346**	0.8273	0.8148	0.7481	0.7415	0.7416
	0.8	*0.8201*	0.8306	0.8382	0.8439	0.8305	0.8252	0.9014	0.8710	0.8364	**0.8141**	0.8139
	0.9	*1.0542*	1.0621	1.0900	1.0997	1.0678	1.0629	1.1580	1.1145	1.0703	**1.0534**	1.0534
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.6649	0.6728	0.6510	**0.6279**	0.6621	0.6523	0.6523	*0.6466*	0.6599	0.6490	0.6491
	0.2	0.7382	0.7433	0.6904	**0.6607**	0.6932	0.6877	0.6928	*0.6870*	0.6931	0.6886	0.6885
	0.3	0.7115	0.6999	0.6758	**0.6596**	0.6864	*0.6741*	0.6918	0.6841	0.6842	0.6744	0.6746
	0.4	0.6975	0.6920	0.6582	**0.6414**	0.6779	0.6620	0.6640	*0.6570*	0.6793	0.6579	0.6580
	0.5	0.6773	0.6834	*0.6441*	**0.6346**	0.6724	0.6591	0.6643	0.6553	0.6708	0.6506	0.6506
	0.6	0.7027	0.7110	0.6822	**0.6739**	0.6895	0.6777	0.6974	0.6908	0.6904	*0.6757*	0.6757
	0.7	0.7375	0.7372	0.7156	0.7117	0.7186	*0.7112*	0.7439	0.7303	0.7171	**0.7102**	0.7103
	0.8	0.7411	0.7471	0.7257	0.7234	0.7339	*0.7202*	0.7519	0.7444	0.7373	**0.7197**	0.7199
	0.9	0.7039	0.7121	0.6955	*0.6913*	0.7029	0.6925	0.7246	0.7102	0.6992	**0.6880**	0.6882

Table S22: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 200$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
$\rho = 0.1$ $\alpha = 0.5$	0.1	0.4662	0.4675	0.4540	0.4663	0.4572	*0.4525*	0.4649	0.4581	0.4592	**0.4508**	0.4508
	0.2	0.4928	0.4920	*0.4713*	0.5011	0.4771	0.4721	0.4884	0.4826	0.4790	**0.4692**	0.4692
	0.3	0.4769	0.4771	0.4638	0.4899	0.4644	*0.4563*	0.4706	0.4619	0.4596	**0.4546**	0.4545
	0.4	0.5189	0.5196	*0.5090*	0.5215	0.5152	0.5137	0.5379	0.5274	0.5187	**0.5050**	0.5052
	0.5	0.5199	0.5274	0.5202	0.5374	0.5137	*0.5084*	0.5409	0.5337	0.5146	**0.5065**	0.5065
	0.6	0.5562	0.5580	0.5509	0.5560	0.5565	*0.5488*	0.5682	0.5585	0.5575	**0.5440**	0.5440
	0.7	0.5644	0.5701	*0.5598*	0.5662	0.5673	0.5641	0.5991	0.5881	0.5692	**0.5563**	0.5562
	0.8	**0.6174**	0.6229	0.6562	0.6626	0.6266	0.6214	0.6734	0.6583	0.6255	*0.6214*	0.6215
	0.9	**0.7385**	0.7443	0.7714	0.7764	0.7477	0.7450	0.7717	0.7669	0.7511	*0.7424*	0.7424
$\rho = 0.1$ $\alpha = 1.0$	0.1	0.5232	0.5243	**0.4925**	0.5023	0.4971	0.4964	0.5023	0.4984	*0.4958*	0.4991	0.4989
	0.2	0.4883	0.4951	0.4721	**0.4708**	0.4785	0.4747	0.4752	*0.4718*	0.4784	0.4725	0.4725
	0.3	0.5029	0.5089	0.4918	0.4935	0.4957	0.4884	0.4934	*0.4883*	0.4954	**0.4868**	0.4868
	0.4	0.5013	0.5024	**0.4845**	0.4866	0.4877	0.4870	0.4951	0.4900	0.4883	*0.4865*	0.4866
	0.5	0.4714	0.4682	*0.4425*	**0.4398**	0.4531	0.4515	0.4497	0.4433	0.4509	0.4459	0.4457
	0.6	0.4716	0.4682	0.4540	0.4580	0.4530	**0.4483**	0.4668	0.4638	*0.4509*	0.4531	0.4531
	0.7	0.5076	0.5082	0.4956	0.4975	0.4897	**0.4861**	0.5022	0.4974	0.4886	*0.4864*	0.4864
	0.8	0.4954	0.4969	0.4953	0.4944	0.4962	*0.4899*	0.5012	0.4962	0.4947	**0.4891**	0.4891
	0.9	0.5015	0.5030	*0.4902*	0.4923	0.4994	0.4912	0.5046	0.4979	0.4984	**0.4870**	0.4870
$\rho = 0.6$ $\alpha = 0.5$	0.1	0.4966	0.4948	0.4793	**0.4761**	0.4841	0.4793	0.4884	0.4841	0.4831	*0.4774*	0.4776
	0.2	0.5485	0.5418	*0.5278*	0.5441	0.5339	0.5289	0.5432	0.5370	0.5348	**0.5257**	0.5257
	0.3	0.5508	0.5531	0.5093	**0.5031**	0.5322	0.5225	0.5128	*0.5086*	0.5326	0.5147	0.5145
	0.4	0.5526	0.5539	*0.5217*	**0.5161**	0.5422	0.5315	0.5431	0.5314	0.5426	0.5242	0.5241
	0.5	0.5640	0.5629	0.5523	0.5519	0.5588	*0.5506*	0.5705	0.5558	0.5566	**0.5434**	0.5434
	0.6	0.6095	0.6160	0.6040	0.6072	0.6113	*0.6034*	0.6231	0.6202	0.6143	**0.5967**	0.5968
	0.7	**0.5672**	0.5779	0.5824	0.5836	0.5782	0.5729	0.6195	0.6123	0.5785	*0.5704*	0.5706
	0.8	*0.6281*	0.6355	0.6351	0.6376	0.6367	0.6307	0.6746	0.6645	0.6379	**0.6270**	0.6270
	0.9	*0.7422*	0.7468	0.7483	0.7557	0.7463	0.7449	0.7793	0.7708	0.7495	**0.7402**	0.7402
$\rho = 0.6$ $\alpha = 1.0$	0.1	0.5302	0.5290	0.5108	**0.4839**	0.5118	*0.5060*	0.5103	0.5067	0.5092	0.5072	0.5072
	0.2	0.5050	0.5063	*0.4827*	**0.4642**	0.4897	0.4849	0.4862	0.4832	0.4893	0.4831	0.4832
	0.3	0.4964	0.4938	*0.4701*	**0.4603**	0.4791	0.4727	0.4821	0.4765	0.4797	0.4732	0.4732
	0.4	0.5005	0.5015	0.4770	**0.4705**	0.4844	0.4774	0.4770	*0.4728*	0.4849	0.4770	0.4771
	0.5	0.4925	0.4964	0.4782	**0.4688**	0.4836	0.4771	0.4883	0.4851	0.4821	*0.4755*	0.4755
	0.6	0.5137	0.5092	0.4978	**0.4903**	0.4988	0.4950	0.5178	0.5092	0.4978	*0.4920*	0.4921
	0.7	0.4857	0.4854	0.4768	0.4759	0.4741	**0.4675**	0.4859	0.4806	0.4746	*0.4697*	0.4697
	0.8	0.5296	0.5345	0.5220	**0.5187**	0.5252	0.5215	0.5438	0.5357	0.5255	*0.5198*	0.5199
	0.9	0.5108	0.5134	0.5105	0.5109	0.5127	**0.5069**	0.5367	0.5279	0.5148	*0.5081*	0.5082

S5.4 Simulation under varying model misspecification

Following the simulation setup in Section 5.1, we generate a finite population according to

$$y_i = \sum_{j=1}^{101} \beta_j x_{ij} + e_i, \qquad i = 1, \ldots, 2000,$$

where $\beta_j = \sqrt{1.5} j^{-1}$ for $j = 1, \ldots, 51$ and $\beta_j = 0$ for $j = 52, \ldots, 101$. We set $x_{i1} = 1$, while the remaining covariates $\{x_{ij}\}$and errors $\{e_i\}$ are independent and identically distributed (i.i.d) as $N(0, 1)$. We again consider the nested candidate models, with the mth model containing the first m regressors. We vary the number of candidate models M across $\{11, 21, 31, 41, 51, 61, 71, 81, 91, 101\}$ to examine different degrees of model misspecification. When $M < 51$, all candidate models are misspecified because they omit relevant regressors. When $M = 51$, the largest candidate model coincides with the true model. We draw a sample of size $n = 300$ from the finite population using πPS sampling, with inclusion probabilities specified as in Section 5.1. The procedure is repeated $D = 1000$ times to compare the model selection and averaging methods (a)–(k). We also report the results for the largest candidate model (LARGE).

Tables S23–S24 show that the proposed methods (SMA1 and SMA2) yield similar results and perform best overall. When the candidate models are severely underfitted ($M = 11$), the largest-model estimator performs well because it incorporates the most information. As M increases, the proposed methods improve and achieve the best performance among the methods considered. When the largest candidate model is overfitted ($M > 51$), its performance gradually deteriorates as M increases, possibly because including irrelevant covariates increases the estimation variance. However, the proposed methods continue to perform best. Moreover, the model averaging methods (SAIC, SBIC, and SMA1) generally outperform their model selection counterparts (AIC, BIC, and C_p). As M increases, the performance of AGG1 improves, ranking second only to the proposed methods in terms of MSE_μ.

Table S23: Values of MSE_μ for various methods in the simulation

	M	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	LARGE	SMA1	SMA2
$\rho = 0.1$	11	*0.1432*	0.1472	0.1631	0.2111	0.1479	0.1466	0.1814	0.1704	0.1483	**0.1426**	0.1472	0.1473
	21	0.1203	0.1279	0.1278	0.1761	0.1262	*0.1172*	0.1752	0.1610	0.1271	0.1181	**0.1113**	0.1114
	31	0.1410	0.1470	0.1206	0.1707	0.1330	*0.1190*	0.1773	0.1631	0.1324	0.1355	**0.1054**	0.1054
	41	0.1672	0.1700	*0.1164*	0.1651	0.1369	0.1205	0.1771	0.1621	0.1349	0.1626	**0.1028**	0.1028
	51	0.1983	0.2023	*0.1176*	0.1646	0.1448	0.1269	0.1797	0.1639	0.1410	0.2052	**0.1059**	0.1058
	61	0.2210	0.2314	*0.1162*	0.1646	0.1434	0.1251	0.1763	0.1616	0.1378	0.2507	**0.1045**	0.1044
	71	0.2377	0.2629	*0.1187*	0.1656	0.1431	0.1260	0.1785	0.1636	0.1389	0.3031	**0.1074**	0.1073
	81	0.2570	0.2974	*0.1184*	0.1650	0.1477	0.1297	0.1789	0.1639	0.1417	0.3636	**0.1088**	0.1086
	91	0.2736	0.3311	*0.1180*	0.1644	0.1465	0.1274	0.1768	0.1617	0.1400	0.4232	**0.1087**	0.1084
	101	0.2858	0.3589	*0.1182*	0.1624	0.1497	0.1303	0.1790	0.1636	0.1402	0.4879	**0.1099**	0.1097
$\rho = 0.6$	11	*0.1442*	0.1471	0.1634	0.2016	0.1487	0.1476	0.1804	0.1700	0.1488	**0.1434**	0.1480	0.1481
	21	0.1194	0.1268	0.1249	0.1607	0.1255	*0.1163*	0.1773	0.1626	0.1263	0.1168	**0.1106**	0.1108
	31	0.1422	0.1475	0.1202	0.1573	0.1338	*0.1191*	0.1771	0.1626	0.1336	0.1358	**0.1055**	0.1056
	41	0.1691	0.1717	*0.1166*	0.1528	0.1389	0.1217	0.1790	0.1637	0.1367	0.1652	**0.1041**	0.1041
	51	0.1995	0.2032	*0.1167*	0.1508	0.1454	0.1279	0.1765	0.1613	0.1416	0.2051	**0.1056**	0.1055
	61	0.2194	0.2307	*0.1162*	0.1517	0.1443	0.1260	0.1776	0.1625	0.1389	0.2491	**0.1064**	0.1063
	71	0.2384	0.2622	*0.1156*	0.1511	0.1457	0.1268	0.1778	0.1619	0.1389	0.3036	**0.1063**	0.1061
	81	0.2547	0.2957	*0.1157*	0.1490	0.1447	0.1270	0.1772	0.1633	0.1390	0.3605	**0.1077**	0.1074
	91	0.2727	0.3282	*0.1158*	0.1485	0.1454	0.1261	0.1768	0.1615	0.1384	0.4233	**0.1083**	0.1080
	101	0.2814	0.3586	*0.1165*	0.1499	0.1469	0.1285	0.1780	0.1622	0.1391	0.4876	**0.1098**	0.1094

Table S24: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation

	M	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	LARGE	SMA1	SMA2
$\rho = 0.1$	11	*0.3354*	0.3364	0.3454	0.3515	0.3363	0.3370	0.3501	0.3483	0.3366	**0.3352**	0.3382	0.3383
	21	*0.2981*	0.3006	0.3051	0.3088	0.3029	0.2992	0.3236	0.3196	0.3049	**0.2976**	0.3005	0.3005
	31	0.3565	0.3567	*0.3447*	0.3487	0.3487	0.3450	0.3618	0.3570	0.3471	0.3541	**0.3414**	0.3414
	41	0.3426	0.3403	*0.3210*	0.3223	0.3293	0.3234	0.3387	0.3331	0.3260	0.3407	**0.3206**	0.3205
	51	0.3287	0.3344	0.3220	0.3272	0.3266	*0.3204*	0.3431	0.3368	0.3214	0.3346	**0.3153**	0.3152
	61	0.3173	0.3262	0.3073	0.3179	0.3093	*0.3048*	0.3249	0.3202	0.3051	0.3318	**0.2999**	0.2999
	71	0.3520	0.3584	*0.3094*	0.3114	0.3197	0.3118	0.3266	0.3233	0.3137	0.3624	**0.3056**	0.3056
	81	0.3524	0.3647	*0.3015*	0.3048	0.3061	0.3021	0.3187	0.3153	0.3063	0.3873	**0.2983**	0.2983
	91	0.3775	0.4071	0.3348	0.3400	0.3366	*0.3333*	0.3534	0.3493	0.3381	0.4249	**0.3314**	0.3313
	101	0.3768	0.3966	0.3236	0.3250	0.3290	*0.3229*	0.3362	0.3329	0.3264	0.4446	**0.3196**	0.3194
$\rho = 0.6$	11	*0.3459*	0.3481	0.3525	0.3508	0.3472	0.3468	0.3576	0.3541	0.3473	**0.3456**	0.3473	0.3473
	21	0.3129	0.3147	0.3134	0.3161	0.3115	**0.3092**	0.3278	0.3222	0.3119	0.3124	*0.3099*	0.3099
	31	0.3216	0.3219	*0.3122*	0.3139	0.3173	0.3126	0.3287	0.3250	0.3155	0.3163	**0.3081**	0.3081
	41	0.3384	0.3414	0.3328	0.3322	0.3388	*0.3305*	0.3554	0.3489	0.3380	0.3407	**0.3230**	0.3231
	51	0.3546	0.3575	0.3299	*0.3282*	0.3429	0.3357	0.3473	0.3441	0.3402	0.3581	**0.3254**	0.3255
	61	0.3544	0.3527	*0.3236*	0.3244	0.3332	0.3243	0.3428	0.3404	0.3325	0.3656	**0.3202**	0.3204
	71	0.3593	0.3656	0.3118	*0.3104*	0.3158	**0.3101**	0.3381	0.3329	0.3134	0.3837	0.3106	0.3106
	81	0.3836	0.4004	0.3333	*0.3330*	0.3447	0.3383	0.3461	0.3441	0.3411	0.4178	**0.3330**	0.3329
	91	0.3752	0.3936	0.3237	0.3292	0.3281	0.3235	0.3373	0.3350	*0.3204*	0.4299	**0.3194**	0.3195
	101	0.3826	0.4056	0.3241	*0.3231*	0.3349	0.3271	0.3454	0.3408	0.3321	0.4561	**0.3211**	0.3212

S5.5 Simulation of weight selection consistency

Following the simulation setup in Section 5.1, we generate a finite population according to

$$y_i = \sum_{j=1}^{11} \beta_j x_{ij} + e_i, \qquad i = 1, \ldots, 2000,$$

where $\beta_j = cj^{-1}$ for $j = 1, \ldots, 6$ and $\beta_j = 0$ for $j = 7, \ldots, 11$. We set $x_{i1} = 1$, and the remaining covariates $\{x_{ij}\}$ and errors $\{e_i\}$ are i.i.d. as $N(0, 1)$. The parameter c controls the R^2 of the model. We again consider nested candidate models, with the mth model containing the first m regressors. Thus, the 6th through 11th candidate models are correct models, as they contain the true model. We draw a sample from the finite population using πPS sampling, with inclusion probabilities specified as in Section 5.1, and repeat the procedure $D = 1000$ times. Figure S1 shows the sum of the estimated weights assigned by SMA1 to the correct models across different sample sizes. This sum approaches one as the sample size increases, numerically verifying the selection consistency of the weights established in Theorem 4. Convergence becomes faster as R^2 increases.

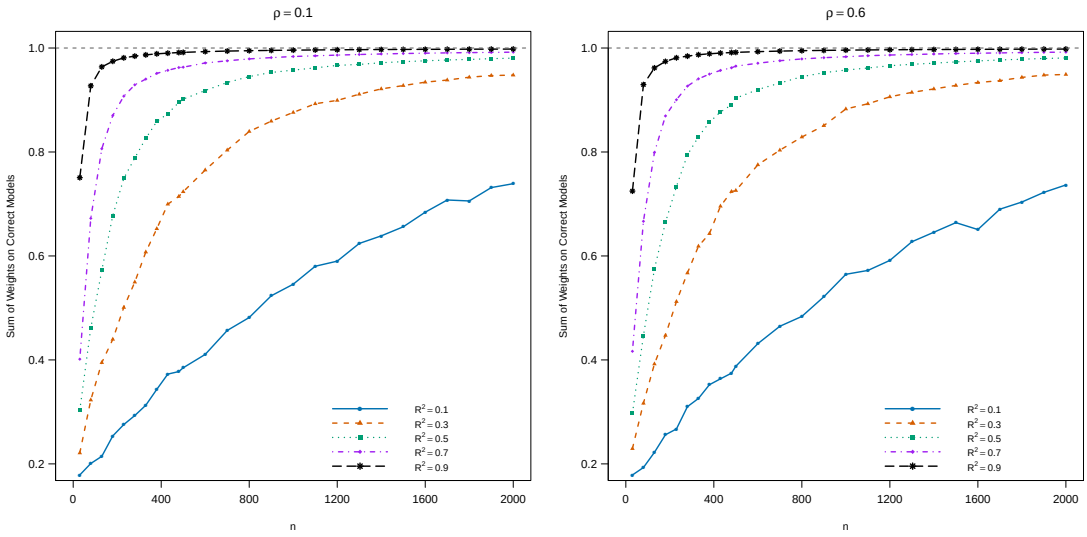


Figure S1: Sum of the estimated weights assigned by SMA1 to correct models across different sample sizes ($\rho = 0.1$, left; $\rho = 0.6$, right).

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